

2D to 3D Translation with Medical Image Applications

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Abstract

To translate between medical images of different modalities, one option is to rely on novel deep learning approaches within the broad task of image-to-image translation. Over the past decade, this problem has spawned many computer vision subtasks, including image inpainting, domain adaptation, semantic segmentation, super-resolution, among others. However, adapting or designing deep learning frameworks for real-world data, such as medical imaging, imposes additional considerations in terms of data availability, structure, and topology.

In particular, 2D to 3D medical image translation entails further complexities:

1. Lack of unpaired datasets makes learning direct mappings difficult, thus precise transformations cannot be guaranteed.
2. Dimensionality mismatch calls for a generative modeling formulation where issues such as poor spatial alignment and impractical data representations can lead to inaccurate translations.
3. Hallucination can prove fatal in medical contexts. Unlike in a “natural image” setting, which can focus on aesthetics or creativity, medical imaging relies on the precise representation of anatomical structures.

The thesis makes advancements in such areas and works towards having positive outcomes in applications like 3D CT reconstruction from 2D X-rays, involving lower radiation doses for patients, cost savings, and increased accessibility. Specifically, the research work presented in this thesis explores different deep learning approaches for the task of 2D to 3D translation, considering necessary data domain and technical aspects. Our proposed methods generally outperform competitive models and introduce novel ideas with potential use in other frameworks or relevant applications. These methods have been implemented in PyTorch and are open source for the research community.

This work aims to contribute to the problem of 2D to 3D image translation using deep learning. To complement existing methods for CT reconstruction from X-rays, which primarily focus on adapting architectures to medical images, we concentrate on studying the data structure and making well-founded design decisions to effectively solve modeling issues and achieve state-of-the-art results. This thesis is also intended to be accessible to researchers without a computer science background, promoting multidisciplinary collaboration with clinicians and biomedical experts.

Declaration

The work in this thesis is based on research carried out at the Department of Computer Science, Durham University, United Kingdom. No part of this thesis has been submitted elsewhere for any other degree or qualification and it is all my own work unless referenced to the contrary in the text.

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Notation

$a, b, c; \alpha, \beta, \omega$ Scalars are lowercase

$\mathbf{x}, \mathbf{y}, \mathbf{z}; \mathbf{x}, \mathbf{y}, \mathbf{z}$ Vectors are bold lowercase

$\mathbf{A}, \mathbf{B}, \mathbf{C}$ Matrices are bold uppercase

X, Y, Z Vector spaces or random variables in Probability Theory

$\mathcal{X}, \mathcal{Y}, \mathcal{Z}$ Sets of random variables

$p(\cdot), q(\cdot)$ Probability density functions

$f(\mathbf{x})$ Function on signal $\mathbf{x}$

$\mathbb{R}^n$ n -dimensional vector space of real numbers

D Number of dimensions; indexed by $d = 1, \dots, D$

N Number of data points; indexed by $n = 1, \dots, N$

$a := b$ a is defined as b

$\mathbf{A} \oplus \mathbf{B}$ Concatenation of matrices $\mathbf{A}$ and $\mathbf{B}$

$I(\cdot)$ Information content

$\mathbb{E}_p[\cdot]$ Expectation operator in the distribution p

$\|\cdot\|$ Norm; Euclidean, unless specified

$D_{KL}(p||q)$ Kullback-Leibler divergence between p and q

$JS(p||q)$ Jensen-Shannon divergence between p and q

$S_\varepsilon(\cdot)$	Sinkhorn divergence
$\boldsymbol{\theta}, \boldsymbol{\phi}$	Parameter vectors
$\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$	Gaussian distribution with mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$
$\boldsymbol{I}$	Identity matrix
$\mathbf{0}$	Matrix of zeros
$\mathbf{1}$	Matrix of ones

List of Abbreviations & Acronyms

i.i.d.	Independent, identically distributed
ADA	Adaptive Data Augmentation
CNN	Convolutional Neural Network
CT	Computed Tomography
DA	Data Augmentation
DAG	Data Augmentation Optimized for GAN
DDPM	Denoising Diffusion Probabilistic Model
DGM	Deep Generative Modeling
DL	Deep Learning
DRR	Digitally Reconstructed Radiograph
FID	Frechet Inception Distance
GAN	Generative Adversarial Network
GRAF	Generative Radiance Fields
GPU	Graphics Processing Unit
GRU	Gated Recurrent Unit
I2I	Image-to-Image
INR	Implicit Neural Representation

KID	Kernel Inception Distance
LPIPS	Learned Perceptual Image Patch Similarity
LSTM	Long Short-Term Memory
MAE	Mean Absolute Error
MCGE	Monte Carlo Gradient Estimator
MCMC	Markov Chain Monte Carlo
MIP	Maximum Intensity Projections
MLP	Multi-Layer Perceptron
MMD	Maximum Mean Discrepancies
MRI	Magnetic Resonance Imaging
MSE	Mean Square Error
NeRF	Neural Radiance Fields
NLL	Negative Log-Likelihood
NLP	Natural Language Processing
OT	Optimal Transport
PDF	Probability Density Function
PMF	Probability Mass Function
PSNR	Peak Signal-to-Noise Ratio
RNN	Recurrent Neural Network
SDF	Signed Distance Function
SGA	Stochastic Gradient Ascent
SGD	Stochastic Gradient Descent
SOTA	State-of-the-art
SSIM	Structural Similarity Index
VAE	Variational Auto-Encoder
VGG	Visual Geometry Group architecture
ViT	Vision Transformer
VQ	Vector-Quantized

Dedication

To my loving family

CHAPTER 1

Introduction

Medical imaging is a valuable tool for clinicians to provide quality healthcare. The ability to support and increase the accuracy of diagnosis is crucial to the decision-making process of clinicians. Furthermore, after diagnosis, many medical conditions require careful follow-up to ensure progress of treatment and patient healing, for which different imaging modalities can provide objective information to related parties.

Various imaging modalities have main uses, including X-ray images to visualize bones, Magnetic Resonance Imaging (MRI) for detailed imaging of soft tissues, Ultrasound for prenatal imaging, and Computed Tomography (CT) for diagnosing traumatic injuries, among others. CT is especially a powerful and widely used diagnostic tool. However, it has several limitations that make it unsuitable for patients with certain characteristics.

The primary concern is the high dose of ionizing radiation exposure compared to conventional X-rays, which can increase the risk of cancer and long-term malignancies [20]. Additionally, CT imaging is generally more expensive than other modalities. Moreover, CT scanners may be limited due to the high costs of the equipment and the need for specialized facilities and personnel.

Deep learning researchers working towards obtaining 3D medical images in a cheaper and non-harmful manner investigate whether it is possible to reconstruct 3D CT scans from lower-dimensional 2D X-ray images [21]. However, they encounter numerous challenges, including bypassing the need to acquire paired datasets of CT scans and X-ray projections for accurate generation, achieving robustness in imaging contrast differences, and reducing hallucination of anatomical structures. In this thesis, I study and design deep learning approaches that aim to work towards these important aspects: data constraints, generation quality, and generalization.

Starting from the fundamentals of deep learning image translation and their use in medical imaging, I focused on designing novel approaches that achieve accurate image generation while considering complex modeling issues such as data dimensionality mismatch (i.e., 3D vs. 2D images) and their structural differences. The goal of this thesis is to contribute to the area of research that aims to translate 2D (X-rays) to 3D (CT) medical images, with potential use in clinical settings. In this context, the following major aspects are identified:

1. **Meeting data constraints:** Overcoming the limitations of medical datasets, which can be small in size and exhibit modality inhomogeneities.
2. **Learning useful data representations:** Disentangling anatomical structures regardless of their modality.
3. **High-quality reconstructions:** Addressing the complex modeling issues inherent in 2D to 3D image translation.

The thesis is dedicated to the generative modeling problem of 2D to 3D image translation with the medical application of CT reconstruction from X-rays. The deep learning approaches proposed here aim to make contributions to the previously mentioned aspects. While the presented frameworks and experiments focus on medical images, they can be adapted to other real-world inspired applications.

1.1 Motivation

Image-to-image (I2I) translation is a broad problem within the realm of deep generative modeling [22, 23]. The origin of this task goes back to the general idea of converting an input image to a target domain while preserving the content of the input. The advent and successful adaptation of convolutions into neural networks [24] facilitated their use for a wide range of image processing tasks, including image translation. As a result, image translation became an integral part of numerous applications in computer vision, such as style transfer [25, 26], super-resolution [27, 28], and image inpainting [29, 30], showcasing the versatility and power of neural networks in handling complex visual transformations.

The first deep learning approaches for I2I translation were based on convolutional neural networks (CNNs) and Generative Adversarial Networks (GANs) [31]. These early methods were applied to tasks like style transfer and super-resolution. Later advancements, such as CycleGAN [6], introduced cycle consistency for bidirectional mappings without paired data, while Pix2Pix [5] used paired datasets for precise translations. These improvements enhanced facial attribute manipulation, image inpainting, and the generation of realistic photographs from semantic layouts. However, these methods were initially not used for medical images, where preserving important diagnostic details is crucial.

Designing and training deep learning approaches for medical imaging is significantly different from the “natural images” setting. The main reasons for this are the continued dependency on data for the success of deep learning frameworks, as their large number of parameters require substantial amounts of data to train effectively. In particular, aspects such as mode coverage [32], preventing mode collapse [33], and generalization [34] are heavily dependent on access to large amounts of quality data. This dependency limits the straightforward adaptation of state-of-the-art (SOTA) deep learning models to real-world datasets, including medical imaging, and has thus given rise to major training strategies in the field such as semi-supervised learning [35], self-supervised learning [36] and unsupervised learning [37].

Accessing medical datasets can be done through public resources from research institutions or via collaboration with clinical experts. The former is often easier, as

these collections of data typically undergo some kind of processing and/or labeling. However, it may involve applying for access, and in most cases, the data format originates from the imaging system used, requiring further processing. In contrast, obtaining data through collaboration with a hospital entails a lengthy process to secure necessary permits and ethical clearance, as well as additional data processing to make it suitable for deep learning purposes. In both cases, these conditions limit the acquisition of large amounts of data.

Training with limited medical data poses significant challenges for deep learning frameworks, including poor mode coverage, which leads to model biases and an inability to capture the entire variability of the problem space. Increased model complexity exacerbates issues such as overfitting or underfitting, critically affecting model performance. Despite the development and improvement of various regularization techniques, such as data augmentation [38] and transfer learning [39], these methods still require careful design to be effective and to indeed enhance results in medical settings. Addressing these limitations is crucial when dealing with the translation between medical images of different modalities.

Translating between medical images of different modalities is a complex generative problem that involves learning two data distributions. Despite sharing common features, solutions often require training two deep learning models, with designs and learning objectives tailored to the structure of the data for content preservation and style transfer. This is a complicated task due to domain-specific constraints for each distribution, which may require additional objectives. Moreover, such models have increased computational complexity, training time, and memory requirements.

The work presented in this thesis deals with the previously mentioned aspects and proposes deep learning frameworks that extend beyond the translation of images with the same dimensionality. This extension aims to contribute to the task of 2D to 3D image translation with particular application in X-rays and CT scans.

1.2 Thesis Contributions

The main contributions of this thesis are as follows:

- An extensive literature review of the fundamental concepts and approaches for image-to-image translation, with a focus on studying the structure of the data and a special emphasis on medical imaging.
- To the best of my knowledge, the MedNeRF work is the first to adapt neural radiance fields for medical imaging. It learns to reconstruct CT projections from a few or even a single-view X-ray through a continuous representation of CT scans by disentangling the shape and volumetric depth of surface and internal anatomical structures from 2D images.
- The introduction of a new 2D to 3D image translation method that allows 3D volume reconstructions to retain valuable information from the 2D inputs. This is achieved through a simple and effective processing technique and training regularization based on optimal transport. The proposed approach significantly improves generalization, training stability, and efficiency.
- A novel method that operates in an information-rich code space, enabling unaligned 2D to 3D synthesis via full-coverage attention across input views. This method demonstrates state-of-the-art performance on two complex volumetric imagery datasets.

1.3 Publications

The work presented in this thesis has been peer-reviewed and published. The list below indicates the chapters associated:

- Corona-Figueroa, A., Frawley, J., Bond-Taylor, S., Bethapudi, S., Shum, H. P., and Willcocks, C. G., (2022). MedNeRF: Medical Neural Radiance Fields for Reconstructing 3D-aware CT-projections from a Single X-ray. In *The 44th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*, pages 3843-3848 (Contributing to Chapter 3).

- Corona-Figueroa, A., Bond-Taylor, S., Bhowmik, N, Gaus, Y. F. A., Breckon T. P., Shum, H. P., and Willcocks, C. G., (2023). Unaligned 2D to 3D Translation with Conditional Vector-Quantized Code Diffusion using Transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 14585-14594 (Contributing to Chapter 5).
- Corona-Figueroa, A., Shum, H.P. and Willcocks, C.G., (2024). Repeat and Concatenate: 2D to 3D Image Translation with 3D to 3D Generative Modeling. In *Workshop on Data Curation and Augmentation in Medical Imaging, CVPR*. Best Paper Award. (Contributing to Chapter 4).

1.4 Thesis Scope and Structure

Chapter 2: medical imaging and deep learning for image translation.

First of all, this thesis focuses on laying the theoretical foundations of the presented work: the space of data and deep learning architectural properties that make them suitable for image-to-image translation tasks. Through a medical imaging perspective, this literature review examines the generation of CT scans from limited X-ray projections, aiming to lower costs and minimize the harmful effects of ionizing radiation on patients. I review seminal works in image-to-image translation for both natural and medical images and explain their motivation, theory and contributions.

Chapter 3: from X-rays to neural representations of CT-projections.

Then I proceed to study implicit representations for generating 3D CT-like images by parameterizing neural radiance fields. The proposed framework has potential applications in other medical imaging modalities where 2D multi-view images are accessible. The published manuscript of this work resulted from a collaboration with Dr Sarath Bethapudi, the lead consultant Musculoskeletal radiologist for County Durham Darlington NHS Foundation Trust (CDDFT).

Chapter 4: reformulation into 3D generative modeling. Here I explore the possibility of achieving correlated translated outputs while reducing the loss of 2D

input information through a processing step that may be equivalent to no encoding. In contrast to the approach presented in the previous chapter, here the proposed method is able to generalize to out-of-distribution inputs without needing subsequent fine-tuning of the model’s parameters but rather with an effective dual optimal transport regularization. The published manuscript of this work received the Best Paper Award at the CVPR 2024 Workshop on Data Curation and Augmentation in Medical Imaging.

Chapter 5: unaligned translation of X-rays to CT scans. In this chapter, I consider a more complex aspect of image translation: removing the need for both paired and aligned datasets. These features are highly desirable, particularly when dealing with real-world datasets, such as medical images. To this end, the proposed framework is based on transformers, absorbing diffusion and vector-quantized representations. Additionally, this chapter presents experiments demonstrating applicability in the complex domain of X-ray security imaging, where we encounter higher variability in data topologies.

Chapter 6: conclusions and future work. Finally, I discuss the conclusions of this thesis and present preliminary directions for future work in terms of alternative data representations for 2D to 3D translation.

CHAPTER 2

Literature Review

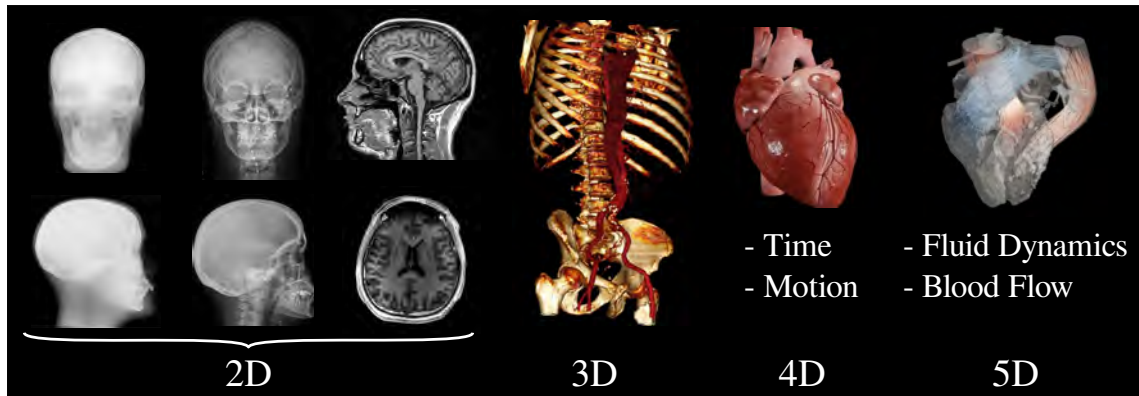


Figure 2.1: **Illustration of various dimensionalities in medical imaging.** There are different degrees of quality and contrast aspects in 2D images often used to visualize bones and abnormalities in the chest. 3D imaging combines multiple 2D projections providing detailed anatomical information including organs and blood vessels. Moving to 4D images, also known as real-time 3D provides dynamic volumetric structure of organs for evaluating their function. While advanced techniques such as cardiac 5D CT can even capture details like blood flow dynamics and tissue deformation. Medical imaging is a well-defined field where physicists, engineers and computer scientists team up to bridge the gap between a patient's anatomy and a physician's diagnosis.

This chapter presents the fundamental concepts and theory behind deep learning for image-to-image translation and reviews seminal works in both medical imaging

and broader applications involving natural images. The general problem of translating data from one space to another requires analyzing the structure of the data from both source and target domains. This analysis is crucial as it involves understanding the shape, geometry (i.e., space), and dimensionality of the data.

First, I introduce the problem of image-to-image translation, which involves representing content from the source domain in an equivalent and realistic manner in the target domain. The complexity of this task varies based on the nature of the domains, data restrictions, and application requirements of the desired output (Figure 2.1). For instance, translating medical images between MRI images and CT scans for tumor segmentation involves integrating information from both modalities to improve diagnosis and planning. However, MRI emphasizes soft tissues, while CT emphasizes bone and dense structures, making direct translation complex. Additionally, high-quality paired datasets are often scarce, and imaging modalities may have different resolutions, noise levels, and artifacts, further complicating the translation process.

Most deep learning models handle image translations within the same dimensionality (e.g., 2D to 2D or 3D to 3D), which poses a more straightforward mapping between source and target domains as it requires only learning the direct mapping and ensuring realism without generating new data. However, translating between data spaces with differing dimensionalities involves generating new data and translating these new data points into the target domain, a task complicated by the model’s limited access to comprehensive input domain information.

To this end, a model capable of translating between different dimensionalities should also handle translations within the same dimensionality. This is a more general problem which falls under the realm of deep generative modeling [40], as it involves transforming an input distribution to an output distribution via neural networks. This process requires learning the underlying distribution of data to generate new, similar samples. Learning both source and output data distributions and translating samples between them is a complex yet fascinating task with diverse applications. In this thesis, I focus primarily on the field of medical imaging with the special application of 2D X-rays to 3D CT image translation.

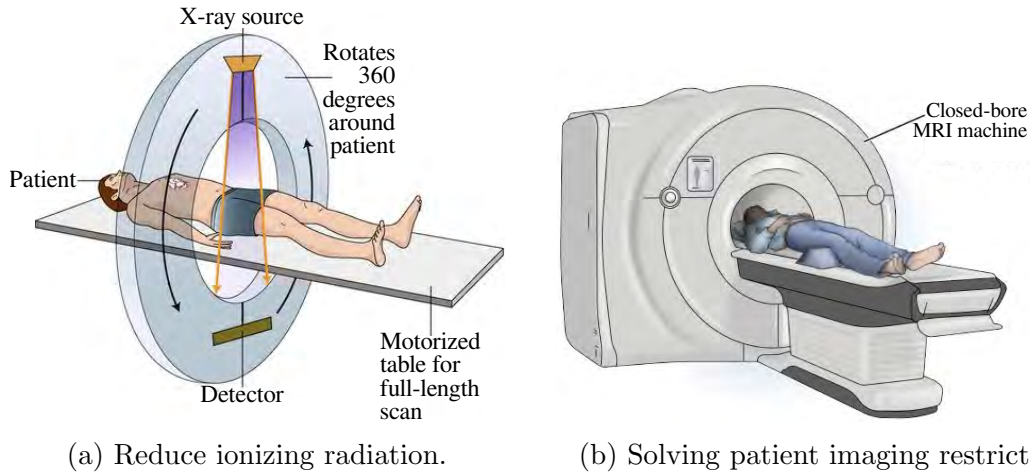


Figure 2.2: **Two example limitations in medical imaging**, from [1] and [2]. In medical settings, **generative modelling** deep learning approaches may be used to (a) reconstruct 3D CT-like images from fewer 2D X-ray projections, and translate images from one imaging modality to another. Over the last decade, deep learning techniques have been developed and published, demonstrating continual improvement of medical image quality.

2.1 Medical Imaging and Current Limitations

Medical imaging is a crucial part of diagnosing, monitoring, and treating medical conditions as it provides images of the interior of the body through various techniques [41, 42]. Unfortunately, due to factors such as the degree of intervention, acquisition time, accuracy, and costs, patients can be limited to certain modalities and thus may not receive the most informative imaging [43].

There are different medical image modalities, each aiding in ruling out or diagnosing organ disorders caused by lesions or sickness. For example, MRI is commonly used in cases of suspected spinal cord compression due to herniated discs or tumors. However, MRI acquisition can be time-consuming, taking several minutes to over an hour, which can be challenging for patients who are claustrophobic or have difficulty remaining still (e.g., children). Moreover, MRI is contraindicated in patients with metallic implants such as pacemakers or surgical clips due to safety concerns related to the strong magnetic fields [44] (Figure 2.2).

In cases where MRI is not feasible or contraindicated, CT scans may be the only imaging modality available to the patient. However, CT scans involve high doses of ionizing radiation, increasing the risk of radiation-induced cancer [45]. Developing

alternative imaging techniques and methodologies could help mitigate the limitations and risks associated with traditional imaging modalities.

2.1.1 Can Deep Learning Help?

Machine learning researchers designing deep learning methods can aid in translating medical images from one modality to another. However, this encounters numerous challenges, from acquiring large amounts of labeled data for training to complying with ethical considerations. Moreover, generating images that are sufficiently accurate to be reliable for experts and clinicians carries many implications.

The deep learning task of image-to-image translation is not limited to natural images but can extend to more general data domains such as medical images. The definition remains the same: learning a mapping function between two different domains of medical images. This goal can be defined as $G : X \rightarrow Y$, where the goal is to transform an input image from domain X to the output image domain Y . Depending on the application requirements, training can be broadly categorized into a supervised or unsupervised settings.

In the supervised setting, paired examples $\{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^N$ are available, and learning G can be achieved by minimizing the straightforward loss function $\mathcal{L}(G(\mathbf{x}_i), \mathbf{y}_i)$, which measures the difference between the generated image and the ground truth output.

In the unsupervised setting, no paired dataset is assumed. Instead, the model learns the underlying distribution of images in both domains, enabling their translation. This approach is more suitable for medical imaging and other real-world settings where data constraints are common. Regularization terms can be applied to ensure that the translated images are not only visually realistic but also semantically meaningful and consistent with the original input images.

Considering data and training aspects could enable the successful application of deep learning (DL) approaches for medical image-to-image (I2I) translation (Figure 2.3). This holds the potential to expand the availability of imaging modalities for patients, thereby improving diagnostic accuracy while reducing risks. Clear examples include translating between imaging modalities of the same dimension-

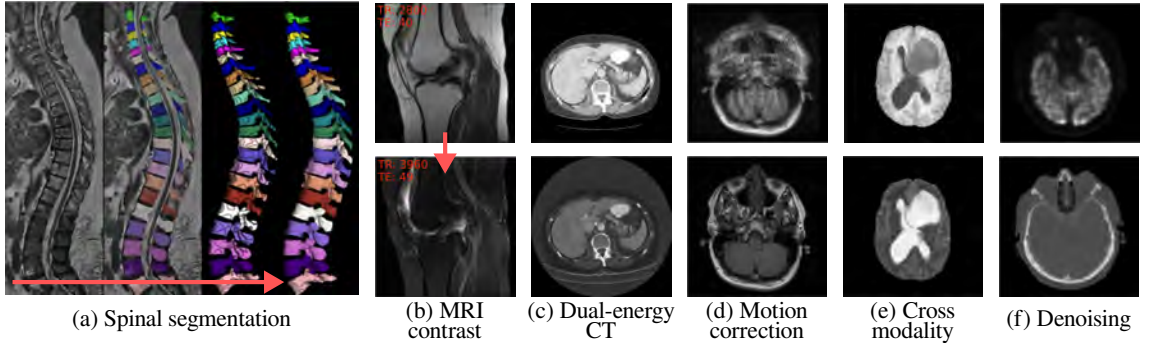


Figure 2.3: **Exemplary medical image deep learning translation tasks.** (a) Automated segmentation of spinal MRI used with 3D visualization techniques can avoid the underprediction of small structures in clinical applications. (b-f) Depict various studied applications where deep learning models are used to improve medical diagnosis by translating between modalities and/or enhancing the quality of images.

ality, such as 2D-2D or 3D-3D transformations like CT-MRI, low-contrast X-ray to high-contrast X-ray with segmentation, super-resolution, and artifact reduction tasks [46].

However, a more interesting problem is translating between modalities with a dimensionality gap, such as 2D-3D translations. In such cases, the task not only involves translating features but also generating additional data, which can be formulated as a generative modeling problem within the context of machine learning. This opens up research avenues and development in the field of medical imaging, with the potential to revolutionize diagnostic capabilities and patient care.

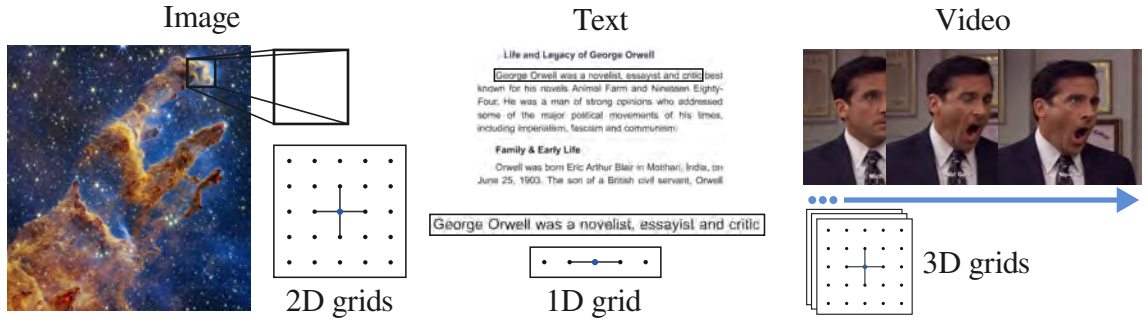


Figure 2.4: **Exploring the dimensionality of data.** From 1D signals to 2D grids of images and into the realm of 3D space, Euclidean data spaces provide the foundation for understanding the structure and relationships of shapes in our world.

2.2 Deep Learning for Data Translation

Deep neural networks can be designed for data translation, where the goal is to learn the mapping between the input data to the output data. Many computer vision tasks originated from this fundamental problem, such as image generation [31, 47–49], image-to-image translation [6, 50, 51], style transfer [52, 53], semantic segmentation [54, 55], image compression [56], image super-resolution [57], etc. The space of the data is an important part of the story, as it allows us to study the data with an underlying structure including its shape and dimensionality. By considering data as functions on a space (e.g. Euclidean) with different dimensions and characteristics, it’s possible to leverage geometrical concepts to explore ways to transform points from one shape into another. Transformations could involve changing rainy scenes to sunny ones, turning sketches into photo-realistic images, or even reconstructing CT scans from X-rays [58].

Considering the space and shape of data provides a general approach to the problem of image-to-image translation. In the context of deep learning, this means that the architectural design choices should align with the structure of both the input and output data spaces. If that is the case, the image translation method developed becomes applicable to data domains beyond its original design, ensuring certain data transformation guarantees (e.g., equivariance and invariance).

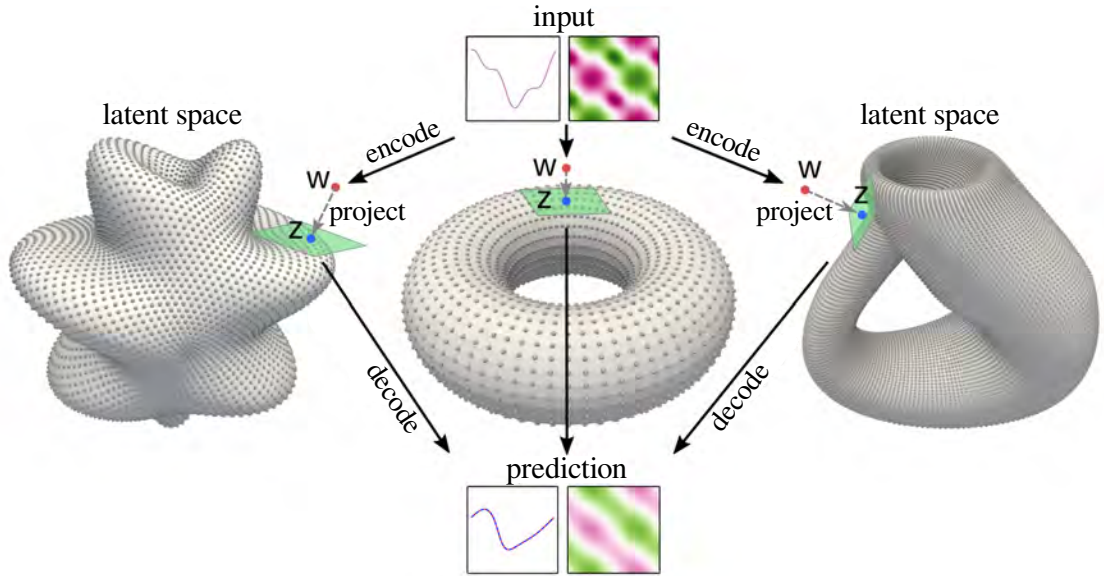


Figure 2.5: **Learning data representations in manifold latent spaces**, adapted from [3]. To learn essential structures within the data, deep learning models make use of the manifold hypothesis, which suggests that high-dimensional data can be described in a low-dimensional manifold. This often relies on encoders that learn compressed representations z of the input, which are then used by decoders to make predictions that improve over time.

2.3 Underlying (Euclidean) Space of Data

Among the geometric landscapes, we find the Euclidean space, which spawns image, video, and speech signals. Properties such as a common system of coordinates, vector space structure, and shift invariance align with our understanding of the physical world. For example, images can be seen as a function mapping pixels in a two-dimensional (2D) Euclidean space to color or intensity values. A video is a sequence of images in a three-dimensional (3D) space; two dimensions for the spatial information (width and height) and a third dimension for time (Figure 2.4). While audio results in a sequence of amplitude values of the sound wave forming a one-dimensional (1D) function representing the signal. Euclidean spaces are used to represent such structures of objects in a geometric fashion, providing a framework for defining transformations from one space to another. In the Euclidean space, however, parallel lines never meet, and the angles within a triangle sum up to 180 degrees [59]. Negating those leads to non-Euclidean geometries, namely hyperbolic geometry and elliptic geometry [60].

2.3.1 Manifolds and Non-Euclidean Domains

In non-Euclidean geometries, we encounter spaces that curve and twist, showcasing diverse and complex structures. These topological spaces, called manifolds, resemble Euclidean spaces at a local scale and can stretch and warp while maintaining smoothness and continuity, enabling rich geometries [61]. The assumption is that data can be effectively captured by a lower-dimensional surface (manifold) within a higher-dimensional space (i.e., the space in which we observe or measure the data) (Figure 2.5). This concept has been extensively utilized in machine learning for complex data representation [62], feature extraction [63], unsupervised learning [64], and improved generalization [65]. The objective is to discover such manifolds by learning the intrinsic features of the data, where the properties of Euclidean geometry are approximately preserved for each data point. Therefore, machine learning models can inherit and utilize Euclidean properties (e.g., distance, angles), enabling neural networks to learn realistic and visually appealing transformation rules for data.

2.4 Deep Learning Architectures

Key points - understanding of the structure of data allows for more effective architectural design choices:

This chapter explores the properties of key deep learning architectures, including convolutional neural networks (CNNs), recurrent neural networks (RNNs), and transformers. By examining these architectures through the lens of the data space, it's possible to determine their suitability for various data structures, such as grids of pixels or token sequences.

1. **Architectural Suitability for Data Structures:** I analyze how different neural networks are designed to handle specific data structures. For example, CNNs are well-suited for grid-like data (e.g., images) due to their translation invariance and equivariance properties, while transformers do well at handling sequences due to their ability to capture global context.
2. **Key Properties for Image-to-Image Translation:** I highlight the importance of these architectural properties in image-to-image translation tasks. CNNs' translation invariance helps in recognizing patterns regardless of their position, and transformers' global context understanding is beneficial for processing sequential data.
3. **Designing Effective Deep Learning Methods:** Understanding these geometric aspects enables researchers to design more efficient deep learning models that tackle specific data requirements.

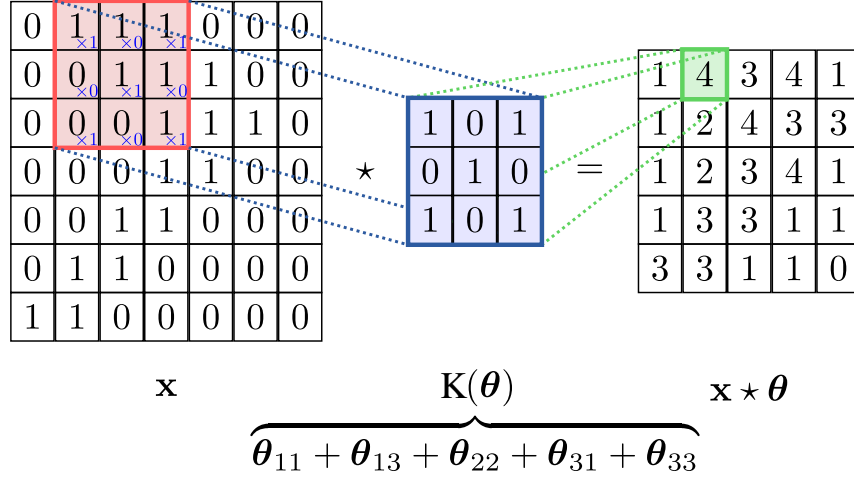


Figure 2.6: The process of convolving a 2D image $\mathbf{x}$ with filter $\mathbf{K}$ with parameters $\boldsymbol{\theta}$ (adapted from [4]).

2.4.1 Convolutional Neural Networks

Understanding the structure of data and the spaces they inhabit allows for building and exploiting those properties by artificial neural networks. Convolutional Neural Networks (CNNs) [24] are perhaps the most well-known architecture. In computer vision applications, CNNs have been particularly successful [66] because of properties including the extraction of hierarchical features that are both translation invariant and equivariant. For example, in image classification, the position of objects should not affect the network’s ability to recognize and classify them.

Typically, a CNN architecture consists of the following three aspects. First, kernels (filters) scan the input, determining which areas contain specific patterns (edges, textures) independently of their location (Figure 2.6). Second, pooling layers have been added by practitioners to downsample the spatial dimension of the feature maps, effectively increasing the receptive fields and level of abstraction [67], where slightly different positions of a feature lead to the same pooled value. Third, activation functions help the network to be more robust to small translations by activating the same feature regardless of its location. Stacking several convolutions summarizes a patch centered around each pixel, thus creating a final representation of the image. Such representations have been notably effective in various tasks, including object detection, semantic segmentation, motion estimation, and more.

Robustness to image transformations is particularly relevant in medical contexts

and when dealing with real-world data. This includes aspects such as being invariant to patient position relative to the imaging system or accidental calibration errors. Deep neural networks can learn geometric transformation robustness through random data augmentations [68–70] added to the training set. Examples include data transformations such as rotation, scale, and shearing; intensity transformations like contrast adjustment and Gaussian blur; and spatial transformations. These augmentation techniques help create a more comprehensive training dataset, enabling not only robustness but also generalization and improved performance of deep neural networks.

Translation Equivariance

Translating from one space to another involves transformations that preserve the relative position and structure of objects. In essence, translation equivariance implies that if the input is shifted, the output will shift by the same amount. For instance, let T_a represent a translation operator that shifts an input by a units, then the function f is translation equivariant if:

$$f(T_a \mathbf{x}) = T_a f(\mathbf{x}) \quad (2.1)$$

CNNs offer this property, meaning that if the input image $\mathbf{I}$ is translated, the output of the convolution is also translated: this translates into an operation being defined as:

$$(\mathbf{I}(x + a, y + b) * \mathbf{K}) = (\mathbf{I} * \mathbf{K})(x + a, y + b), \quad (2.2)$$

where $(\mathbf{K}$ is a filter and $*$ denotes the convolution operator.

It’s worth noting that if the input image is shifted by some amount, the corresponding feature map after the convolution will also be shifted by the same amount. However, the content of the feature map remains the same. This property from CNNs has been highly successful for pattern detection (such as edges or textures) regardless of their position in the image. This ensures that the features learned by the filter are translationally invariant, enabling the network to effectively recognize objects regardless of their position within the input data [71, 72].



Figure 2.7: **From Vermeer to Dali.** “The Lacemaker” by Vermeer is known for its delicate depiction of a woman making lace, emphasizing fine details and a soft color palette. Transforming such painting in the style of Salvador Dali, known for his surreal and dreamlike paintings, the transformed image would likely feature more vivid colors, exaggerated forms, and possibly surreal elements that deviate from the realistic style of Vermeer. Learning the artistic style of a painting and apply it to another image is one of the popular image-to-image translation tasks.

Translation invariance is a desirable property for translating images as it helps the model focus on the content and structure of the data rather than its spatial location. This is handy for adapting to different arrangements where the relative positions of objects may change from one geometric space to another. Consider the interesting example of translating Johannes Vermeer paintings into Salvador Dali style (Figure 2.7). A model would then learn to map the highly realistic objects of Vermeer to the artistic world of Dali, where accuracy and realism turn into bizarre and distorted imagery. While mapping from Euclidean space to a non-Euclidean space is mathematically complex, in the context of machine learning and representation learning, preserving important topological features during the transition may suffice for an artistic reinterpretation.

2.4.2 Recurrent Neural Networks

Sequential data, such as video and text, pose a key difficulty for traditional neural networks in capturing contextual and temporal dependencies between elements. The lack of a mechanism to handle such dependencies and the need to accommodate variable-length inputs introduced Recurrent Neural Networks (RNNs) [73]. Maintaining information about previous elements is the solution for capturing dependencies over time. In particular, we are interested in how the hidden states work

in RNNs. This concept is governed by the internal representations that evolve as the model processes each element in the input sequence $\mathbf{x}$, where each hidden state $\mathbf{h}$ is updated via a weighted combination of the current input at t and the previous hidden state $t - 1$, passed through an activation function

$$\mathbf{h}_t = \text{activation}(\mathbf{W}_h \cdot \mathbf{h}_{t-1} + \mathbf{U}_h \cdot \mathbf{x}_t), \quad (2.3)$$

where the weight matrix $\mathbf{W}_h$ determines the influence of the previous hidden state $\mathbf{h}_{t-1}$ on the current hidden state $\mathbf{h}_t$, while the weight matrix $\mathbf{U}_h$ determines the influence of the current input $\mathbf{x}_t$ on $\mathbf{h}_t$. The equation encodes information not only about the current input $\mathbf{x}_t$ but also the entire history of inputs up to time step $t - 1$, creating a chain of dependencies [74].

RNNs are not only sequential but also flexible. Their step-by-step behavior enables padding of shorter sequences to match the length of the longest in the dataset. In the padded portions, masking can be applied to effectively ignore them by assigning a weight of zero. This mechanism generates a hierarchical representation where the model learns information regardless of scale [75]. In contrast, traditional feedforward networks expect fixed-size input vectors, making them less adaptable in terms of input length.

While RNNs are conceptually intuitive, their effectiveness can be limited as they can suffer from vanishing gradients and struggle to capture long-range dependencies [76]. These challenges were eventually addressed with subsequent variants that include a gating mechanism, such as Long Short-Term Memory (LSTM) networks [77] and Gated Recurrent Units (GRUs) [78]. This selection operation allows both retention and discarding of information from previous steps. Here, rather than dissecting the specific requirements for each, we focus on the fundamental ideas behind RNNs that were used in deep learning translation tasks.

2.4.3 Transformers

Many machine learning tasks operate on data that do not follow a grid-like structure, such as graphs and sets, used to model systems of relations and interactions from

biology to physics. The desirable property to satisfy is permutation invariance. Consider a set as a collection of nodes with no edges. Then, applying a permutation matrix, the outcomes of the same function operating on the original set and $p(\mathbf{x})$ are identical. This property provides a global set-wise output; however, we might also be interested in functions that act locally, in a node-wise manner. An important consequence is that a function updating the features of every node is no longer permutation invariant unless the order of rows in the output is tied to the order of rows in the input. If so, the model also possesses permutation equivariance, as the output permutes wherever the input permutes. For example, a model with both properties could predict the average value of a set of numbers. Changing the order of the numbers in the set would change the order of the predictions for individual numbers, but the overall average remains the same.

In many language applications, we encounter sequential data where spatial relationships or local patterns are no longer relevant (opposed to CNNs). The introduction of Transformers [79] enabled both superior permutation invariance and equivariance compared to CNNs [80,81] by introducing the self-attention mechanism and the use of positional encodings [82]. Models such as BERT [83], GPT [84–86], and various other Transformer-based architectures [87,88] have consistently achieved state-of-the-art outcomes across a broad spectrum of tasks [89].

Self-Attention Mechanism & Permutation Invariance

The key reason for the success of transformers is their self-attention mechanism [79], which enables both permutation invariance and global context [90,91]. Specifically, given an input sequence $\mathbf{X} = \{\mathbf{x}_0, \dots, \mathbf{x}_n\}$, the vectors Query, Key, and Value are obtained via linear transformations from the weight matrices $\mathbf{W}_Q$, $\mathbf{W}_K$, and $\mathbf{W}_V$, respectively. We compute the attention scores between all pairs of elements in the sequence using the dot product of Query and Key vectors:

$$S(\mathbf{x}_i, \mathbf{x}_j) = \langle \mathbf{q}_i | \mathbf{k}_j \rangle. \quad (2.4)$$

The weight matrices transformations project the input sequence into a higher-dimensional space, allowing to effectively capture the correlation between representations of $\mathbf{x}_i$ and $\mathbf{x}_j$.

Creating a probability distribution of the attention weights can be achieved by applying the softmax function to normalize the scores, ensuring they sum to 1. Then we compute the weighted sum of the Value vectors using the attention weights. The final attention expression is:

$$\text{attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right) \mathbf{V} \quad (2.5)$$

Since the order of the elements does not affect the final output of the softmax operation, the model is permutation invariant to the order of elements. Furthermore, the weighted sum operation aggregates information from all elements regardless of their position. However, transformers lack inherent knowledge of the orders of elements in the sequence. Adding positional encodings (PE) introduces information about the position of each element, thus allowing changes in the input to align with the order of changes in the output. In the original paper “Attention is all you need” [79] by Vaswani *et al.* they used sine and cosine functions:

$$\text{PE}(p, 2i) = \sin(p \cdot \omega), \quad (2.6)$$

$$\text{PE}(p, 2i + 1) = \cos(p \cdot \omega), \quad (2.7)$$

where p is the position of the token in the sequence, $\omega = \frac{1}{10000^{2i/d}}$, d the dimensionality of the input embedding, and i is the index of the embedding dimension. By using both sine and cosine functions, the positional encoding can represent different phase shifts of the patterns, providing unique positional information for each token in the sequence. Other options of positional encodings include learned, relative, and dynamic [92].

Since transformers do not assume any particular order of the input tokens, they offer flexibility in input representation, making them highly scalable [93]. In computer vision, images can be broken down into patches (similar to tokens in NLP) and processed simultaneously, unlike CNNs that process data sequentially. Vision trans-

formers (ViTs) have been adapted to various I2I translation tasks [94], leveraging permutation invariance and global context. These properties are particularly relevant to medical imaging, which often involves large and high-dimensional data [95].

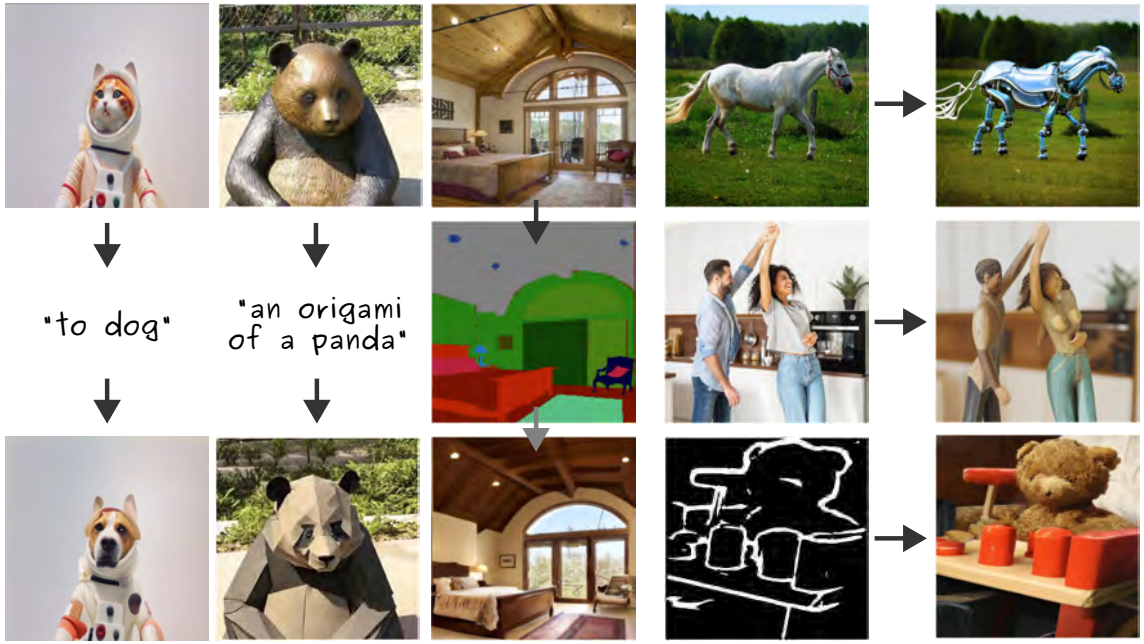


Figure 2.8: **Showcasing natural image-to-image translation.** From image segmentation to transforming images with text prompts, these examples illustrate the diverse applications and transformative potential of image-to-image translation deep learning models in creative works and automated content creation.

2.5 Image-to-Image Translation

Key points - Image-to-Image translation is a central and broad problem in generative modeling:

This chapter introduces the problem of translating an image from one source domain to a target domain, which underpins the broad category of image-to-image translation tasks in computer vision. Key aspects include:

1. The nature of training, whether paired or unpaired, is determined by data availability. Paired methods rely on corresponding source-target image pairs, while unpaired methods operate without such pairs. The desired output influences the stochasticity of the mapping to learn.
2. Some methods support translations in both directions (source to target and vice versa), although this bidirectionality is not always guaranteed.
3. Proper data alignment facilitates straightforward content preservation and translation while effectively disentangling style differences between the source and target domains.

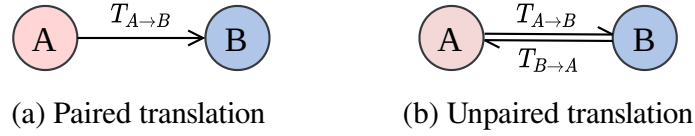


Figure 2.9: **Comparing Paired vs. Unpaired Image-to-Image Translation**
(a) Paired translation requires matched datasets for direct one-to-one mapping, achieving high accuracy with models like Pix2Pix [5]. While (b) leverages unpaired datasets, learning bidirectional mappings with models like CycleGAN [6], offering flexibility and broader applicability.

Image-to-image (I2I) translation involves learning the mapping of a given image to a target image. This task requires an understanding of shared underlying features, which can be domain-independent or domain-specific [96]. Preserving domain-independent features in the mapping while allowing for changes in domain-specific ones ensures the preservation of intrinsic source content and the transfer of extrinsic target style. However, this task presents several challenges, including the absence of paired data and scenarios where an input image needs to map to multiple outputs.

The concept of translation, as articulated by Hertzmann *et al.* in “*Image analogies*” [97] introduces a framework for processing images through example, also referred to as “image analogies”. This framework typically operates with paired datasets of images, where one set represents unfiltered images and the other set represents their filtered versions. The objective is to learn the transformation or filter from one image to its corresponding version and then apply it to another image to achieve an equivalent result, expressed as $A : A' :: B : B'$. The goal is to find an analogous image B' that relates to B in the same manner as A' relates to A . However, many generative modeling problems extend beyond simple image transformations [40]. Tasks such as image inpainting [98] and 2D-3D image translation [21] involve data generation rather than solely learning a set of filters. These complexities highlight the diverse range of challenges and applications within the field of image-to-image translation.

2.5.1 Paired & Unpaired Methods

The characteristics of the data and application requirements play a crucial role in determining the appropriate approach for image translation. With a paired dataset,

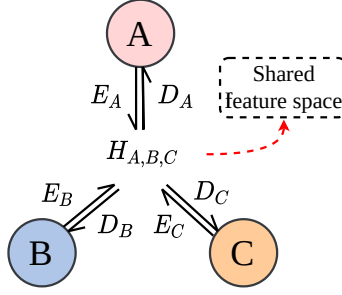


Figure 2.10: **Shared Latent Space for Unpaired Image-to-Image Translation.** Domains A, B, and C each have encoders that map images to a shared latent space, and decoders that reconstruct images from this space. This shared latent space allows for effective translation and consistency across multiple domains.

source images and corresponding images are provided as a training set of image pairs; for each image $\mathbf{x}_A \in \mathcal{X}_A$, there exists a corresponding image from another domain $\mathbf{x}_B \in \mathcal{X}_B$. Conversely, unpaired methods learn the joint distribution of images in different domains using marginal distributions from the individual domains, even when datasets lack direct links between examples (Figure 2.9). However, according to the coupling theory, the unpaired setting is highly under-constrained given the existence of an infinite set of joint distributions that can match the given marginal distributions in general [99].

Paired: samples are drawn from a joint data distribution $(\mathbf{x}_A, \mathbf{x}_B) \sim p_{\mathcal{X}_A, \mathcal{X}_B}$.

Unpaired: samples are drawn from the marginal distributions $\mathbf{x}_A \sim p_{\mathcal{X}_A}$ and $\mathbf{x}_B \sim p_{\mathcal{X}_B}$.

Seminal papers have leveraged paired datasets in supervised training settings for non-parametric texture synthesis [100], sketch based image synthesis [101], generation of realistic outdoor scenes [102], and the pix2pix model spawning various I2I translation tasks, such as photo enhancement, image colorization, and semantic segmentation [5]. However, as previously mentioned, paired datasets are difficult to collect, and thus latest works focus on semi-supervised [103] and unsupervised settings.

Shared-Latent Space

To tackle the ill-posed nature of the unpaired setting, the UNIT framework [104] operates under the assumption that a pair of corresponding images from different domains can be mapped to the same latent representation within a shared-latent space (Figure 2.10). This approach builds upon the Coupled GAN work [105] and models each image domain using a VAE-GAN. Alongside the adversarial training objective, a weight-sharing constraint is incorporated to promote the shared-latent space, facilitating the generation of corresponding images across the two domains.

Shared-latent space constraint assumes there exists a shared latent code $\mathbf{z}$, such that both images can be recovered from this code, and also we can compute this code from each of the two images, that is:

$$\mathbf{z} = E_A(\mathbf{x}_A) = E_B(\mathbf{x}_B), \quad (2.8)$$

$$\mathbf{x}_A = G_A(\mathbf{z}), \quad (2.9)$$

$$\mathbf{x}_B = G_B(\mathbf{z}) \quad (2.10)$$

where E_A and E_B denotes encoding functions for domains A and B , respectively, and similarly for G_A and G_B for generator functions.

From the previous we can derive the bidirectional mappings of both domains as:

$$\mathbf{x}_B = F_{A \rightarrow B}(\mathbf{x}_A) = G_B(E_A(\mathbf{x}_A)), \quad (2.11)$$

$$\mathbf{x}_A = F_{B \rightarrow A}(\mathbf{x}_B) = G_A(E_B(\mathbf{x}_B)) \quad (2.12)$$

learning $F_{A \rightarrow B}$ and $F_{B \rightarrow A}$ implies the cycle consistency constraint:

$$\mathbf{x}_A = F_{B \rightarrow A}(F_{A \rightarrow B}(\mathbf{x}_A)), \quad (2.13)$$

$$\mathbf{x}_B = F_{A \rightarrow B}(F_{B \rightarrow A}(\mathbf{x}_B)). \quad (2.14)$$

The UNIT approach specifically enforces the common-latent space constraint by weight-sharing of the last few layers of E_A and E_B , responsible for extracting high-level representations (i.e., content features) from input images in the two do-

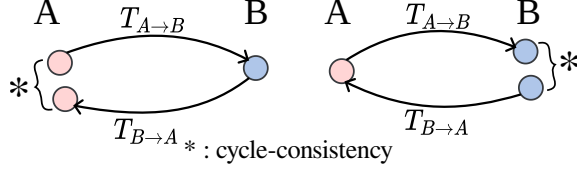


Figure 2.11: **Cycle-Consistency.** Domain A is translated to domain B using $T_{A \rightarrow B}$, and back to A using $T_{B \rightarrow A}$ ensuring the original image is reconstructed. Similarly, domain B is translated to domain A and back, maintaining consistency and preserving the original image features. Cycle-consistency is one of the most popular techniques for unpaired I2I, enabling effective learning from unpaired data.

mains. Similarly, the weights of the first few layers of G_A and G_B are shared. This facilitates the translation of an image $\mathbf{x}_A$ in $\mathcal{X}_A$ to an image in $\mathcal{X}_B$ through $G_B(\mathbf{z}_A \sim q_A(\mathbf{z}_A|\mathbf{x}_A))$, where the latent code $\mathbf{z}_A$ is determined by $q_A(\mathbf{z}_A|\mathbf{x}_A) := \mathcal{N}(\mathbf{z}_A|E_{\mu,A}(\mathbf{x}_A), \mathbf{I})$. Utilizing the reparameterization trick [106], the sampling operations of $\mathbf{z}_A \sim q_A(\mathbf{z}_A|\mathbf{x}_A)$ and $\mathbf{z}_B \sim q_B(\mathbf{z}_B|\mathbf{x}_B)$ can be realized as $\mathbf{z}_A = E_{\mu,A}(\mathbf{x}_A) + \eta$ and $\mathbf{z}_B = E_{\mu,B}(\mathbf{x}_B) + \eta$, respectively, where η is a random vector following a multivariate Gaussian distribution $\eta \sim \mathcal{N}(\eta|0, \mathbf{I})$. This auxiliary random variable transforms the non-differentiable sampling operation into a differentiable one.

Ensuring the existence of such pairs of corresponding images enables images to be mapped to the same latent code, where a single latent code is decoded to yield a pair of corresponding images $(\mathbf{x}_A, G_B(\mathbf{z}_A \sim q_A(\mathbf{z}_A|\mathbf{x}_A)))$. This approach allows the framework to learn two image translation streams $\mathcal{X}_A \rightarrow \mathcal{X}_B$ and $\mathcal{X}_B \rightarrow \mathcal{X}_A$ in an unsupervised manner.

Cycle Consistency

Assuming a deterministic translation implies unimodal translation, where joint distribution matching is achieved through the cycle consistency constraint (CycleGAN [6]). This constraint ensures that translating an image from the source domain to the target domain using the generator and then back to the source domain should recover the original image (Figure 2.11).

The image translation model aims to ensure that the marginal distributions match the true data distributions, represented by $p(\mathbf{x}_A)$ and $p(\mathbf{x}_B)$, along with the

conditionals defined by the generators $p_G(\mathbf{x}_A|\mathbf{x}_B)$ and $p_G(\mathbf{x}_B|\mathbf{x}_A)$:

$$p(\mathbf{x}_A) = \int p_G(\mathbf{x}_A|\mathbf{x}_B)p(\mathbf{x}_B)d\mathbf{x}_B, \quad (2.15)$$

$$p(\mathbf{x}_B) = \int p_G(\mathbf{x}_B|\mathbf{x}_A)p(\mathbf{x}_A)d\mathbf{x}_A \quad (2.16)$$

Under the cycle consistency constraint, for every $\mathbf{x}_A^* \in \mathcal{X}_A$, $\mathbf{x}_B^* \in \mathcal{X}_B$, the following conditions are expressed:

$$\mathbb{E}_{\mathbf{x}_B \sim p_G(\mathbf{x}_B|\mathbf{x}_A^*)} [p_G(\mathbf{x}_A|\mathbf{x}_B)] = \delta(\mathbf{x}_A - \mathbf{x}_A^*), \quad (2.17)$$

$$\mathbb{E}_{\mathbf{x}_A \sim p_G(\mathbf{x}_A|\mathbf{x}_B^*)} [p_G(\mathbf{x}_B|\mathbf{x}_A)] = \delta(\mathbf{x}_B - \mathbf{x}_B^*). \quad (2.18)$$

These conditions assert that when the generator has learned to translate images between domains, the conditional distributions collapse to deterministic delta functions. This ensures the preservation of cycle consistency in the image translation process, reinforcing the fidelity of the generated images to their original counterparts.

In practice, the cycle consistency loss is used to make sure the learned mappings $G : \mathcal{X}_A \rightarrow \mathcal{X}_B$ and $F : \mathcal{X}_B \rightarrow \mathcal{X}_A$ are inverses of each other:

$$\begin{aligned} \mathcal{L}_{\text{cyc}}(G, F) = & \mathbb{E}_{\mathbf{x}_A \sim p_{\text{data}}(\mathbf{x}_A)} [\|F(G(\mathbf{x}_A)) - \mathbf{x}_A\|_1] \\ & + \mathbb{E}_{\mathbf{x}_B \sim p_{\text{data}}(\mathbf{x}_B)} [\|G(F(\mathbf{x}_B)) - \mathbf{x}_B\|_1] \end{aligned} \quad (2.19)$$

Such constraint has been a crucial component in unpaired I2I translation tasks for improving the quality in terms of realism and coherence, and stability of translation models

An alternative method inspired from CycleGAN, proposed to reuse the discriminator for encoding images in the target domain via a decoupling training strategy. The model named NICE-GAN [107] results in a more compact architecture as it eliminates the need for a separate encoding component while achieving SOTA unsupervised translations.

Dual Learning

Similar to the CycleGAN framework, the DualGAN model [108] operates with two GANs, enabling images from either the input or output domains to be translated and then reconstructed. Specifically, a primal GAN learns the generator $G_A : U \rightarrow V$, which maps an image $\mathbf{u} \in U$ to an image $\mathbf{v} \in V$, while the dual GAN learns an inverse generator $G_B : V \rightarrow U$ along with their respective discriminators D_A and D_B .

The DualGAN training objective is very similar to the identity loss in CycleGAN, incorporating an L_1 distance as a reconstruction error between translated samples and original images of the target domain distribution:

$$\begin{aligned}\mathcal{L}_{\text{DualGAN}}(\mathbf{u}, \mathbf{v}) = & \lambda_U \|\mathbf{u} - G_B(G_A(\mathbf{u}, \mathbf{z}), \hat{\mathbf{z}})\|_1 \\ & + \lambda_V \|\mathbf{v} - G_A(G_B(\mathbf{v}, \hat{\mathbf{z}}), \mathbf{z})\|_1 \\ & - D_A(G_B(\mathbf{v}, \hat{\mathbf{z}})) - D_B(G_A(\mathbf{u}, \mathbf{z})),\end{aligned}$$

where λ_U and λ_V are weighting scalars, and both $\mathbf{z}$ and $\hat{\mathbf{z}}$ are sampled from a Gaussian normal distribution.

Although the authors conducted extensive experiments and demonstrated the robustness of their method across multiple image domains, they also noted limitations compared to the conditional GAN (cGAN) [109], particularly in tasks involving semantic-based labels. Additionally, their method was unable to outperform the success and popularity of the CycleGAN approach.

Joint Distribution Matching

Generating images in domain B from domain A and vice versa, given input images $\mathbf{x}_A$ and $\mathbf{x}_B$, can be represented by the conditional distributions $p(\mathbf{x}_{A \rightarrow B} | \mathbf{x}_A)$ and $p(\mathbf{x}_{B \rightarrow A} | \mathbf{x}_B)$. Modelling such conditionals may suffice when a paired dataset of domains A and B is available, as the existence of image pairs is already fulfilled by the data. However, in an unpaired setting, the joint distributions $p(\mathbf{x}_A, \mathbf{x}_{A \rightarrow B})$ and $p(\mathbf{x}_{B \rightarrow A}, \mathbf{x}_B)$ need to be modelled, as these capture the probability of observing pairs of images where the first image belongs to one domain and the second image is

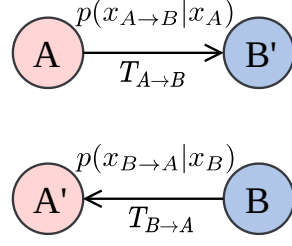


Figure 2.12: **Joint Distribution Matching for I2I** ensures that translated images match the joint distributions of their target domains. Generators $T_{A \rightarrow B}$ and $T_{B \rightarrow A}$ handle the translation between domains A and B based on the input images.

its translation to the other domain (Figure 2.12). This constraint is crucial for unsupervised training, as achieving optimality implies that $p(\mathbf{x}_A, \mathbf{x}_{A \rightarrow B}) = p(\mathbf{x}_{B \rightarrow A}, \mathbf{x}_B)$, ensuring consistency between these distributions. Essentially, they approximate the same underlying joint distribution $p(\mathbf{x}_A, \mathbf{x}_B)$.

Mutual Information Maximization

When translating images from one domain to another, it’s desirable to ensure that both latent representations retain as much shared information as possible. This can be achieved by adding a mutual information (MI) maximization regularization term to the model, which focuses on capturing the content and structure of both domains (Figure 2.13).

The TUNIT model, as described in [110], proposes a three-step process for unpaired image-to-image (I2I) translation. First, images are clustered based on set-level characteristics. Second, their individual content and style are encoded. Lastly, the mapping function among the estimated domains is learned. Let’s denote E_C and E_S as the encoder branches providing domain labels and style codes, respectively. The clustering method is differentiable and maximizes the MI between an image $\mathbf{x}$ and its randomly augmented version $\mathbf{x}^+$ with the data transformation f . The objective is to maximize entropy $H(\mathbf{p})$ and minimize conditional entropy $H(\mathbf{p}|\mathbf{p}^+)$. Here, $\mathbf{p} = E_C(\mathbf{x})$ represents the softmax output from E_C , indicating the probability

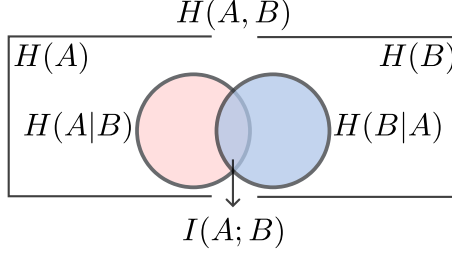


Figure 2.13: **The Mutual Information Maximization approach** learns a common latent spaces where mutual information between domains A and B is maximized, promoting effective translation by preserving shared features.

vector of $\mathbf{x}$ over K domains.

$$\mathcal{L}_{MI} = \mathbf{I}(\mathbf{p}, \mathbf{p}^+) = \mathbf{I}(\mathbf{P}) = \sum_{i=1}^K \sum_{j=1}^K \mathbf{P}_{ij} \ln \frac{\mathbf{P}_{ij}}{\mathbf{P}_i \mathbf{P}_j}, \quad (2.20)$$

$$\text{s.t. } \mathbf{P} = \mathbb{E}_{\mathbf{x}^+ \sim f(\mathbf{x}) | \mathbf{x} \sim p_{\text{data}}(\mathbf{x})} [E_C(\mathbf{x}) \cdot E_C(\mathbf{x}^+)^T]. \quad (2.21)$$

Where $\mathbf{P}_i = \mathbf{P}(\mathbf{p} = i)$ denotes the K -dimensional marginal probability vector, and $\mathbf{P}_{ij} = \mathbf{P}(\mathbf{p} = \text{argmax}(i), \mathbf{p}^+ = \text{argmax}(j))$ denotes the joint probability. Such formulation encourages E_C to assign the same domain label to the pair $\mathbf{x}$ and $\mathbf{x}^+$, and thus preserving characteristics of the original images.

Contrastive Learning

Contrastive learning is a key technique in unsupervised or self-supervised learning for many computer vision tasks to learn representations of data by contrasting positive and negative samples. Similar to the MI approach, in the context of I2I, this technique is useful for learning meaning representations of images that capture the underlying structure of the data.

Specifically, positive samples are pairs of data points considered similar or related, while negative samples are pairs of data points that are considered dissimilar. During the process of image translation it's desired to preserve content or style of the original images, this makes constrastive learning relevant as positive samples can be considered from the same domain before and after translation in an unpaired setting. Similarly, negative samples could be pairs of images from different domains or simply random pairs of images that are not related (Figure 2.14).

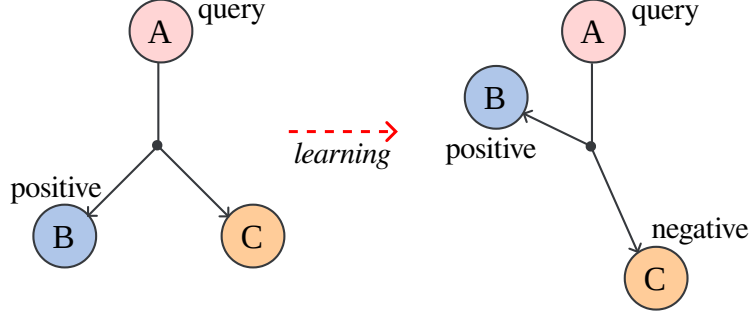


Figure 2.14: **Contrastive Learning** in the context of I2I translation is useful in the unpaired setting. In this setup, domain A represents the query or input images that need to be translated, while domain B consists of positive examples, which are images from the target domain. Domain C includes negative samples, which are images that do not belong to the target domain. This framework aims to pull the latent representation of the input image closer to the latent representations of the positive examples while pushing it away from the latent representations of the negative samples.

The CUT model introduced in [111] propose a contrastive loss function, instead of the cycle-consistency loss, to learn an embedding that associates corresponding patches to each other. In particular, this approach is used at a multilayer, patchwise fashion, allowing the model to capture both low-level and high-level features of the images. Additionally, authors found that drawing negative patches from within the input image itself, rather than from other images in the dataset helps to preserve the content of the input.

In their method, a “query”, its “positive”, and N “negatives” are mapped to K -dimensional vectors $\mathbf{v}, \mathbf{v}^+ \in \mathbb{B}^K$ and $\mathbf{v}^- \in \mathbb{R}^{N \times K}$, respectively. Then authors formulate it as an $(N + 1)$ classification problem, where the distances between the query and other examples are scaled by a temperature τ and are passed as logits. They calculate the cross-entropy loss, representing the probability of the positive example being selected over negatives:

$$\mathcal{L}(\mathbf{v}, \mathbf{v}^+, \mathbf{v}^-) = -\log \left[\frac{\exp(\mathbf{v} \cdot \mathbf{v}^+ / \tau)}{\exp(\mathbf{v} \cdot \mathbf{v}^+ / \tau) + \sum_{n=1}^N \exp(\mathbf{v} \cdot \mathbf{v}_n^- / \tau)} \right]. \quad (2.22)$$

Where the goal is to associate the input and output data.

By using this contrastive learning approach, the authors are able to achieve successful results in image-to-image translation, where the output takes on the appear-

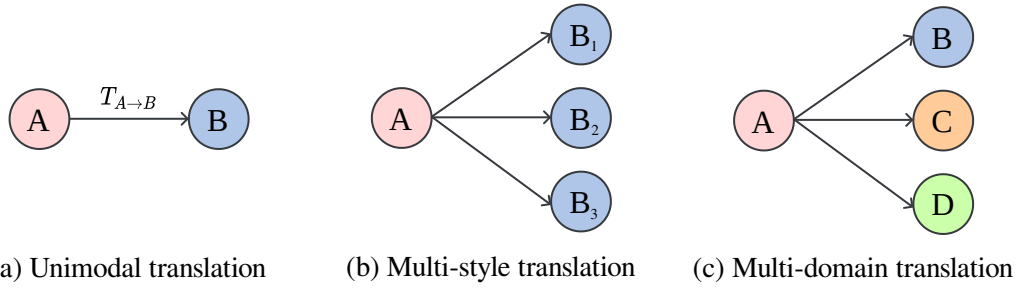


Figure 2.15: **Stochasticity of the I2I translation mapping.** (a) Focuses on translating images from a source domain to a target domain in a one-to-one fashion, while (b) multi-style translation allows generating multiple stylistic variations of the input image within the target domain by introducing a style latent code that can be manipulated to produce diverse outputs. Instead of limiting translation to a single target domain, (c) multi-domain translation can translate an input image from the source domain to several distinct target domains, each with unique characteristics.

ance of the target domain while preserving the structure or content of the specific input. This is a more straightforward and effective way of maintaining correspondence in content and appearance compared to traditional cycle-consistency methods as they only employ two networks; a generator and an encoder.

2.5.2 Unimodal & Multimodal Translation

The stochasticity of the mapping in image translation can be derived into two distinct approaches: unimodal and multimodal translation. Unimodal translation involves a straightforward one-to-one mapping, where the model learns to produce a deterministic output (i.e., no stochasticity). For instance, applications like surveillance, medical imaging, and digital photography often necessitate enhancing the quality of images through a direct mapping between a low-resolution image and its high-resolution counterpart [112].

Conversely, multimodal translation refers to a one-to-many mapping, where the model learns to generate diverse outputs for a single input (i.e., stochastic mapping). This concept is relevant in various tasks, such as fashion design, where the objective might be to produce multiple plausible designs or color variations for a given handbag image.

However, capturing the full distribution of possible outputs in multimodal translation presents a more intricate challenge than merely injecting noise to promote

stochasticity [113]. Multimodal approaches strive to introduce variability by assuming that the latent space of images can be disentangled into a content space and a style space. In this framework, images across different modalities share a common content but diverge in style (Figure 2.15).

One model that embodies this principle is MUNIT [113], which achieves image translation to a target domain by recombining its content latent code with a random style code from the target style space. By sampling different style codes, MUNIT can generate diverse and multimodal outputs, showcasing the potential of such approaches in tackling complex image translation tasks.

Style-Augmented Cycle Consistency

The cycle consistency constraint, although effective for unimodal image translation, is often considered too stringent for a multimodal framework. To address this limitation, the MUNIT model proposed a more flexible form of cycle consistency known as style-augmented cycle consistency.

In this framework, denoted by $\mathbf{h}_A$ and $\mathbf{h}_B$, representing points in the joint spaces of image and style, a unimodal or multimodal image translation is defined by introducing the style code $\mathbf{s}$. Modifying the deterministic mapping $F_{A \rightarrow B}$ from $\mathcal{H}_A$ to $\mathcal{H}_B$ (and vice versa) allows for the incorporation of style information:

$$F_{A \rightarrow B}(\mathbf{h}_A) = F_{A \rightarrow B}(\mathbf{x}_A, \mathbf{s}_B) := (G_B(E_A^c(\mathbf{x}_A), \mathbf{s}_B), E_A^s(\mathbf{x}_A)), \quad (2.23)$$

which ensures $F_{A \rightarrow B} = F_{B \rightarrow A}^{-1}$. By replacing the original style code with a different one, the model can effectively recombine content from one image with style from another. This mechanism bears resemblance to classical style transfer algorithms.

By embracing style-augmented cycle consistency, the MUNIT model strikes a balance between preserving cycle consistency and allowing for diverse outputs, thereby advancing the capabilities of multimodal image translation frameworks.

Domain Classification

A different approach explores the incorporation of an auxiliary classifier on top of the discriminator, which indirectly conditions the image generation toward a specific domain. This approach was introduced in the StarGAN framework [114] for multi-domain image-to-image (I2I) translation using a single GAN. In their method, the goal is to learn the mapping function $G : X \times \{1, 2, \dots, N\} \rightarrow Y_1 \cup Y_2 \cup \dots \cup Y_N$ that translates an input image $\mathbf{x} \in X$ into an output image $\mathbf{y}_i$ corresponding to a specified target domain from output space domains $Y_1, Y_2, \dots, Y_N$. Authors also added a mask vector that enabled the model to ignore unspecified labels and focus on the ones provided by a particular dataset.

2.5.3 Directional & Bidirectional Mappings

The nature of image translation determines whether the model performs solely directional translation, converting images from one domain to another, or bidirectional translation, capable of both conversion directions. For example, a directional image-to-image (I2I) translation model trained to convert horse images into zebra images might lack the ability to revert zebras back into horses. Bidirectional translation entails a more complex architecture and additional training objectives compared to directional translation, as it necessitates learning both mappings. Consequently, a bidirectional model would possess the capability to convert images of zebras back into images of horses.

Within the MUNIT framework, bidirectional image translation is enabled through the introduction of a bidirectional reconstruction loss, which learns pairs of encoder and decoder inverses. Specifically, this loss promotes reconstruction in both image directions, from image to latent and from latent to image:

$$\mathcal{L}_{\text{recon}}^{\mathbf{x}_A} = \mathbb{E}_{\mathbf{x}_A \sim p(\mathbf{x}_A)} [\| G_A(E_A^c(\mathbf{x}_A), E_A^s(\mathbf{x}_A)) - \mathbf{x}_A \|_1], \quad (2.24)$$

$$\mathcal{L}_{\text{recon}}^{\mathbf{c}_A} = \mathbb{E}_{\mathbf{c}_A \sim p(\mathbf{c}_A), \mathbf{s}_B \sim q(\mathbf{s}_B)} [\| E_B^c(G_B(\mathbf{c}_A, \mathbf{s}_B)) - \mathbf{c}_A \|_1], \quad (2.25)$$

$$\mathcal{L}_{\text{recon}}^{\mathbf{s}_B} = \mathbb{E}_{\mathbf{c}_A \sim p(\mathbf{c}_A), \mathbf{s}_B \sim q(\mathbf{s}_B)} [\| E_B^s(G_B(\mathbf{c}_A, \mathbf{s}_B)) - \mathbf{s}_B \|_1], \quad (2.26)$$

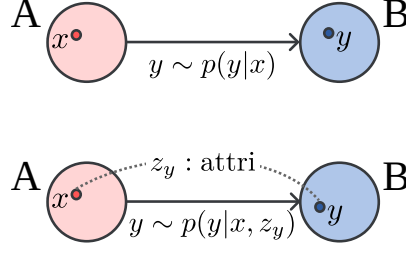


Figure 2.16: **Conditional approaches** guide the translation by the input image from the source domain or in addition to a $\mathbf{z}$ code which can represent attributes such as style or content, enabling flexible and controllable translations.

where $q(\mathbf{s}_B)$ represents the prior $\mathcal{N}(0, \mathbf{I})$, $p(\mathbf{c}_A)$ is given by $\mathbf{c}_A = E_A^c(\mathbf{x}_A)$, with $\mathbf{c}$ and $\mathbf{s}$ denoting content and style features, respectively. Similar loss terms $\mathcal{L}_{\text{recon}}^{\mathbf{x}_B}$, $\mathcal{L}_{\text{recon}}^{\mathbf{c}_B}$, and $\mathcal{L}_{\text{recon}}^{\mathbf{s}_A}$ can be defined accordingly.

This bidirectional reconstruction loss facilitates the training of MUNIT models to perform both forward and inverse translations, enhancing their versatility and enabling a broader range of applications in image-to-image translation tasks.

2.5.4 Conditional Approaches

Image translation problems formulated via unstructured losses such as regression or per-pixel classification do not explicitly capture relationships between pixels in the output image, treating each pixel as an independent prediction task [50]. Consequently, this approach can lead to a lack of global coherence in the generated images, where features like objects appear in random locations without any spatial consistency, as observed in real examples from the target domain. In contrast, a conditional generative model is better suited for image-to-image translation tasks as it penalizes the joint configuration of the output, enabling realistic spatial arrangements of image features (Figure 2.16).

The first conditional approach to undertake I2I translation tasks was the pix2pix model introduced in “Image-to-Image Translation with Conditional Adversarial Networks” [50]. This method is based on conditional GANs (cGANs), where both the generator and discriminator are conditioned on input images and the model learns the direct mapping between paired images in different domains, for instance, translating satellite images between its boundary representations (edges) and photograph-

like full satellite images. That process often involves using zero level sets to simply complex scenes by highlighting only on the zero level contours.

The conditional GAN objective learns the mapping from observed images $\mathbf{x}$ and the random noise vector $\mathbf{z}$, to output images $\mathbf{y}$, that is, $G : \{\mathbf{x}, \mathbf{z}\} \rightarrow \mathbf{y}$:

$$\mathcal{L}_{\text{cGAN}}(G, D) = \mathbb{E}_{\mathbf{x}, \mathbf{y}} [\log D(\mathbf{x}, \mathbf{y})] + \mathbb{E}_{\mathbf{x}, \mathbf{y}} [\log (1 - D(\mathbf{x}, G(\mathbf{x}, \mathbf{z})))] , \quad (2.27)$$

where G and D are parameterized by a U-Net and a Patch discriminator, respectively. The incorporation of $\mathbf{z}$ is not given at the input of the generator in addition to $\mathbf{x}$, but rather as in the form of dropout on several layers. Authors find this prevents the generator ignoring the noise.

In addition to the conditional GAN objective, the authors augment it with a reconstruction term based on the L1 distance, as it encourages less blurring:

$$\mathcal{L}_{L1}(G) = \mathbb{E}_{\mathbf{x}, \mathbf{y}} [\| \mathbf{y} - G(\mathbf{x}, \mathbf{z}) \|_1] , \quad (2.28)$$

then the final objective is expressed as:

$$G^* = \arg \min_G \max_D \mathcal{L}_{\text{cGAN}}(G, D) + \lambda \mathcal{L}_{L1}(G) \quad (2.29)$$

Since the release of the pix2pix model, hundreds of users have employed it artistic works. However, this model demonstrated limited stochasticity in the outputs, and thus unable to capture the full entropy of the conditional distributions they intend to model.

2.5.5 Data Alignment

A restrictive assumption for unsupervised I2I translation is that data from the input and output domains are aligned. Data alignment implies that there is a straightforward correspondence in terms of the number of instances and the spatial arrangement of objects between examples in both the input and output domains. For instance, translating human faces to anime characters can yield meaningful results if both datasets contain similar poses and only male or female faces as it's more



Figure 2.17: **Example of data alignment on human-anime faces.** Alignment of objects within images from both source and target domains can affect the effectiveness of the translation. This example depicts well-aligned translation where positions of the head, eyes, nose, and mouth in the human face image closely match those in the anime face image. Alignment can ensure structural integrity of the objects within images during the translation process, resulting in coherent and visually appealing outputs.

likely that their physical features can have a direct correspondence (Figure 2.17). However, collecting aligned datasets or carefully selecting and preprocessing samples to align them can be laborious and time-consuming.

A better approach is to simply consider the setting with unaligned domains as it would enable the translation of more imaging modalities in a cost-effective manner. One solution is to implement a detector to extract and select examples from both input and output datasets via bounding boxes [115]. Nevertheless, without human supervision, there may still be unwanted and unaligned examples present, which can negatively affect the translation quality. An alternative method is to embed images into two spaces: a domain-invariant content space and a domain-specific attribute space, showing diverse translations and mitigating mode collapse [116].

More general approaches rely on the idea of feature matching, where features are extracted using a neural network and correspondences are established in a latent space. Many other works utilize cycle-consistency loss to enforce alignment in an implicit manner. These techniques improve the generated translations by ensuring they are visually coherent and realistic, even in the absence of explicit alignment between the input and output domains.

2.6 Deep Learning Approaches

Key points - The potential of combined Deep Learning approaches, or why GANs are not the panacea:

1. While GANs have been central to image-to-image translation, producing high-quality and visually appealing outputs, they can suffer from well-studied limitations such as mode collapse, training instability, lack of interpretability, and overfitting.
2. To address the shortcomings of GANs, various deep learning approaches with solid theoretical foundations have been developed. Notably, denoising diffusion models have emerged as a breakthrough in generative modeling, enabling a wide array of tasks within a single framework.
3. Advances in data representation, particularly implicit neural representations, have furthered the capabilities of image-to-image translation. The successful combination of diverse deep learning techniques, including vector quantization and pretraining, has enhanced the effectiveness and integration of these methods.

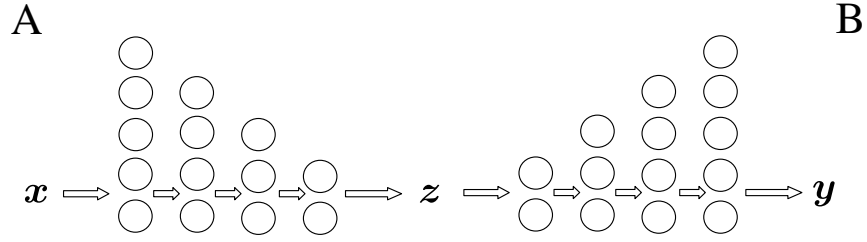


Figure 2.18: **Encoder-Decoder Architectures.** The encoder reduces the spatial dimensions of the input image while increasing feature channels, compressing the image into a latent representation $\mathbf{z}$. The decoder reconstructs the output image from this latent representation. U-Net and general encoder-decoder networks are widely used in applications such as image segmentation, translation, and reconstruction.

2.6.1 U-Net & Encoder-Decoder Networks

Encoder-decoder architectures represent a broad category in Deep Neural Networks, comprising a compressing stage and a reconstruction stage. Examples include autoencoders, variational autoencoders (VAEs), and conditional generative models, commonly employed in various generative modeling tasks.

The compressing stage learns a latent representation of the input data through multiple down-sampling layers with increasing channel numbers (encoder), while the reconstruction stage generates an output from the latent representation using multiple up-sampling layers with decreasing channel numbers (decoder) (Figure 2.18). The performance of encoder-decoder architectures is typically assessed by a reconstruction loss (e.g., an L^p distance) that measures the disparity between the input $\mathbf{x}$ and the reconstructed output $\hat{\mathbf{x}}$. At each step, the model parameters $\boldsymbol{\theta}$ are adjusted to refine abstract features and better represent the information in the input data via a lower-dimensional representation $\mathbf{z}$:

$$\mathbf{z} = \text{Encoder}(\mathbf{x}; \theta_{\text{enc}}), \quad (2.30)$$

$$\hat{\mathbf{x}} = \text{Decoder}(\mathbf{z}; \theta_{\text{dec}}). \quad (2.31)$$

This design is highly adaptable, spawning various variants such as fully-connected, convolutional, recurrent, and attention-based auto-encoder architectures, each tailored to address specific problems.

One noteworthy variant is the U-net architecture, which enhances encoder-decoder

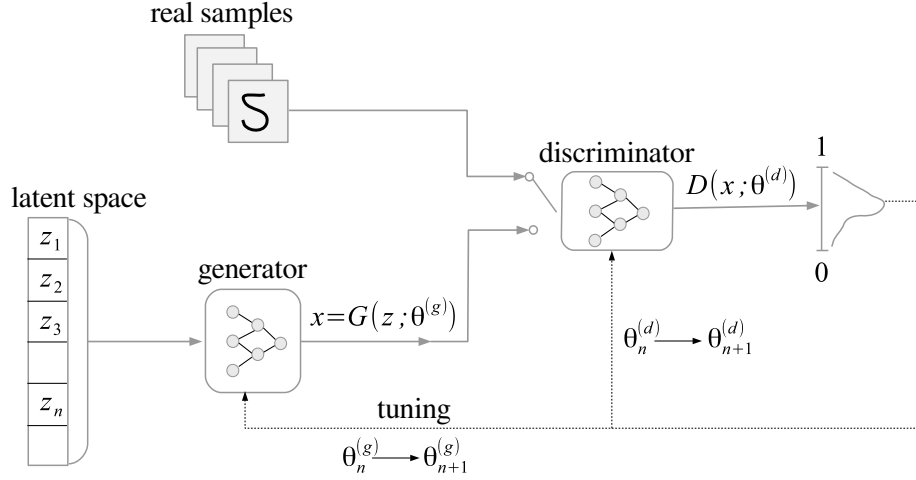


Figure 2.19: **Schematic of Generative Adversarial Networks (GANs).** (Adapted from [7]). The generator network G takes the latent vector $\mathbf{z}$ and transforms it through multiple layers into a fake sample that resembles the real data distribution. The discriminator, denoted by D is another neural network, and it takes an input sample and outputs a probability score. The training proceeds iteratively, with G and D continuously improving their performance in an adversarial manner. G learns to produce increasingly realistic samples, while D becomes better at distinguishing real samples from fake ones. This adversarial training framework is the core principle behind GANs, enabling the generation of high-quality synthetic data.

networks with skip-connections, facilitating information transfer between corresponding layers in the encoder and decoder. Initially designed for biomedical image segmentation tasks, the U-Net architecture has found application across various domains due to its ability to preserve fine details and alleviate the vanishing-gradient issue. Skip connections are represented by the concatenation ($\oplus$) of feature maps $\mathbf{h}$ from the first down-sampling (encoder) layer i to the input of the last up-sampling (decoder) layer j , and similarly for subsequent layers:

$$\mathbf{h}_{\text{skip}} = \mathbf{h}_{\text{enc}}^i \oplus \mathbf{h}_{\text{dec}}^j. \quad (2.32)$$

This integration of skip connections enhances the flow of information across the network, enabling more effective feature propagation and facilitating better reconstruction performance.

2.6.2 Generative Adversarial Networks

Generative Adversarial Networks (GANs) [31] establish a zero-sum game between two neural networks: a generator and a discriminator. The generator’s objective is to produce images that resemble the data distribution, while the discriminator aims to distinguish between samples from the true data distribution and those generated by the generator. GANs strive to find the Nash equilibrium between the generator and discriminator (Figure 2.19). Initially heralded as the state-of-the-art approach for generative modeling due to their capacity to generate highly realistic samples through adversarial training, GANs can be trained in an unsupervised manner, rendering them applicable to a wide array of tasks. However, they can suffer from issues such as mode collapse, training instability, and a lack of explicit control over generated samples.

So far, I have reviewed many image-to-image (I2I) translation approaches based on GANs, which were once considered the leading method in many generative modeling tasks. However, due to the limitations previously mentioned, GANs are not the ultimate solution. Task-specific considerations have led to several advancements, including VAEs, DDPMs, and transformer-based architectures. In particular, techniques that emphasize reconstruction accuracy and uncertainty quantification, such as probabilistic models, are preferred over GANs.

2.6.3 Variational Autoencoders

The VAE model [106] consists of an encoder $q_\phi(\mathbf{z}|\mathbf{x})$ to approximate the posterior distribution $p(\mathbf{z}|\mathbf{x})$ and a generative model (decoder) $p_\theta(\mathbf{x}|\mathbf{z})$ to map the latent variable $\mathbf{z}$ to the data $\mathbf{x}$. Then the VAE trains its generative model to learn a distribution $p(\mathbf{x})$ to be near the given data by maximising the lower bound of the data log-likelihood:

$$\log p_\theta(\mathbf{x}) = \sum_{i=1}^N \log p_\theta(\mathbf{x}_i), \quad (2.33)$$

$$\log p_\theta(\mathbf{x}_i) = \log \int p_\theta(\mathbf{x}_i|\mathbf{z})p(\mathbf{z})d\mathbf{z}. \quad (2.34)$$

However, directly maximizing this likelihood function using Stochastic Gradient Ascent (SGA) combined with the naive Monte Carlo gradient estimator (MCGE) can result in highly skewed samples $p_\theta(\mathbf{z}|\mathbf{x})$ with very high variance. Maximizing the log-likelihood does not explicitly encourage the model to learn a meaningful latent space structure.

The VAE therefore introduces the $q_\phi(\mathbf{z}|\mathbf{x})$ as multivariate Gaussian distribution with a diagonal covariance structure:

$$q_\phi(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\mathbf{z}|\mu_{\mathbf{z}}(\mathbf{x}, \phi), \sigma_{\mathbf{z}}^2(\mathbf{x}, \phi)\mathbf{I}), \quad (2.35)$$

then the objective can be rewritten as

$$\log p_\theta(\mathbf{x}_i) = \mathcal{L}(\mathbf{x}_i, \theta, \phi) + D_{KL}[q_\phi(\mathbf{z}|\mathbf{x}_i) \parallel p_\theta(\mathbf{z}|\mathbf{x}_i)] \quad (2.36)$$

where D_{KL} denotes the non-negative Kullback-Leibler (KL) divergence, for which a variational lower bound can be obtained

$$\log p_\theta(\mathbf{x}_i) \geq \mathcal{L}(\mathbf{x}_i, \theta, \phi). \quad (2.37)$$

Then the final VAE objective is

$$\mathcal{L}(\mathbf{x}_i, \theta, \phi) = \mathbb{E}_{\mathbf{z} \sim q_\phi(\mathbf{z}|\mathbf{x}_i)} [\log p_\theta(\mathbf{x}_i|\mathbf{z})] - D_{KL}[q_\phi(\mathbf{z}|\mathbf{x}_i) \parallel p_\theta(\mathbf{z}|\mathbf{x}_i)]. \quad (2.38)$$

VAEs offer a more stable training approach compared to GANs and more efficient sampling mechanisms than autoregressive models. However, the main drawback of variational methods is their unsatisfactory trade-off between sampling and reconstruction quality, which can result in blurry or low-quality samples.

2.6.4 Vector-Quantization

The deep generative modelling techniques I just presented originally learn continuous latent representations and have yield impressive samples and applications in I2I. At the same time, challenging training settings such as unsupervised learning,

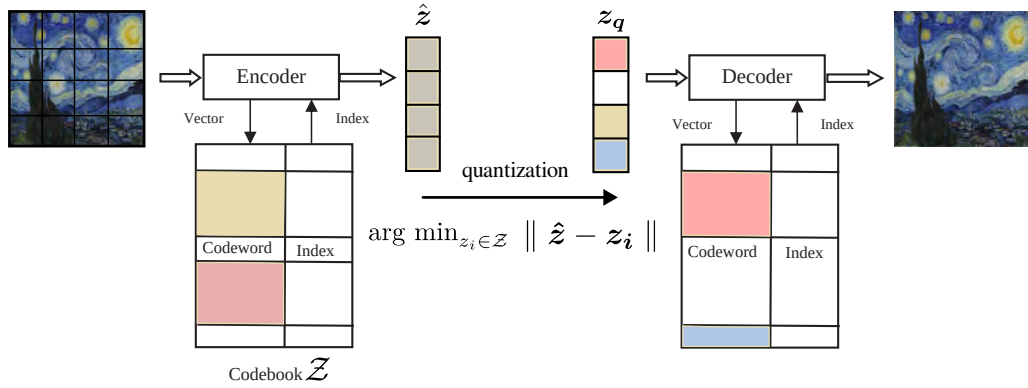


Figure 2.20: **Vector Quantization in Computer Vision.** An input image is divided into blocks, and a feature vector is extracted from each block. These feature vectors are clustered to generate a codebook $\mathcal{Z}$. During quantization, each feature vector is compared to the codewords in the codebook. The result is a compressed representation of the input image, where each block is represented by an index corresponding to a codeword. This quantized representation can be used to reconstruct an approximate version of the original image by replacing each index with the corresponding codeword from the codebook and reassembling the blocks.

unaligned data and complex topologies heavily rely on learning structured and more interpretable data representations, but the usefulness of generic continuous representations from previous work has proven to be inefficient in such cases. Later, a major shift in learning useful representations is the incorporation of the vector quantization (VQ) method as a way of learning compressed and symbolic latent representations of the data.

The quantization term comes from strong historical ties between information theory and signal processing [117]. The design of a vector quantizer is motivated by the information theory insight that one can always obtain better performance by coding vectors instead of scalars. Where coding refers to the conversion or mapping of a continuous sequence into a sequence of discrete quantities. Such concept has since found applications in various fields, including data compression for communication or storage, signal processing and machine learning approaches.

From a deep learning perspective, vector quantization is closely related to image compression with neural networks, such as scalar quantization to compress activations [118] and compressing images into a compact representation using an autoregressive architecture [119]. Learning to generate efficient compressed representations of images allows researchers high-quality reconstructions. None of those methods,

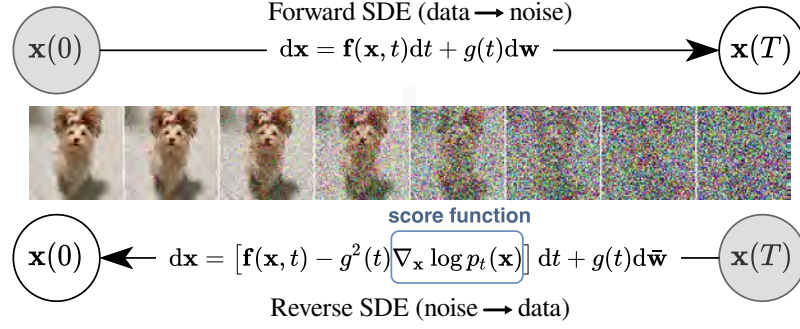


Figure 2.21: Schematic of score matching and denoising diffusion in computer vision (adapted from [8]).

however, were close in performance with continuous latent variables back then as the effective design of representative vectors (codewords) for quantization was particularly challenging.

A vector-quantizer (VQ) consists of an encoder that maps each continuous-valued input vector $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\}$ to a channel symbol $\mathbf{c}_i$ from a set known as the codebook $\mathbf{C} = \{\mathbf{c}_0, \mathbf{c}_1, \dots, \mathbf{c}_k\}$, and a decoder that reconstructs the original vector $\mathbf{x}$ using the corresponding codeword $\mathbf{c}_{i^*}$. Mathematically, vector-quantization can be represented as the mapping $\mathbf{x} \rightarrow \mathbf{c}_{i^*}$, where the quantizer's performance relies on a distortion measure (Figure 2.20).

The distortion measure d evaluates the cost of reproducing any input vector $\mathbf{x}$ as a reproduction vector $\hat{\mathbf{x}}$. A quantizer is considered effective if the resulting average distortion is small. This measure can be defined using various metrics, such as the squared Euclidean distance:

$$d(\mathbf{x}, \hat{\mathbf{x}}) = \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2 \quad (2.39)$$

This is achieved by identifying the index i that minimizes the distortion between $\mathbf{x}$ and $\mathbf{c}_i$:

$$i^* = \arg \min_i \|\mathbf{x} - \mathbf{c}_i\|_2^2. \quad (2.40)$$

2.6.5 Score Matching & Denoising Diffusion

In statistical and machine learning contexts the normalization constant is an essential concept as it ensures that the probability mass function (PMF) integrates or sums to 1 over the entire domain. Estimating a statistical model when the probability density function (PDF) of the data is known up to a normalization constant is a difficult problem as the normalization can be computationally infeasible. This is often the case when dealing with complex distributions.

Given a random vector $\mathbf{x} \in \mathbb{R}^n$ with probability density function $p_{\mathbf{x}}(\cdot)$, and a parametrized density model $p(\cdot; \boldsymbol{\theta})$. The objective is to estimate the parameter vector $\boldsymbol{\theta}$ from $\mathbf{x}$. However, we can only compute the density function given the model up to a multiplicative constant $Z(\boldsymbol{\theta})$, as follows:

$$p(\boldsymbol{\xi}; \boldsymbol{\theta}) = \frac{1}{Z(\boldsymbol{\theta})} q(\boldsymbol{\xi}; \boldsymbol{\theta}), \quad (2.41)$$

where the computation of Z involves an integral that is often analytically intractable:

$$Z(\boldsymbol{\theta}) = \int_{\boldsymbol{\xi} \in \mathbb{R}^n} q(\boldsymbol{\xi}; \boldsymbol{\theta}) d\boldsymbol{\xi}. \quad (2.42)$$

Score Matching

Traditional methods for estimating such models typically involve Markov Chain Monte Carlo (MCMC) methods, which can be computationally expensive and may not always provide accurate estimates, especially in high-dimensions.

An alternative method was suggested by Hyvärinen in 2005 [120] that consisted in minimizing the expected squared distance between the gradient of the log-density given by the model and the gradient of the log-density of the observed data, effectively removing the dependency to $Z(\boldsymbol{\theta})$ (Figure 2.21). The author referred to the gradient of log-density as the score function denoted by:

$$\psi(\boldsymbol{\xi}; \boldsymbol{\theta}) = \nabla_{\boldsymbol{\xi}} \log q(\boldsymbol{\xi}; \boldsymbol{\theta}), \quad (2.43)$$

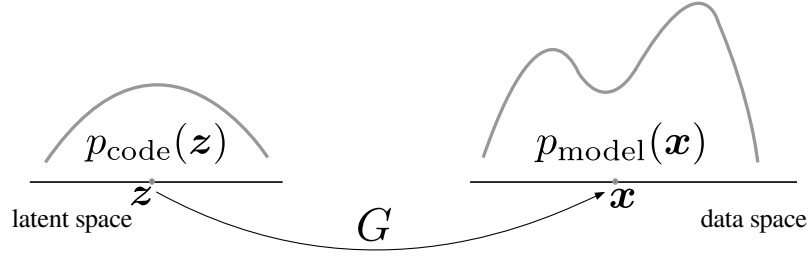


Figure 2.22: The transformation G between the latent space and data space (adapted from [7]).

similarly, the score function of the distribution of the observed data $\mathbf{x}$ is denoted as:

$$\psi_{\mathbf{x}}(\cdot) = \nabla_{\boldsymbol{\xi}} \log p_{\mathbf{x}}(\cdot). \quad (2.44)$$

With this method, the model is estimated by minimizing the expected squared distance between $\psi(\cdot; \boldsymbol{\theta})$ and $\psi_{\mathbf{x}}(\cdot)$, defined as:

$$J(\boldsymbol{\theta}) = \frac{1}{2} \int_{\boldsymbol{\xi} \in \mathbb{R}^n} p_{\mathbf{x}}(\boldsymbol{\xi}) \|\psi(\boldsymbol{\xi}; \boldsymbol{\theta}) - \psi_{\mathbf{x}}(\boldsymbol{\xi})\|_2^2 d\boldsymbol{\xi}, \quad (2.45)$$

and the score matching estimator of $\boldsymbol{\theta}$ is given by:

$$\hat{\boldsymbol{\theta}} = \arg \min_{\boldsymbol{\theta}} J(\boldsymbol{\theta}). \quad (2.46)$$

Such computation of the score function can be directly obtained from q , and without depending on the computation of the normalization scalar Z .

A Brief History of Deep Generative Models

While score matching was introduced in 2005 as a promising technique for estimating the gradient of the log-density function of a probability distribution, it took time for researchers to develop suitable architectures to effectively incorporate this idea into generative models [121].

The large-scale research of deep image synthesis models is tractable since 2013, when Kingma and Welling introduced Variational Autoencoders (VAEs) [106], the first deep learning models designed for unsupervised learning and data generation by leveraging probabilistic modeling to capture the underlying distribution of the

data and generate new samples from it. A year later in 2014, Goodfellow *et al.*, revolutionized the field of image synthesis with Generative Adversarial Networks [31], consisting of two neural networks, a generator and a discriminator, trained adversarially to generate realistic images. Shortly after, a notable variant from Radford *et al.* in 2015, introduced deep convolutional architectures in the GAN framework [122], resulting in improved stability and performance compared the to original simpler architecture based on feedfoward neural networks for both generator and discriminator.

Then autoregressive generative models such as PixelCNN [123] and PixelRNN [48] proposed by van den Oord *et al.* generated images pixel by pixel, conditioning on previously generated pixels, allowing for fine-grained control over the generation process. Since then other variants of GANs such as progressive growing [124] and StyleGAN versions [125–127] brought additional improvements for increased resolution and quality.

An important breakthrough was the adaptation of transformers for image synthesis in the paper “*Taming Transformers for High-Resolution Image Synthesis*” by Esser *et al.* in 2021. [128] Since then, various influential works have emerged, showcasing the versatility of the transformer architecture beyond natural language processing (NLP) domains. A notable example is the TransGAN model [129], which completely replaced convolutions with transformers to create a GAN capable of scaling to high-resolution image generation while capturing semantic contexts and low-level textures simultaneously.

Another significant advancement was DALL-E [130], which demonstrated a transformer language model trained to generate high-resolution images directly from a text prompt consistently. This model and its subsequent versions have found widespread use in creative and artistic tasks [131, 132]. The CLIP approach [133] introduced a novel pre-training task where the model learns to predict the caption for each image and demonstrated its ability to transfer knowledge to downstream tasks, including action recognition, geo-localization, and object classification. Despite the success of transformers and their attention mechanisms in capturing long-range dependencies and diversifying image synthesis tasks, they also present computational

challenges and are often preferred for text-to-image synthesis tasks.

Denoising Diffusion Models

Generative models offer distinct advantages that make them more or less suitable for various image synthesis tasks (Figure 2.22). Models providing an explicit probabilistic framework allow for probabilistic inference and uncertainty quantification. However, they often require estimating posterior distributions or conducting Bayesian inference, which involves more computationally expensive techniques such as MCMC or variational inference. The development of denoising diffusion probabilistic models (DDPMs) [134] were the first to effectively apply the concept of score matching to generative modeling tasks, resulting in high-quality results. In contrast to vision transformers, denoising diffusion models directly estimate the conditional distribution of clean images given noisy inputs, enabling better interpretability of the generation process. This probabilistic nature arises from their diffusion steps, which consist of a sequence of stochastic transformations, and the model is trained to maximize the likelihood of the observed data under the diffusion process.

In a nutshell, denoising diffusion is a process that involves adding noise to a signal and then gradually removing it to recover the original signal. It has been used in various applications such as image restoration, super-resolution, temporal data modeling, natural language generation, etc [135–137]. Formally, the term diffusion refers to the gradual spread of information or energy through a medium. In the context of denoising diffusion, this term denotes the transformation of the input data into a sequence of noisy versions of itself. Given a noise distribution $p_{\mathbf{z}}(\mathbf{z})$, samples from it undergo a sequence of invertible transformations T_t to approximate samples from the true data distribution $p_{\text{data}}(\mathbf{x})$.

The model’s objective is to denoise each noisy version and recover the original data using a neural network, often parameterized as a U-Net-like architecture or a transformer-based one [138]. Denoising Diffusion Probabilistic Models (DDPMs) frame the denoising process as a probabilistic inference problem. Given a sequence of noisy observations, the model learns the conditional distribution of the original data given the noisy observations, denoted as $p(\mathbf{x}_t|\mathbf{x}_{t-1})$, where $\mathbf{x}_t$ represents the data at

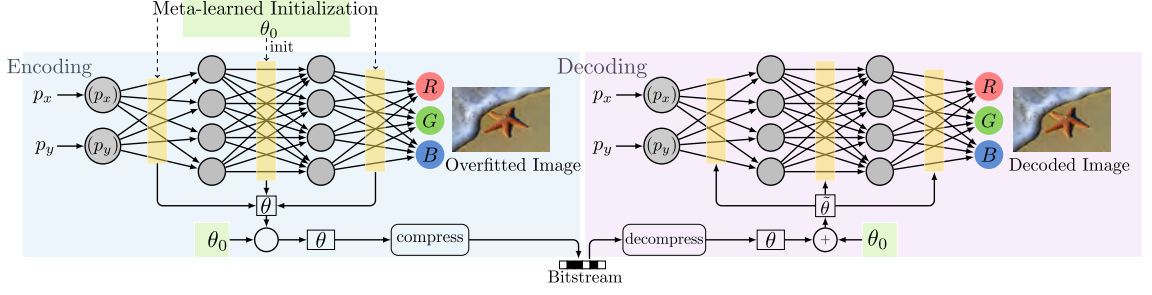


Figure 2.23: **Implicit Neural Representations in a nutshell.** (adapted from [9]) INRs are a relative novel way of modelling complex functions or data without explicit parameterization. Essentially, implicit functions map input coordinates to scalar values using neural networks. Trained on pairs of input coordinates and corresponding function values, these networks learn to approximate desired functions accurately. Applications include surface reconstruction and rendering, offering flexibility, generalization, and efficiency.

time step t . Then, the model is trained to minimize the negative log-likelihood of the data under the learned conditional distribution.

Capturing the conditional distribution is often modeled autoregressively, where the joint distribution of the data is decomposed into a product of conditional distributions. This implies that each element $\mathbf{x}_t$ is conditionally dependent on the corresponding element $\mathbf{x}_{t-1}$:

$$p(\mathbf{x}_t | \mathbf{x}_{t-1}) = \prod_{i=1}^D p(\mathbf{x}_{t,i} | \mathbf{x}_{t-1,1:i}) \quad (2.47)$$

However, not all data conforms to such a sequential manner (e.g., images). To this end, non-autoregressive diffusion models the conditional distribution $p(\mathbf{x}_t | \mathbf{x}_{t-1})$ without assuming dependencies between elements of the data. This results in a more computationally efficient process but may sacrifice some modeling capacity.

For discrete latent representations, we have absorbing diffusion [139], inspired by Conditional Masked Language Models in text. In this approach, a transition matrix is employed to determine whether a particular token remains unchanged or transitions to an absorbing state called [MASK] with some probability β_t in each forward step. The reverse process gradually unveils these masks, effectively removing any relationships between categories.

2.6.6 Implicit Neural Representations

Having a model that handles high-dimensional data with lower memory and computational resources is a crucial characteristic for real-world applications. The curse of dimensionality is a clear and related phenomenon, particularly in machine learning and data analysis. The main challenges include increased complexity in both distance calculations and optimization algorithms, which could potentially impact scalability. Moreover, the number of samples required to represent the data distribution increases exponentially with the dimensionality of the data. This becomes critical in medical settings due to limited data availability compared to natural images. Furthermore, this problem exacerbates the model's capacity to generalize, as with a large number of features, models are more prone to memorizing noise.

Research on alternative and more efficient data representations gave rise to the idea of learning functions from data, employing an implicit representation rather than learning the explicit geometry of the data. In the context of machine learning, these are referred to as implicit neural representations (INRs). An INR is a mathematical model that defines a continuous function from data through training neural networks (NNs). Let $f : \mathcal{X} \rightarrow \mathbb{R}$ be the unknown function we want to approximate, where $\mathcal{X}$ is the input space. An INR $\phi_{\theta} : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}$ is parameterized by a neural network ϕ_{θ} with parameters θ that maps input-output pairs $(x, y) \in \mathcal{X} \times \mathcal{Y}$ to real-valued outputs, where $\mathcal{Y}$ is the output space (Figure 2.23).

The neural network is trained to minimize the discrepancy between the predicted output and the true output for a given input x :

$$\phi_{\theta}(x, f(x)) \approx 0. \quad (2.48)$$

INRs are commonly employed in machine learning tasks where the underlying function is complex and direct parametrization is unclear. They offer various advantages, such as the flexibility to handle arbitrary input and output spaces, and compact representations.

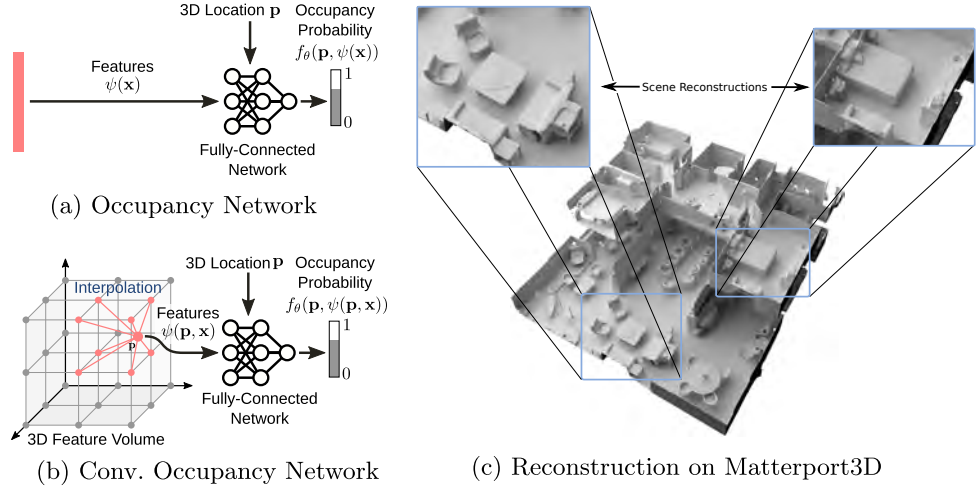


Figure 2.24: **Representation and Reconstruction with Occupancy Networks.** (Adapted from [10]) (a) Occupancy networks predict the likelihood of points being inside surfaces in 3D space. (b) 3D convolutional networks process volumetric data to extract hierarchical features. (c) These methods enable detailed 3D reconstructions of environments like those in the Matterport3D dataset, useful for applications such as virtual tours and robotics.

Encoding & Decoding Processes

Learning INRs follows an encoding and decoding process where the objective is to capture relevant features in the input data and then accurately synthesize outputs from such latent representations. In the encoding process, a latent representation $\mathbf{z}$ is obtained from the input data $\mathbf{x}$ using an encoder neural network, where $E_\phi(\mathbf{x}) = \mathbf{z}$. Typically, the encoder network consists of multiple convolutional or fully connected layers combined with non-linear activations such as ReLU or sigmoid.

In the decoding process, the latent representation $\mathbf{z}$ is mapped to a continuous function f using a decoder neural network, denoted as $D_\varphi(\mathbf{z}) = f(\mathbf{x})$. Depending on the task and the format of the input data, the decoder can take the form of a convolutional network, a MLP, or RNN. The function f then represents the output synthesized by the INR for the input data $\mathbf{x}$. Such output can be an image, a 3D shape, or any other form of structured or unstructured data.

Occupancy Networks

The representation of many real-world 3D datasets are based on voxel grids or surface meshes. However, such representations are discrete and may suffer from resolution limitations or sparse and unstructured data. Efforts to provide more flexible and accurate modeling of complex geometry include the occupancy networks approach [140]. These offer a compact and efficient representation of 3D shapes by directly predicting occupancy probabilities instead of explicit surface parameterization.

Occupancy networks represent a 3D shape as a continuous occupancy function $O : \mathbb{R}^3 \rightarrow [0, 1]$. For a given 3D point $\mathbf{x}$, $O(\mathbf{x})$ outputs an occupancy probability, indicating whether the point lies inside the surface of the object (occupied) or outside (unoccupied) (Figure 2.24).

Learning the occupancy function is performed from a set of 3D shapes represented as voxel grids or point clouds. They follow an encoder-decoder process, as mentioned in the previous section, and the decoder’s goal is to predict the occupancy probability of K randomly sampled points in the 3D volume by evaluating the mini-batch loss:

$$\mathcal{L}_{\mathcal{B}}(\boldsymbol{\theta}) = \frac{1}{b} \sum_{i=1}^b \sum_{j=1}^K \mathcal{L}(O(\mathbf{x}_{ij}), O_{\text{GT}}(\mathbf{x}_{ij})). \quad (2.49)$$

Here, $\mathcal{L}$ represents the loss function, $O(\mathbf{x}_{ij})$ is the predicted occupancy probability for the j -th point in the i -th shape, and $O_{\text{GT}}(\mathbf{x}_{ij})$ is the ground truth occupancy probability. The loss is computed over the mini-batch $\mathcal{B}$ containing multiple shapes.

This approach addresses many difficulties in modeling complex geometry by not being constrained by the discretization of the 3D space. Furthermore, its introduction extensively demonstrates its representation power with various tasks, including reconstruction of 3D shapes, point cloud completion, voxel super-resolution, and unconditional mesh generation.

Signed Distance Functions

Although occupancy predictions from the previous approach can be indirectly affected by the interior shape, these are not explicitly modeled in the near surface of

the sampled points. To address this limitation, a different implicit representation named Signed Distance Functions (SDFs) was introduced shortly after by Park *et al.* in 2019 [141]. In this approach, both the shape’s surface and its interior are encoded, making it more adequate for rendering techniques. SDFs enable the representation of more complex topologies, including concavities, self-intersections, and sharp features. Since they are continuous and differentiable, gradient-based optimization can be used for shape and surface fitting. For these reasons, SDFs are widely used in physics simulations like collision detection and rigid body dynamics.

SDFs encode the signed distance from each point in space to the surface of the object, defined as $\phi : \mathbb{R}^3 \rightarrow \mathbb{R}$. Let $\mathbf{x}$ be a point in 3D space, then we can define:

$$\begin{cases} \text{negative,} & \text{if } \mathbf{x} \text{ is inside the shape} \\ 0, & \text{if } \mathbf{x} \text{ is on the surface of the shape} \\ \text{positive,} & \text{if } \mathbf{x} \text{ is outside the shape} \end{cases} \quad (2.50)$$

where the surface of the shape is defined as the zero-level set of the SDF, $\{\mathbf{x} \in \mathbb{R}^3 \mid \phi(\mathbf{x}) = 0\}$.

Authors demonstrated its applicability in shape representation and completion tasks of complex topologies and closed surfaces. However, a clear limitation is that it assumes models are in a canonical pose, making them unsuitable for completion in-the-wild, as it would require optimization over the special Euclidean group $SE(3)$ transformation space (rotation and translation) to align the input shape.

Neural Radiance Fields

Novel view synthesis is a long-standing problem in graphics and computer vision, involving the rendering of new views of an object or scene from a given set of input images, with or without camera poses as priors. The groundbreaking Neural Radiance Fields (NeRF) method, introduced in 2021 by Mildenhall *et al.* [142], proposed representing 3D scenes as continuous functions learned from posed 2D images. This representation has been widely used and inspired many influential works in the field of computer graphics [143].

Let $\mathbb{R}^3$ represent the 3D space and S^2 denote the unit sphere parameterized by azimuth and elevation angles. The NeRF framework defines a volumetric function $F : \mathbb{R}^3 \times S^2 \rightarrow \mathbb{R}^3$. This continuous 5D function outputs the radiance (color) in each viewing direction $\mathbf{d} : (\theta, \phi)$ at each given 3D point $\mathbf{x} : (x, y, z)$. NeRF learns the volumetric function from a set of posed images $(I_i, \mathbf{p}_i)$, where I_i represents the observed image and $\mathbf{p}_i$ denotes its corresponding camera pose. These images are normally generated by rendering synthetic views of the scene using computer graphics techniques. The final objective is to learn the parameters of F that predict the observed radiance values for all points and viewing directions.

The parameterization of F consists of two components; a density network $\sigma(\mathbf{x}; \theta_{\text{sigma}})$ and a color network $\mathbf{c}(\mathbf{x}, \mathbf{d}; \theta_{\text{color}})$. The volume density σ function computes the occupancy probability at a point $\mathbf{x}$ optimizing the parameters θ_{sigma} . In contrast, the RGB color $\mathbf{c}$ function is predicted from both location and viewing direction.

Putting everything together, the rendering process starts with a novel viewing direction $\mathbf{d}$ and a corresponding camera pose $\mathbf{p}$. NeRF then renders the scene by integrating along rays cast from the camera $C(\mathbf{p}, \mathbf{d})$, defined as $\mathbf{r}(t) = \mathbf{p} + t\mathbf{d}$, with near and far planes t_{near} and t_{far} :

$$C(\mathbf{p}, \mathbf{d}) = \int_{t_{\text{far}}}^{t_{\text{near}}} \sigma(\mathbf{p} + t\mathbf{d})\mathbf{c}(\mathbf{p} + t\mathbf{d}, \mathbf{d})dt. \quad (2.51)$$

NeRF approximates this rendering integral using Monte Carlo integration during training, and both sets of parameters θ_{sigma} and θ_{color} are optimized to minimize the difference between the predicted and observed radiance values in the training images.

The NeRF method has attracted significant attention due to its remarkable success in enabling high-quality novel view synthesis and scene manipulation by providing a neural network-based representation of 3D scenes.

2.6.7 Hybrid Approaches for I2I Translation

Hybrid approaches for image-to-image (I2I) translation have leveraged many of the techniques previously reviewed. Here, I describe the most relevant ones in terms

of impact and in a chronological manner. It’s worth noting that most of these works incorporate various regularization strategies such as gradient penalty or self-supervised auxiliary tasks. While I do not go into the details of these strategies here, I refer readers to a comprehensive survey in [144].

GANs initially led the way. The UVCGAN model [145] introduced a CycleGAN variant where the generator incorporates a U-Net with a pixel-wise vision transformer (ViT), enhancing the capture of global structure. Input data is encoded through a series of transformer encoder blocks and flattened along spatial dimensions to form a token sequence. This approach effectively handles long-range spatial dependencies during the translation process, with attention weights highlighting relevant regions in the source images.

Shortly after hybrids started to emerge. After the introduction of diffusion models in computer vision, researchers explored the combination of these and adversarial training to enhance the quality of unpaired image-to-image (I2I) translation. The PITI model [146] is a notable example, adapting a pretrained diffusion model for various I2I tasks, resulting in high-quality and diverse image synthesis. The method first pretrains on a large dataset for downstream tasks to learn a rich latent space, the model then fine-tunes translated images by conditioning the model with text tokens produced by a transformer, leading to improved perceptual quality.

VQ approaches didn’t stay behind. Later, a vector-quantized approach showcased various I2I translation applications [147], including unconditional image generation, image extension, and transitional stylization. The unified VQ framework enables unconditional distribution sharing among different domains, incorporating two encoders to learn joint domain-invariant content and domain-specific style, along with domain-specific decoders. The content distribution is learned autoregressively using a transformer, while the translation process employs an attention window to extend the content map into different styles.

What about auxiliary pretraining? The success of deep learning in text gave rise to the Plug-and-Play Diffusion model [148], a novel text-guided I2I translation method, allowing users to manipulate generated content using both an image and a target text prompt. The input text prompt conditions a pretrained text-to-image diffusion model to generate images aligned with the text, while preserving the semantic layout of the guidance image with an auxiliary semantic segmentation network. Results showcased the versatility and fine-grained control of the model in tasks such as translating sketches, rough drawings, and animations into realistic images, and altering the class and appearance of objects in a given image.

Leveraging existing models. In contrast, the pix2pix-zero model [149] offered a training-free and prompt-free diffusion-based approach for I2I translation on the fly. Users specify the edit direction from source domain to target domain (e.g., sketch to oil pastel), and GPT-3 generates groups of diverse sentences whose embeddings are obtained from a CLIP model. Translated images are then generated via a DDPM that obtains a noise map guided by the closest embedding direction to the edit direction. This approach, equipped with cross-attention, allows for smooth edits of the original image, effectively preserving realism and content compared to previous methods.

2.7 Designing 2D-3D Image Translation Models

While numerous works have been developed for I2I translation [22,23], there remains a notable gap in terms of dimensionality. The works I just reviewed deal with only 2D-2D image translation, and despite they demonstrate robustness and diversity in various datasets, it is unclear if they can adapt and scale to 2D-3D image translation.

On the other hand, we also find many works working with 3D-3D image translation, in particular for medical imaging purposes [150,151]. It’s evident that learning a mapping from images of same dimensionality is less complex compared to learning a mapping where there is a dimensionality gap (e.g., 1D-2D, 2D-3D). The reason is that filling that gap involves a generative modeling problem with special constraints

including input conditionals, content preservation, perceptual realism, without mentioning concerns of hallucinated outputs for certain applications (e.g., medical).

Designing a model that solves the 2D-3D image translation problem entails considering that the deep learning architectures that handle one data dimensionality work and learn differently in another one. For instance, in the case of CNNs, 2D kernels operate on 2D feature maps by convolving the input along spatial dimensions (height and width). 3D kernels, however, are far more computationally and memory expensive because of the additional depth (or temporal) dimension present in the 3D input data like video sequences or 3D medical scans. The problem is more complicated than simply re-purposing 2D or 3D architectures to handle both geometries.

The generative modeling problem involves reconstructing data to fill the dimensionality gap. More formally, this refers to learning the underlying probability distributions of both input and output image domains. This would allow the model to generate samples that reflect the original datasets, and thus capture the patterns and structures in the data. All these, while preserving information and spatial relationships from the original data, that is, the generative model should learn to faithfully represents aspects in the translated images and at the same time, distinguish domain differences.

MedNeRF: Medical Neural Radiance Fields for Reconstructing 3D-aware CT-Projections from a Single X-ray

3.1 Introduction

3D medical imaging often involves joining multiple 2D slices from CT or Magnetic Resonance Imaging (MRI), and part of their workflow consists of specifying values for the position of the patient, the imaging source, and the detector. The quality and accuracy of a CT 3D representation require hundreds of X-ray projections with a thin slice thickness [152]. Moreover, this process exposes patients to more ionising radiation than typical X-rays and requires the patient to remain immobile for up to more than 1 hour, depending on the type of test [153]. Continuous 3D representations would give radiologists optics of every point in the internal anatomy captured. While such representations are useful, there are practical challenges in CT due to the increased radiation exposure, angle-dependent structures, and time consumption [20].

Earlier approaches in medical image reconstruction used analytic and iterative methods [154, 155] on given input data. However, they often encounter mismatches between the mathematical model and physical properties of the imaging system.

Instead, several recent approaches leverage deep learning [156] for sparse view reconstruction [157–159], 3D CT reconstruction from 2D images [160], and anomaly detection [161]. These deep learning approaches solved the mismatches between the mathematical model and imaging system and reported improved reconstructions by fine-tuning state-of-the-art architectures. However, they require a large amount of training data, which may be difficult to meet in the medical domain where acquiring expert annotations is both cost and time prohibitive.

The Neural Radiance Fields (NeRF) [142] model is a *recent* reformulation for estimating a 3D volumetric representation from images. Such representations encode the radiance field and density of the scene in the parameters of a neural network. The neural network learns to synthesize new views via volume rendering from point samples along cast rays. However, these representations are often captured in controlled settings [162]. First, the scene is taken by a set of fixed cameras within a short time frame. Second, all content in the scene is static and real images often need masking. These constraints prohibit the direct application of NeRF to the medical domain, where the imaging system greatly differs from conventional cameras, and the images are captured over a long time frame hampering the patient’s stillness. Moreover, the overlapping of anatomical structures in medical images hinders the definition of edges which cannot be easily solved with masking. These aspects explain why the NeRF approach especially shows successes for “natural images”.

To address these challenges, this chapter proposes MedNeRF, a model that adapts Generative Radiance Fields (GRAF) [163] in the medical domain to render CT projections given a few or even a single-view X-ray. The proposed approach not only synthesizes realistic images, but also capture the data manifold and provides a continuous representation of how the attenuation and volumetric depth of anatomical structures vary with the viewpoint without 3D supervision. This is achieved via a new discriminator architecture that provides a stronger and more comprehensive signal to GRAF when dealing with CT scans.

Closest to the goal of this thesis are [158, 159], which both train a coordinate-based network in sinograms of low-dose CT of phantom objects and apply it to the sparse-view tomography reconstruction problem. In contrast to [158], the proposed

method learns multiple representations in a single model by randomly feeding data of different medical instances instead of separately optimizing for each collection of images. For testing [159] reconstruction ability, they integrate it into reconstruction methods and use at least 60 views. Unlike previous methods, this approach does not rely on additional reconstruction algorithms, and requires multiple views only during training.

CT projections of datasets of digitally reconstructed radiographs (DRR) from chest, head-neck and knee are rendered. Both qualitative and quantitative evaluations demonstrate high-fidelity renderings, comparing this approach to other recent radiance field-based methods. Furthermore, CT projections are generated for a medical instance given a single-view X-ray, showing the effectiveness of the proposed model to cover surface and internal structures.

3.2 Related Work

Medical image synthesis and reconstruction

In this chapter we’re particularly interested in the problem of inferring the internal anatomical structure and view-dependent appearance given a set of medical input data. Prior work reasons over analytic and iterative methods [154, 164–166] on the underlying measurement data. However, these often encounter mismatches between the mathematical model and physical properties of the imaging system.

Instead, several recent DL approaches solved the mismatches between the mathematical model and imaging system and reported improved reconstructions by fine-tuning state-of-the-art architectures. However, they require a large amount of training data, which may be difficult to meet in the medical domain where the expert annotation is both cost and time-prohibitive [167].

Generative Adversarial Networks (GAN)

In the literature, there are several examples of GANs for applications in radiology, which all have achieved remarkable results in various tasks, for example, image

reconstruction and denoising [168, 169], data augmentation [170], transfer between modalities [171], and image segmentation. While these GANs can be used for cross-modality image transfer, they are tested to deal with the input and output data having the same dimension.

Closest to our goal is Ying *et al.* [160], where they build a generative adversarial network (GAN) over digitally reconstructed radiographs (DRR) from the chest to synthesize a 3D computed tomography (CT) and include CycleGAN for style transfer. Different from their methods, the proposed method uses a generative radiance field over a medical category and render novel views conditioned on latent codes for the shape and appearance of a medical instance. This approach not only synthesizes realistic images as a traditional GAN, but it also provides a continuous representation of how the object varies with the viewpoint, and it’s able to estimate its depth without any 3D supervision.

Neural representations and rendering

The modern NeRF [142] reformulation for estimating a 3D volumetric representation from images has been explored in non-medical domains for the tasks of representing shapes [172] and novel scene geometries [173]. These approaches proved that volumetric rendering is inherently differentiable, allowing the formulation of an analytic expression of the gradients, which solves the significant challenge for inverse rendering tasks. However, they require object masks and camera information for view-consistent results.

A later work allowed the representation of both shapes and scenes [174], and provided controllability of camera viewpoint and object poses. However, it’s unclear how to estimate internal anatomies textures and more complex structure properties such as density. In this work, the objective is to learn a 3D continuous representation of the anatomical structure (e.g., chest, head-neck and knee) in a CT and the overall radiance of the scan.

Self-supervised learning

Over the last decade, self-supervised techniques have been applied for capturing both low-level and high-level visual features [175]. Within a self-supervised framework, CNNs are trained on pretext tasks such as image inpainting [176], colorizing grayscale images [177], image super-resolution [178], or even playing image puzzles [179], etc. This proposed task generates a helpful supervisory signal from the data itself leveraging its structure and therefore doesn't require additional human annotations. Hendrycks *et al.* [180] showed how self-supervised techniques achieve better data representations and improve the model robustness.

Most appropriate to our generation task are image inpainting and image super-resolution, which learn visual features that would result in more realistic samples. More specifically, the proposed method is inspired by [181] work, which synthesizes few-shot high-quality images by training a self-supervised discriminator as a feature-encoder. The authors showed how image inpainting (cropping) and the ability to scale to higher resolutions lead to a more descriptive feature extraction than random initialization. The hypothesis is that the discriminator can cover more information from the input, here the proposed approach simplifies and adapts [181], and tests whether these self-supervisions would also boost medical image synthesis.

Data Augmentation

Data augmentation (DA) transformations have often been applied in applications for the medical domain as an attempt to alleviate the extremely expensive process of data collection. The goal is to increase the size of the dataset by adding its transformed version, which would result in an improved generalization ability of the model and a reduced chance of over-fitting. [182] adopted two augmentation techniques to a GAN to denoise low-dose CT images and outperformed similarity metrics compared to iterative reconstruction methods. The choice of augmentation techniques were not factors of analysis. This left question whether a different choice would bring further boost.

Noguchi *et al.* [183] furthered this idea for whole-body CT bone segmentation

by testing three DA frameworks: (1) conventional method, (2) mixup, and (3) random image cropping and patching (RICAP). Although (2) and (3) were initially proposed for image classification tasks, the authors used them for image segmentation, for which they had a small dataset to train a U-Net. They found that (2) and (3), in particular, further improved segmentation performance under different scan conditions.

These works hinted that different DA might work better for certain data. A recent work analyses the classical approach of DA in GANs and showed that many generated examples were out of the original data distribution. This suggested that some techniques may cause the generator to learn a rather infrequent or sometimes non-existent data distribution [184].

3.3 Methods

3.3.1 Dataset Preparation

To train the models, DRRs are generated instead of collecting paired X-rays and corresponding CT reconstructions, which would expose patients to more radiation. Furthermore, DRR generation removes patient data and enables control in capture ranges and resolutions. DRRs were generated by using 20 CT chest scans from [185, 186], 20 CT head-neck scans from [186, 187] and five CT knee scans from [166, 188]. These scans cover a diverse group of patients at different contrast types showing both normal and abnormal anatomy. The radiation source and imaging panel are assumed to rotate around the vertical-axis, generating a DRR of 128×128 resolution at every five degrees, resulting in 72 DRRs for each object. During training the whole set of 72 DRRs (a fifth of all views within a full 360-degree vertical rotation) are used for all patients and at inference the model renders the rest. This is in contrast to traditional NeRF approaches that fit a single neural representation per object or scene. Instead, the generative network is trained with all available data and is able to produce diverse neural representations conditioned on the given input views at inference. The presented work did not involve experimental procedures on human subjects or animals and thus did not require Institutional Review Board

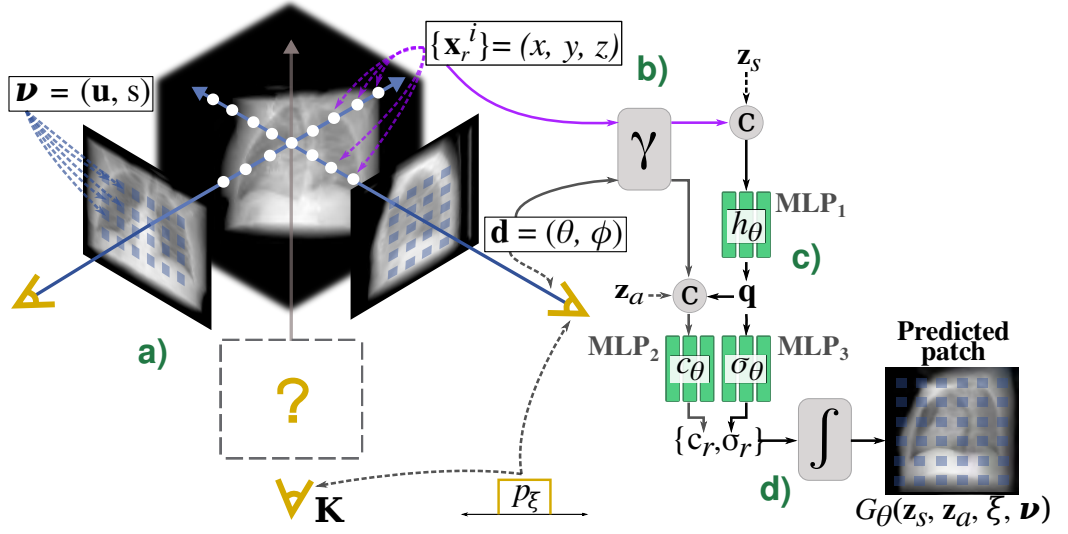


Figure 3.1: An overview of GRAF’s generator.

approval.

3.3.2 GRAF Overview

GRAF [163] is a model that builds from NeRF and defines it within a Generative Adversarial Network (GAN). This model produces 3D views of objects based on implicit representation of 3D shapes and textures. It consists of a generator G_θ that predicts an image patch $\mathbf{P}_{\text{pred}}$ and a discriminator D_ϕ that compares the predicted patch to a patch $\mathbf{P}_{\text{real}}$ extracted from a real image. GRAF has shown an effective capacity to disentangle 3D shape and viewpoint of objects from 2D images alone, in contrast to the original NeRF [142] and similar approaches such as [189]. Therefore, we aim to translate GRAF’s methods to our task, and in subsection 3.3.3 I describe the new discriminator architecture, which allows to disentangle 3D properties from DRRs.

Let’s consider the experimental setting to obtain the radiation attenuation response instead of the color used in natural images. To obtain the attenuation response at a pixel location for an arbitrary projection $\mathbf{K}$ matrix with pose ξ that describes the position and orientation of the X-ray source. First, the sampling pattern $\nu = (\mathbf{u}, s)$ is defined, where $\mathbf{u}$ is the center position of a patch in the 2D image and s is the scale of the patch. From ν , R X-ray beams are sampled within a $K \times K$ image-patch $\mathbf{P}$. Each X-ray beam $\mathbf{r}$ originates from a pixel location within the

patch and travels through the 3D object, intersecting a series of points. Then, to obtain the attenuation response along an X-ray beam, $\{\mathbf{x}_r^i\}_{i=1}^N$ points are defined. These are ordered between two planes: the near plane and the far plane of the projection, which represent the front and back boundaries of the region we’re imaging. The total attenuation response at a pixel location is obtained by accumulating the attenuation values from all sampled points along each beam. (Fig. 3.1a).

The object representation is encoded in a multi-layer perceptron (MLP) that takes as input a 3D position $\mathbf{x} = (x, y, z)$ and a viewing direction $\mathbf{d} = (\theta, \phi)$ that represents from which the point is being observed. The MLP produces two main outputs: density σ and the attenuation response c . σ indicates whether a point is inside or outside the object, while c represents X-ray attenuation intensity. To capture fine details within the 3D anatomy, it’s necessary to learn high-frequency features, for this, we make use of positional encoding. Before feeding the input to the MLP, each input coordinate p where p can represent x, y, z, θ , or ϕ is mapped into a higher-dimensional space. Specifically, the encoding function $\gamma(p)$ transforms each coordinate into a $2L$ -dimensional vector (Fig. 3.1b):

$$\gamma(p) = [\dots, \cos(2^j \pi p), \sin(2^j \pi p), \dots] \quad (3.1)$$

where the input is transformed using sine and cosine functions at different frequencies (from coarse to fine) where each frequency is $2^j \pi$ for $j = 0, \dots, m - 1$.

For modeling the shape and appearance of anatomical structures, let $\mathbf{z}_s \sim p_s$ and $\mathbf{z}_a \sim p_a$ be the latent codes sampled from a standard Gaussian distribution, respectively (Fig. 3.1c). To obtain the density prediction σ , the shape encoding $\mathbf{q}$ is transformed to volume density through a density head σ_θ . Then, the network $g_\theta(\cdot)$ operates on a shape encoding $\mathbf{q} = (\gamma(\mathbf{x}), \mathbf{z}_s)$ that is later concatenated with the positional encoding of $\mathbf{d}$ and appearance code $\mathbf{z}_a$ (Fig. 3.1c):

$$\mathbf{q} = \text{MLP}_1(\gamma(\mathbf{x}), \mathbf{z}_s) \quad (3.2)$$

$$c = \text{MLP}_2(\mathbf{q}(\mathbf{x}, \mathbf{z}_s) \gamma(\mathbf{d}), \mathbf{z}_a) \quad (3.3)$$

$$\sigma = \text{MLP}_3(\mathbf{q}(\mathbf{x}, \mathbf{z}_s)) \quad (3.4)$$

The final pixel response c_r is computed by the compositing operation (Fig. 3.1c):

$$c_r = \sum_{i=1}^N c_r^i \alpha_r^i \exp \left(- \sum_{j=1}^{i-1} \sigma_r^j \delta_r^j \right) \quad (3.5)$$

where $\alpha_r^i = 1 - \exp(-\sigma_r^i \delta_r^i)$ is the alpha compositing value of sampled point i and $\delta_r^i = \| \mathbf{x}_r^{i+1} - \mathbf{x}_r^i \|_2$ is the distance between the adjacent sampled points. The exponential factor accounts for the cumulative attenuation of the ray as it travels through previous points. The alpha compositing value α_r^i indicates how much of the ray is blocked by the point i due to its density, contributing to the ray’s total opacity.

In this way, both the density and pixel values are computed at each sampled point along the beam r with network g_θ . Finally, combining the results of all R beams, the generator G_θ predicts an image patch $\mathbf{P}_{\text{pred}}$, as illustrated in Fig. 3.1d.

3.3.3 MedNeRF

This work explores the adaptation of GRAF to the medical domain, applying it to render a volumetric representation from DRRs. Leveraging a large dataset, GRAF’s discriminator D_ϕ is able to continuously provide useful signals to train the generator G_θ . However, medical datasets like those considered in our problem are generally small, which causes two sequential issues:

The lack of real information to the generator: In GRAF (and in GAN in general), the only source of features of the training data contributing to the generator is the indirect gradient transferred from the discriminator. Findings suggests that a single convolutional feedback from GRAF’s discriminator poorly conveys refined features from DRRs resulting in inaccurate volumetric estimation.

Brittle adversarial training: With a limited training dataset, the generator or discriminator may fall into ill-posed settings such as mode collapse, which would lead to generating a limited number of instances and consequently, a suboptimal data distribution estimation. While some works have applied data augmentation techniques to leverage more data in the medical domain, some transformations could mislead the generator to learn the infrequent or even non-existent augmented data distribu-

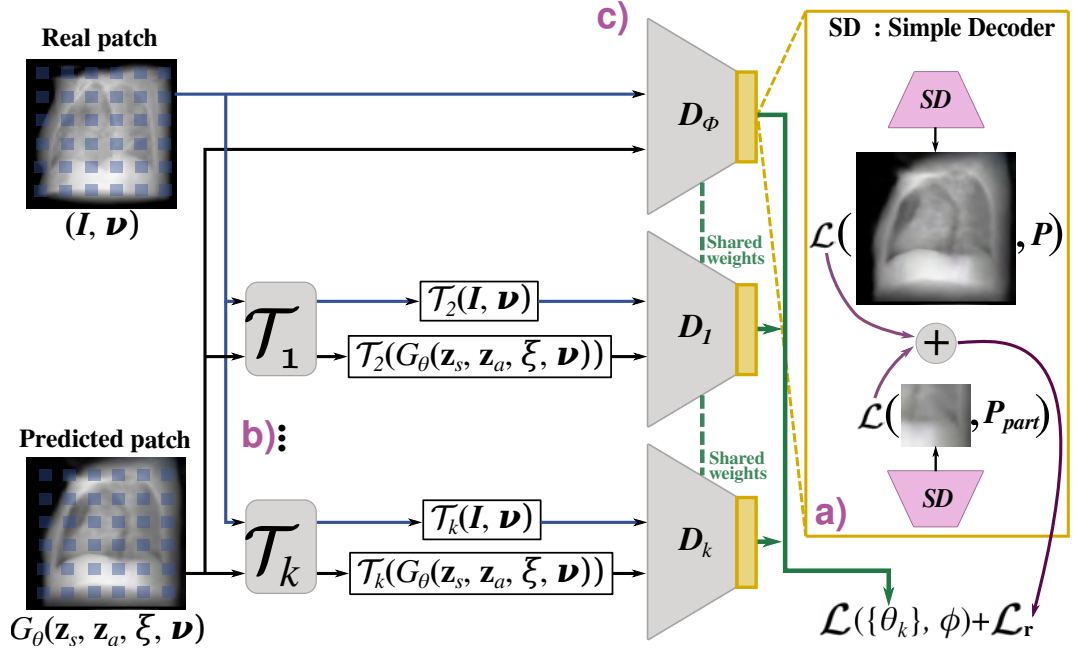


Figure 3.2: An overview of the proposed discriminator with self-supervised learning and DAG.

tion [184]. Ablations suggest that naively applying classic data augmentation works less favorably than the adopted framework.

Self-supervised Learning for High-Fidelity Synthesis

To allow *richer* feature-maps covering from the DRRs such that it produces more comprehensive signals to train G_θ , GRAF’s discriminator architecture is replaced with recent advancements in self-supervised approaches. Then D_ϕ is able to learn useful global and local features training it on a pretext task, in particular, the self-supervision method based on auto-encoding [181]. Different from [181], only two decoders are used for the feature-maps on scales: $\mathbf{f}_1$ on 32^2 and $\mathbf{f}_2$ on 8^2 (Fig. 3.2a). Results demonstrates that this choice allows better performance and enables a correct volumetric depth estimation. D_ϕ must therefore not only discriminate $\mathbf{P}_{\text{pred}}$ predicted from G_θ but also extract comprehensive features from real image patches $\mathbf{P}_{\text{real}}$ that enable the decoders to resemble the data distribution.

To assess global structure in decoded patches from D_ϕ , the Learned Perceptual Image Patch Similarity (LPIPS) metric [190] is used. The weighted pairwise image distance between two VGG16 feature spaces is computed, where the pretrained

weights are fit to better match human perceptual judgments. The additional discriminator loss is therefore:

$$\mathcal{L}_r = \mathbb{E}_{\mathbf{f} \sim D(\mathbf{p}), \mathbf{p} \sim \mathcal{P}} \left[\frac{1}{whd} \| \phi_i(\mathcal{G}(\mathbf{f})) - \phi_i(\mathcal{T}(\mathbf{p})) \|_2 \right] \quad (3.6)$$

where $\phi_i(\cdot)$ denotes the i th layer output of a pretrained VGG16 network, and w , h , and d stand for the width, height and depth of a feature space, respectively. Let $\mathcal{G}$ be the processing on the intermediate feature-maps $\mathbf{f}$ from D_ϕ , and $\mathcal{T}$ the processing on real image patches. When coupled with this additional reconstruction loss, the network learns representations that transfer across tasks.

Improving Learning via Data Augmentation

The learning of G_θ and D_ϕ is improved by adopting the Data Augmentation Optimized for GAN (DAG) framework [184] in which a data augmentation transformation $\mathcal{T}_k$ (Fig. 3.2b) is applied using multiple discriminator heads $\{D_k\}$. To further reduce memory usage, all layers of D_ϕ are shared except the last layers corresponding to each head (Fig. 3.2c). Because applying differentiable and invertible data augmentation transformations $\mathcal{T}_k$ has the Jensen-Shannon (JS) preserving property [184]:

$$\text{JS}(p_d^{\mathcal{T}_k} \| p_g^{\mathcal{T}_k}) = \text{JS}(p_d \| p_g) \quad (3.7)$$

where $p_d^{\mathcal{T}_k}$ is the transformed training data distribution and $p_g^{\mathcal{T}_k}$ the transformed distribution captured by G_θ . By using a total of four transformations combining flipping and rotation, optimization to the original data distributions encouraged, which also brings the most performance boost. These choices allow the proposed model to benefit from not only $\text{JS}(p_d \| p_g)$ but also $\text{JS}(p_d^{\mathcal{T}_k} \| p_g^{\mathcal{T}_k})$, thereby improving the learning of G_θ and generalization of D_ϕ . Furthermore, using multiple discriminators with weight-sharing provides learning regularization of D_ϕ .

Replacing GRAF’s logistic objective with a hinge loss, then the overall loss is defined as bellow:

$$\mathcal{L}(\theta, \{\phi_k\}) = \mathcal{L}(\theta, \phi_0) + \frac{\lambda}{n-1} \sum_{k=1}^n \mathcal{L}(\theta, \phi_k) \quad (3.8)$$

$$\begin{aligned}\mathcal{L}(\theta, \phi_k) = & \mathbb{E}_{\mathbf{z}_s \sim p_s, \mathbf{z}_a \sim p_a, \boldsymbol{\xi} \sim p_\xi, \boldsymbol{\nu} \sim p_\nu} [f(D_\phi(G_\theta(\mathbf{z}_s, \mathbf{z}_a, \boldsymbol{\xi}, \boldsymbol{\nu})))] \\ & + \mathbb{E}_{\mathbf{I} \sim p_D, \boldsymbol{\nu} \sim p_\nu} [f(-D_\phi(\mathbf{I}, \boldsymbol{\nu}))] + \mathcal{L}_r\end{aligned}\quad (3.9)$$

where $f(u) = \max(0, 1 + u)$. This loss is optimized with $n = 4$, where $k = 0$ corresponds to the identity transformation and $\lambda = 0.2$ (as in [184]).

Volumetric Rendering from a Single View X-ray

After training a model, the complete X-ray projections within a full vertical rotation is reconstructed for a medical instance given a single view X-ray. The proposed method follows the relaxed reconstruction formulation in [191], which fits the generator to a single image. Then, the parameters of the generator G_θ are *slightly* fine-tuned along with the shape and appearance latent vectors $\mathbf{z}_s$ and $\mathbf{z}_a$. Empirically, fine-tuning time can range from as little as 2 minutes to as long as 15 minutes, depending on how different the input X-ray views are from the learned distribution. The distortion and perception tradeoff is well known in GAN methods [192] and therefore the generation objective is modified by adding the distortion Mean Square Error (MSE) loss between the generated image and the target image, which incentivises a balance between blurriness and accuracy:

$$\mathcal{L}_{\text{gen}} = \lambda_1 \mathcal{L}_r + \lambda_2 \mathcal{L}_{\text{MSE}} + \lambda_3 \mathcal{L}_{\text{NLLL}} \quad (3.10)$$

where the NLLL term minimizes the negative log-likelihood loss of the latent variables $\mathbf{z}_s$ and $\mathbf{z}_a$, in addition to the tuned hyperparameters $lr = 0.0005$, $\beta_1 = 0$, $\beta_2 = 0.999$, $\lambda_1 = 0.3$, $\lambda_2 = 0.1$ and $\lambda_3 = 0.3$.

Once the model locates an optimal combination of $\mathbf{z}_s$ and $\mathbf{z}_a$, they are replicated and used to render the rest of the X-ray projections by continuously controlling the angle viewpoint.

3.4 Results

Here I provide an evaluation of MedNeRF on the described datasets. The proposed model’s results are compared to the ground truth, two baselines, perform an ablation

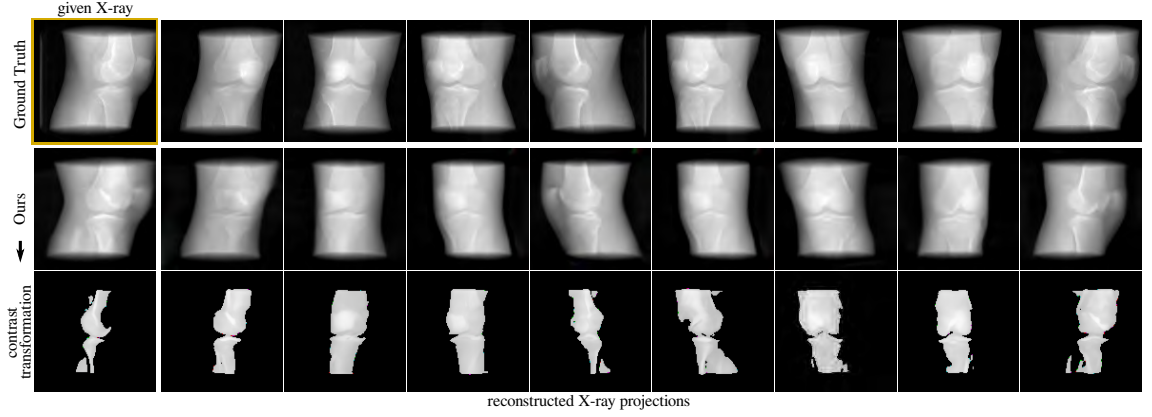


Figure 3.3: Knee renderings from continuous viewpoint rotations showing tissue and bone. Given a single-view X-ray from a CT, it’s possible to generate the complete set of CT-projections within a full vertical rotation by *slightly* fine-tuning a pretrained model along with the shape and appearance latent codes.

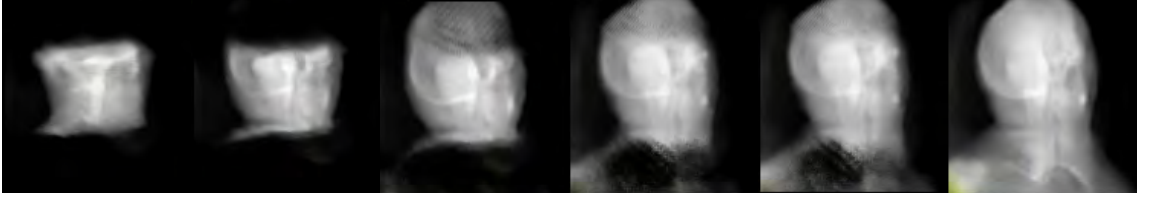


Figure 3.4: Head-neck reconstruction.

study, and show qualitative and quantitative evaluations. All models were trained for 100,000 iterations with a batch size of 8. Projection parameters (u, v) are chosen to evenly sample points on the surface of a sphere, specifically a slight horizontal elevation of 70-85 degrees and $u_{\min} = 0$, $u_{\max} = 1$ for a full 360-degree vertical rotation. However, only a fifth of the views (72-views each at five degrees) were provided during training and let the model render the rest.

For cross-validation, the chest and head & neck datasets, each containing 20 instances, used the first five numbered patients for testing, with the remaining instances allocated for training. For the knee dataset, due to its smaller size, only the first instance was reserved for testing while the rest were used for training. Reported results in the tables include sigma values to reflect variability from random model parameter initialization, rather than cross-validation across datasets.

Table 3.1: Quantitative results based on PSNR and SSIM of rendered X-ray projections with single-view X-ray input.

Dataset	$\uparrow$ PSNR (dB) ($\mu \pm \sigma$)	$\uparrow$ SSIM ($\mu \pm \sigma$)
Knee	30.17 $\pm$ 1.93	0.670 $\pm$ 0.040
Chest	28.54 $\pm$ 0.79	0.462 $\pm$ 0.082

3.4.1 Reconstruction from Single View X-ray

The model’s ability to synthesize 3D-aware DRR representations given a single-view X-ray as input is evaluated. Despite the implicit linear network’s limited capacity, the proposed model can disentangle 3D anatomy identity and attenuation response of different medical instances, which are retrieved through the described reconstruction reformulation in 3.3.3. The model can also facilitate distinguishing bone from tissue via a contrast transformation, as it renders a brighter pixel value for denser structures (e.g. bone) (Fig. 3.3). From Figure 3.4 it’s possible to observe that the model can also infer anatomical structure. Although parts from the skull and shoulders are missing, the proposed approach can reconstruct them with some interior ambiguities.

Table 3.1 summarises the results based on the peak signal-to-noise ratio (PSNR) and structural similarity (SSIM), which measure the quality of reconstructed:

$$\text{PSNR} = 10 \cdot \log_{10} \left(\frac{\text{MAX}_I^2}{\text{MSE}} \right), \quad (3.11)$$

where MAX_I is the maximum possible pixel value of the image. In practice, the Pytorch Ignite PSNR implementation was used with a data range equal to 2.0 due to a data normalization between -1.0 and 1.0.

Evaluations also make use of the structural similarity index metric (SSIM), which aims to measure human subjective similarity:

$$\text{SSIM}(g, r) = \frac{(2\mu_g\mu_r + C_1)(2\sigma_{gr} + C_2)}{(\mu_g^2 + \mu_r^2 + C_1)(\sigma_g^2 + \sigma_r^2 + C_2)} \quad (3.12)$$

where g and r denote the generated and real values, respectively. μ_g and μ_r are the mean intensities of g and r , in a similar fashion σ_g^2 and σ_r^2 are the variances, σ_{gr}) and, $C_1 = (k_1L)^2$ and $C_2 = (k_2L)^2$ are constants to avoid instability, with L

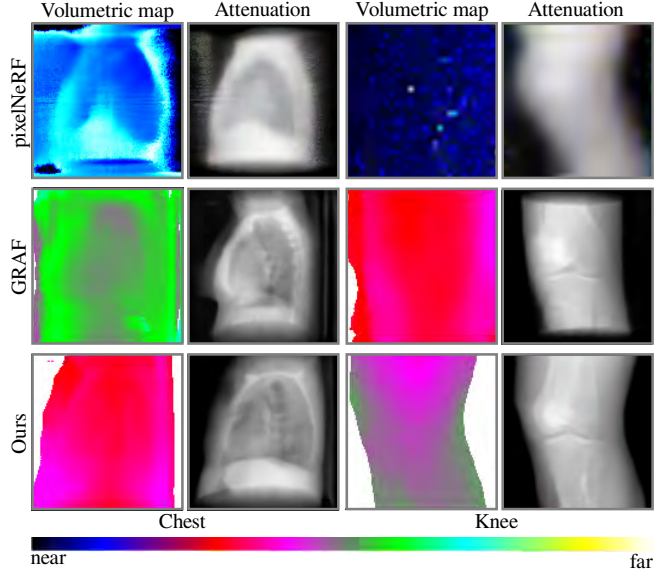


Figure 3.5: Volumetric maps and attenuation renderings on the used dataset.

as the dynamic range of pixel values (e.g., 255 for 8-bit images) and typical values $k_1 = 0.01$ and $k_2 = 0.03$.

The proposed generative loss can achieve a reasonable perception-distortion curve in renderings and show consistency with the location and volumetric depth of anatomical structures at continuous viewpoints compared to the ground truth.

3.4.2 2D DRR Rendering

The proposed model is now evaluated on the task of 2D rendering and compare it to pixelNeRF [189], and GRAF [163] baseline, wherein the original architecture is used. The proposed model can more accurately estimate volumetric depth compared to GRAF and pixelNeRF (Fig. 3.5). For each category, an unseen target instance with a similar view direction and shape is found. Volumetric depth estimation is given by bright colors (far) and dark colors (near). Lacking a perceptual loss, GRAF is not incentivized to produce high-frequency textures. In contrast, the proposed model renders a more detailed internal structure with varied attenuation. GRAF produces a consistent attenuation response, but seems to be unable to distinguish the anatomical shape from the background. The self-supervised discriminator enables the generator to disentangle shape and background by rendering a brighter color for the background and a darker color for the shape, while GRAF renders a bright or

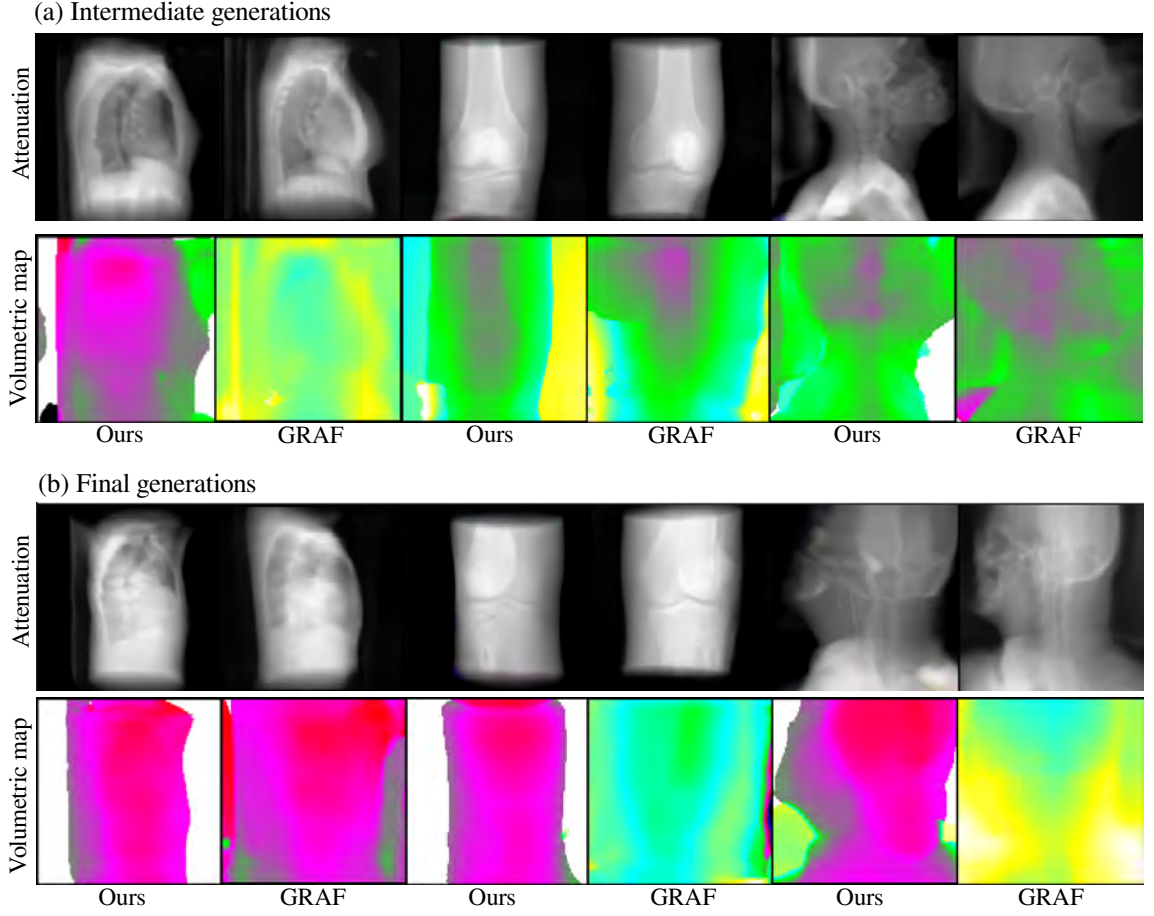


Figure 3.6: Intermediate and final depth maps and radiation renderings for GRAF Baseline and MedNeRF on the proposed datasets.

dark color for both.

To further evaluate MedNeRF to the GRAF baseline, visualizations of intermediate (at half of training) and final (at the end of training) rendered views and predicted depth maps (Figure 3.6) are presented. Findings suggest that the proposed model renders a more detailed internal structure with varied radiation tones in (a) for the head & neck, and chest categories. In contrast, the GRAF baseline struggles to render depth mainly due to the detector, which can be observed as a light gray color in the background of the radiation rendering.

The pixelNeRF model produces blurred attenuation renderings for all datasets, and volumetric maps tend to exhibit strong color shifts (Fig. 3.5). These artifacts are due to the see-through nature of the dataset, compared to solid-like natural objects on which NeRFs are trained. This data characteristic impairs not only volumetric maps but also fine anatomical structures. In contrast, the proposed model

	Chest dataset		Knee dataset		Head-neck dataset	
	↓ FID	↓ KID	↓ FID	↓ KID	↓ FID	↓ KID
GRAF	68.25	0.053	76.70	0.058	55.20	0.033
pixelNeRF	112.96	0.084	166.40	0.158	101.40	0.077
Ours	60.26	0.041	76.12	0.052	63.05	0.038

Table 3.2: FID and KID analysis comparing other methods.

	Chest dataset	
	↓ FID ($\mu \pm \sigma$)	↓ KID
MedNeRF - 3 SD, logistic loss, classic DA	84.85±1.02	0.069
MedNeRF - 2 SD, logistic loss, classic DA	67.73±0.71	0.051
MedNeRF - 2 SD, hinge loss, classic DA	65.34±0.35	0.045
MedNeRF - 2 SD, hinge loss, DAG	60.26±0.32	0.041

Table 3.3: FID and KID analysis of ablations of the proposed model.

is better able to render both volumetric depth and attenuation response. Moreover, pixelNeRF is sensitive to slight changes in projection parameters, hampering optimization for the knee category. In contrast, MedNeRF produces a consistent 3D geometry and does not rely on explicit projection matrices.

Table 3.2 compares image quality based on Frechet Inception Distance (FID) and Kernel Inception Distance (KID) metrics, in which lower values mean better. FID evaluates the quality and diversity of images generated by a generative model by comparing their feature distributions to those of real images. It’s based on the Wasserstein-2 distance between the multivariate Gaussian distributions of real and generated image features, often extracted from an intermediate layer of a pre-trained Inception network.

The FID score between real images with mean μ_r and covariance Σ_r and generated images with mean μ_g and covariance Σ_g is given by:

$$\text{FID} = \|\mu_r - \mu_g\|^2 + \text{Tr} \left(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{\frac{1}{2}} \right) \quad (3.13)$$

where μ and Σ are the mean and covariance of the feature representations of images, Tr denotes the trace of a matrix, which is the sum of its diagonal elements and, $(\Sigma_r \Sigma_g)^{\frac{1}{2}}$ represents the matrix square root of the product of the covariance matrices.

In the case of the KID metric, it also aims to evaluate the quality of images

generated by generative models, similar to FID but using a different statistical approach. KID compares the distributions of real and generated images based on the Maximum Mean Discrepancy (MMD), which uses kernel embeddings rather than assuming a Gaussian distribution, as FID does. This approach can provide a more robust evaluation, especially for smaller sample sizes, which is often the case in medical datasets. The KID score between real images with feature representations X_r and generated images with feature representations X_g , using a polynomial kernel $k(x, y) = (x^\top y / d + 1)^3$ where d is the feature dimension, is given by:

$$\text{KID}(X_r, X_g) = \frac{1}{m(m-1)} \sum_{i \neq j} k(x_i, x_j) + \frac{1}{n(n-1)} \sum_{i \neq j} k(y_i, y_j) - \frac{2}{mn} \sum_{i,j} k(x_i, y_j) \quad (3.14)$$

where $(X_r = \{x_1, x_2, \dots, x_m\})$ and $(X_g = \{y_1, y_2, \dots, y_n\})$ are the feature sets for real and generated images, respectively. m and n are the number of samples for real and generated images, respectively, and, $(k(x, y))$ is a polynomial kernel function that maps input vectors into a higher-dimensional space. Unlike FID, KID is an unbiased estimator and can provide more reliable results for datasets with fewer images.

Optimizing pixelNeRF on the proposed datasets leads to particularly poor results that are unable to compete with the GRAF baseline and the model presented. In contrast, the proposed model outperforms the baselines on FID and KID metrics for all datasets.

3.4.3 Ablation Study

To further evaluate MedNeRF functioning, three ablations (Table 3.3) are presented: wherein an additional simple decoder (SD) is included; the adversarial logistic loss is replaced by its hinge version; and wherein the non-classical DAG approach is adopted. Findings indicate that that the DAG approach brings the most performance boost compared to naively applying classical DA, while the use of a hinge loss performs slightly better than its logistic version. However, an additional decoder in the proposed self-supervised discriminator can lead to a significant drop in performance.

3.5 Availability & Acknowledgments

MedNeRF has been implemented using PyTorch on the GPU and is available at <https://github.com/abrilcf/mednerf> along with the preprocessed datasets and weights to the proposed models. This work was supported by Durham University and the Mexican Council of Humanities, Sciences and Technologies (CONAHCyT).

3.6 Conclusion

This chapter presented a novel Deep Learning architecture based on Neural Radiance Fields for learning a continuous representation of CT scans. The proposed method learns a medical category encoding of the attenuation response of a set of 2D DRRs in the weights of a generator. Furthermore, results indicate a stronger and more comprehensive signal from the proposed discriminator allows generative radiance fields to model 3D-aware CT-projections. Experimental evaluation demonstrates significant qualitative and quantitative reconstructions and improvements over other Neural Radiance Field approaches. Whilst the proposed model may not replace CT entirely, the functionality of generating 3D-aware CT-projections from X-rays has great potential for clinical use in osseous trauma, skeletal evaluation in dysplasia and for orthopaedic pre-surgical planning. This could cut down on the radiation dose given to patients, with significant economic implications such as bringing down the cost of investigations.

Repeat & Concatenate: 2D to 3D Image Translation with 3D to 3D Generative Modelling

4.1 Introduction

Existing methods approach 2D to 3D medical image translation through asymmetrical architectures and/or incorporate various regularization techniques to enforce plausible reconstructions [14, 15] (Fig. 4.1). While these techniques may generate high-quality outputs, even with high-frequency details, they often lack correlation to the input data. This means that the model becomes overly reliant on the latent prior, potentially ignoring the original 2D signal. The situation is further exacerbated if the training data is limited (few hundred images), so when the model is presented with inputs significantly different from the training dataset, such as out-of-distribution samples, it is likely to perform poorly making it unusable for practical scenarios.

The proposed method consists of a simple 2D to 3D mapping translation framework that addresses the correlation issue, ensuring highly-associated outputs even in limited datasets. This is achieved by a preprocessing step that preserves 2D information content throughout the network transformations without relying on other

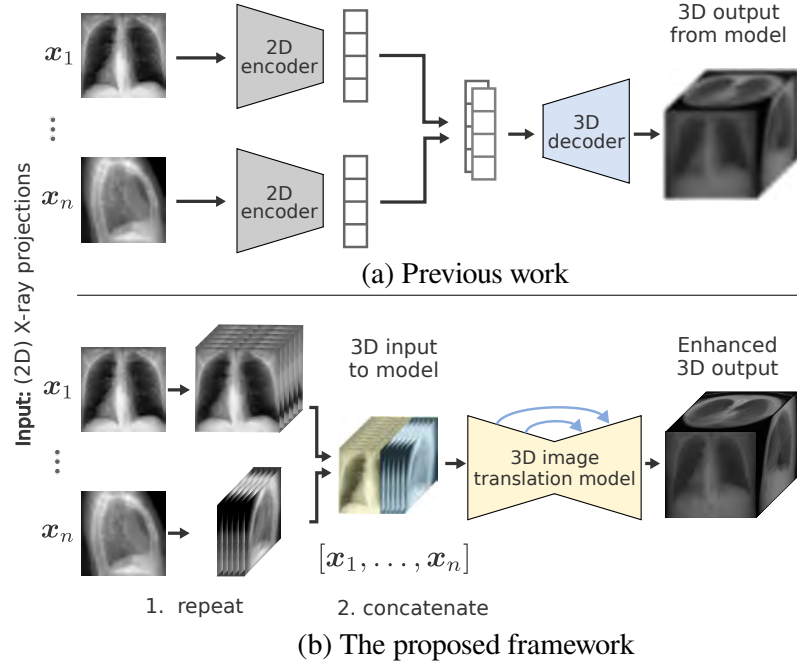


Figure 4.1: (a) Previous approaches focus on 2D to 3D mapping, often employing asymmetric architectures and compressed latent encoding. (b) In contrast, here a 3D to 3D mapping from repeated and concatenated inputs is proposed, enabling faster training with highly correlated outputs without latent compression, even with small datasets (a few hundred images).

priors. First, the various 2D input projections are repeated to match the output 3D target depth (Fig. 4.1). These 3D volumes are then concatenated into a single higher-channel 3D volume. Then a 3D to 3D conditional generative modeling approach that applies neural optimal transport with the de-biased Sinkhorn divergence is proposed. Such an approach effectively allows mass splitting; this particularly suitable for the task, as the original 2D inputs contain valuable information content, requiring each area of the 2D inputs to contribute to entire 3D regions of details in the synthesized 3D image (Fig. 4.2).

The approach was found to give significant improvement in terms of generalization, while being stable to train with only a few hundred images (ideal for medical/real-world datasets). Evaluations showed that the method generates plausible reconstructions after only 2,000 training optimization steps, which can generalize to out-of-distribution inputs after approximately 28 hours of training.

4.2 Related Work

2D to 3D image translation

A line of work involves asymmetric autoencoder architectures, where a 2D encoder extracts spatial features, which are then expanded to fit into a 3D decoder [193–197]. To mitigate information loss at the bottleneck, additional objectives are learned, enforcing a more interpretable latent space or adding an additional 2D encoder [14]. Recent approaches explore implicit neural representations [198–200] or two-stage vector-quantized approaches combined with diffusion [201]. In contrast, this chapter explores a simpler approach that generalizes from small datasets ($\sim 1k$ images) in a 2D to 3D setting without relying on more complex asymmetric architectures.

Regularization and prior knowledge

Generalizing beyond the examples in the training set is a fundamental aspect of machine learning models for medical applications. Key problems are mode collapse and overfitting, which arise due to the limited availability and real-world complexity of medical datasets. This is further exacerbated by data noise and heterogeneity among different imaging modalities. Fortunately, regularization techniques such as parameter constraints or augmentation can mitigate overfitting [144, 202] and help alleviate data shortages.

Data augmentation, in particular, is an established solution to mitigate overfitting [203, 204] by introducing semantic-preserving data transformations such as rotation, translation, and contrast adjustments. However, selecting and combining augmentations is a delicate process, especially to prevent undesirable outcomes such as the model learning to generate the augmented distribution and deviating from its original objective [69]. In this context, incorporating prior knowledge from initial assumptions about the data distribution can guide learning for improved generalization, such as additional conditional information in generative modeling or via additional terms to the main objective functions [205–207]. In the proposed approach, it’s assumed that 2D X-rays capture the underlying 3D objects over a range of depths, acting as a prior that motivates the proposed repeat and concatenate

mapping strategy. Results indicate this facilitates improved generalization for both in- and out-of-distribution inputs, even when trained on small datasets.

Deep generative modeling

2D to 3D image translation involves modeling the probability of unobserved 3D regions which is therefore a generative modeling problem. Deep generative modeling (DGM) is a very large field, from which mainstream approaches include GANs, VAEs, normalizing flows, autoregressive models and probabilistic diffusion models [40]. These methods classically balance an empirically observed trilemma of modeling quality, mode coverage and the number of ‘steps’ taken during sampling [208]. VAEs and normalizing flows are well-known for lower quality sampling, while GANs suffer from mode collapse, capturing only part of the distribution during training. Various strategies, such as weight clipping, gradient penalties, and spectral normalization, have been proposed to mitigate these issues, but they often fail to address fundamental challenges rooted in optimal transport (OT) theory.

Neural optimal transport

In contrast, neural optimal transport offers a more nuanced approach for mapping probability distributions, which is especially relevant for handling the intricacies of medical imaging data. Traditional methods like f -divergences, commonly employed in GANs, perform poorly with smaller, high-dimensional datasets, failing to provide meaningful gradients for effective training [209]. Capturing variability in limited medical data is crucial, and leveraging regularity in human datasets is one approach. For example, organs and bones consistently occupy locations in the human chest, allowing neural networks to learn the mapping between X-ray and 3D CT volumes. OT is relevant in this setting by providing a rigorous distance metric that respects the geometric properties of the distributions. It computes the minimal cost required to transform one distribution into another, offering a coherent and robust measure of similarity that is both intuitive and stable [210]. This geometrically-aware approach enables a more direct and meaningful comparison of medical images across different

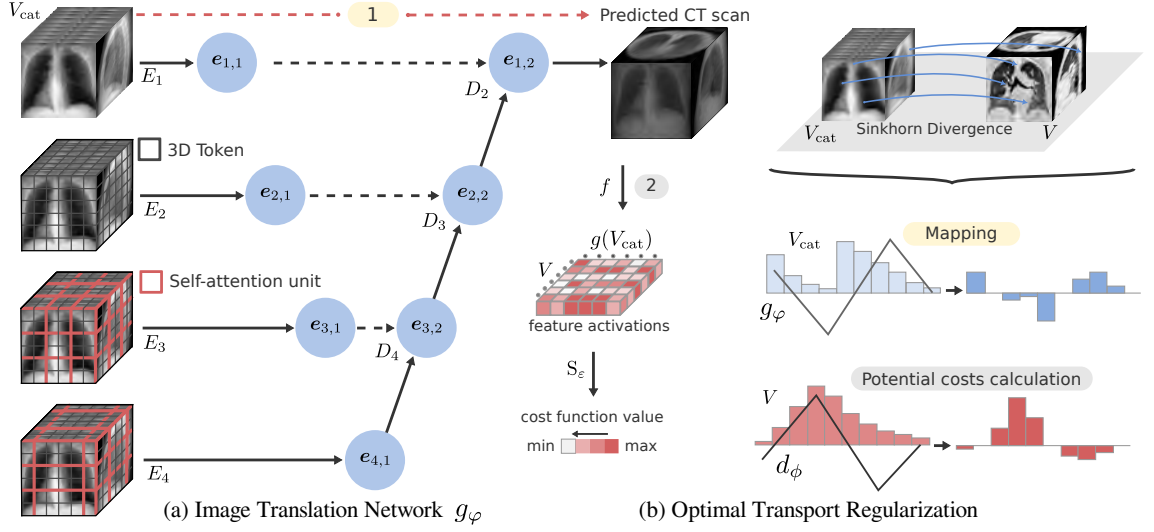


Figure 4.2: Proposed 2D to 3D image translation approach. (a) The mapping between 2D inputs and their corresponding 3D representation is learned by adapting a Swin UNe-Transformer g_ϕ [11] for image translation. (b) Optimization is based on the dual optimal transport regularization between networks g_ϕ and d_ϕ , comparing data points from the two image spaces using the Sinkhorn divergence S_ϵ on the activations of a feature extractor f .

modalities, potentially facilitating the development of more accurate and reliable diagnostic tools.

4.3 Method

In this subsection, the proposed 2D to 3D processing approach is detailed (Fig. 4.1), and then the introduction to the 3D to 3D generative modeling method regularized with optimal transport (Fig. 4.2).

4.3.1 Processing pipeline

At a high-level, this chapter to investigates whether multi-view 2D information can be combined for accurate 3D synthesis without latent encoding. Previous approaches integrate multi-view information in the latent space, as it typically captures a degree of spatial and rotational invariance. However the latents, even with information-rich modeling, fail to retain the full signal from the input 2D views. While, through modern deep generative modeling, they may synthesize high-quality signals with

high-frequency details, these are typically prone to over-hallucination which can potentially pose fatal in real-world medical application. In this application, we prioritize outputs with high correlation to the input 2D views, favoring a degree of blurriness or uncertainty over artificial high-frequency details.

To achieve this objective, this work proposes increasing the dimensionality at the 2D inputs to ensure the signal can influence an entire region of the 3D volume, and concatenating these inputs ensuring that information content loss is reduced within 3D to 3D residual architectures.

More formally, the proposed processing pipeline for N input X-ray projections. Let $I \in \mathbb{R}^{C \times H \times W}$ be an input 2D X-ray, representing a single view with height H , width W and channels C . For a set of N views, we have $\{I^i\}_{i=1}^N$, each aiming to contribute to the reconstruction of a 3D CT volume $V \in \mathbb{R}^{C \times H \times W \times D}$, where D is the depth.

The 2D inputs are ‘stretched’ to match the dimensions of the target 3D volume; then each view is repeated D times by $\Gamma : \mathbb{R}^{C \times H \times W} \rightarrow \mathbb{R}^{C \times H \times W \times D}$ and coarsely align them by transposing views that differ by 90-degrees (leaving the others unchanged, see the experiments section on *Views alignment* for further discussion on this). This is applied for each of the N views, which are concatenated across the channels (denoted by $\oplus$) yielding a composite volume $V_{\text{cat}} \in \mathbb{R}^{NC \times H \times W \times D}$, where

$$V_{\text{cat}} = \bigoplus_{i=1}^N \Gamma(I^i), \quad (4.1)$$

and the reconstructed 3D volume is modeled by a residual U-Net architecture $g_{\varphi}^{\text{unet}} : \mathbb{R}^{NC \times H \times W \times D} \rightarrow \mathbb{R}^{C \times H \times W \times D}$.

4.3.2 Mapping network and generative modeling

For the proposed 3D to 3D mapping network g_{φ} with parameters φ first a residual U-Net [55] equipped with self-attention was considered, following the Swin UNet-Transformer (Swin UNETR) model [11], applied for image translation instead of image segmentation. Based on experimenting with a variety of off-the-shelf U-Net backbones, the experiments settled on Swin UNETR as it empirically generated

higher image quality outputs across all metrics.

Modeling with neural optimal transport

While f -divergences are useful for comparing probability distributions from large, complex datasets in GANs, they can saturate with small, high-dimensional datasets, offering no useful gradients for training [209]. Capturing variability in limited medical data is crucial, and leveraging regularity in human datasets is one approach. For instance, organs and bones consistently occupy locations in the human chest, allowing neural networks to learn the mapping between X-ray and 3D CT volumes. Optimal transport (OT) is relevant here as it induces a well-characterized distance measure between distributions, capturing their geometric structure by minimizing the cost to move mass from one set of points from one distribution to another from a different domain. Unlike f -divergences, which directly quantify density ratio differences, OT can offer a less stringent notion of closeness [210].

The regularized OT distance between points from $\mathcal{X}$ and $\mathcal{Y}$ is defined as:

$$\text{OT}_\varepsilon(\alpha, \beta) = \inf_{\gamma \in \Gamma(\alpha, \beta)} \int_{\mathcal{X} \times \mathcal{Y}} C(\mathbf{x}, \mathbf{y}) d\gamma(\mathbf{x}, \mathbf{y}), \quad (4.2)$$

where α and β can be samples or measures (i.e., in a feature space) from data distributions $\mathcal{X}$ and $\mathcal{Y}$, respectively; $\Gamma(\alpha, \beta)$ describes the set of all joint distributions γ between α and β ; and $C(\cdot)$ is the cost function. This modified version of the classical OT distance incorporates regularization to stabilize computations, especially in scenarios involving discrete distributions or high-dimensional spaces (as it's case for 3D data).

The so-called OT Kantorovich dual problem [211] is also defined for regularized OT distances, enabling efficient computation and optimization as it involves solving a convex optimization problem. Such formulation captures the cost of transporting mass by assigning values to points in $\mathcal{X}$ and $\mathcal{Y}$ via a dual function:

$$C(\mathbb{P}, \mathbb{Q}) = \sup_{\omega} \left(\int_{\mathcal{X}} \omega(\mathbf{x}) d\mathbb{P}(\mathbf{x}) + \int_{\mathcal{Y}} \omega^c(\mathbf{y}) d\mathbb{Q}(\mathbf{y}) \right) \quad (4.3)$$

subject to the constraint $\omega(\mathbf{x}) + \omega^c(\mathbf{y}) \geq c(\mathbf{x}, \mathbf{y})$ for all $(\mathbf{x}, \mathbf{y}) \in \mathcal{X} \times \mathcal{Y}$, where ω is

the potential function, and its c -transform $\omega^c(\mathbf{y}) = \min_{\mathbf{x} \in \mathcal{X}} \{c(\mathbf{x}, \mathbf{y}) - \omega(\mathbf{y})\}$. The lower bound constraint ensures that the potential function ω contributes positively to the transportation cost and prevents the potentials $\omega(\mathbf{x})$ and $\omega(\mathbf{y})$ from taking arbitrarily low values, and thus avoids degenerate solutions. Such formulation has computational advantages, as it typically leads to a convex optimization, and allows the incorporation of regularization techniques.

Sinkhorn Divergence

For the generative modeling task, the dual regularized optimal transport (OT) learning approach [211] was applied using the geometric de-biased Sinkhorn divergence $S_\varepsilon(\cdot)$ [212] as the cost function. The Sinkhorn divergence is suitable for this task as it is an efficient yet positive and definite approximation of OT that interpolates between Maximum Mean Discrepancies (MMD) and OT. Specifically, a feature extractor f was incorporated, which reduces the dimensionality of V_{cat} and V through a mapping into a reduced dimensional vector as suggested in [213]. f is parametrized as a neural network and calculate the cost function in the latent space,

$$S_\varepsilon(\alpha, \beta) := \text{OT}_\varepsilon(\alpha, \beta) - \frac{1}{2}\text{OT}_\varepsilon(\alpha, \alpha) - \frac{1}{2}\text{OT}_\varepsilon(\beta, \beta), \quad (4.4)$$

where α represents the distribution of model features and β the distribution of ground truth 3D CT features extracted from f . While calculating the cost in the data space might suffice in some cases, results suggest that incorporating the feature extractor f is beneficial with 3D data as it reduces its dimensionality and makes the mapping network less susceptible to mode collapse.

This modeling approach is similar to the min-max objective in GANs where the Swin UNETR mapping network g_φ (Fig. 4.2a) replaces the generator and a residual discriminator d_ϕ assigns transportation costs to the produced samples $g_\varphi(V_{\text{cat}})$. The networks are trained using:

$$\mathcal{L}_g = \mathbb{E}_{(V_{\text{cat}}, V) \sim p_{\text{data}}} [S_\varepsilon(g(V_{\text{cat}}), V) - \lambda d(g(V_{\text{cat}}))], \quad (4.5)$$

$$\mathcal{L}_d = \mathbb{E}_{(V_{\text{cat}}, V) \sim p_{\text{data}}} [d(g(V_{\text{cat}})) - d(V)]. \quad (4.6)$$

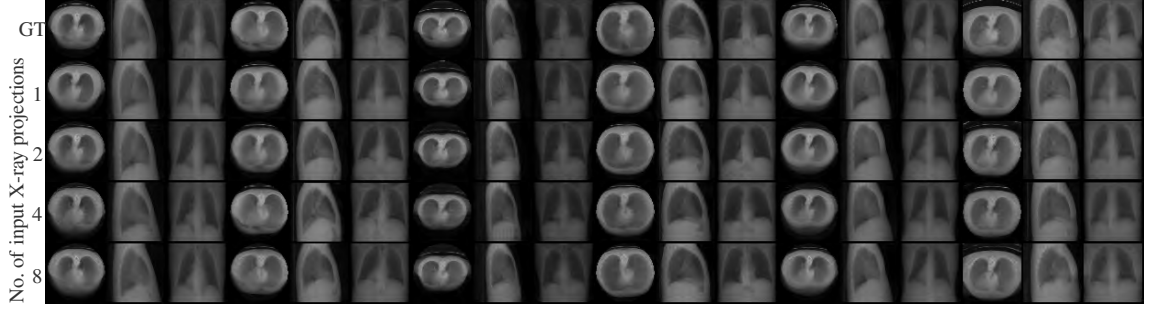


Figure 4.3: Example projections from generated 3D CT volumes from inputs using one, two, four and eight X-rays; obtained from the proposed 3D-3D translation approach with Swin UNETR backbone. Testing instances from LIDC-IDRI dataset were used.

LIDC-IDRI dataset (in-distribution inputs)				
Experiment	$\uparrow$ SSIM	$\uparrow$ PSNR	$\downarrow$ MSE	$\downarrow$ MAE
2D-3D AE	0.1545	3.804	995.2582	5.81
2D-3D AE + L2 Norm.	0.2086	10.833	197.2649	2.5103
3D-3D U-Net	0.4787	22.063	58.7546	0.9264

GT	3D-3D U-Net	2D-3D AE + L2 Norm	2D-3D AE	train	test	train	test
				$1 \oplus z \sim \mathcal{N}(0, \mathbf{I})$ (mode collapse) 3D-3D U-Net		$1 \oplus z_{\text{fixed}} \sim \mathcal{N}(0, \mathbf{I})$ (overfit) 3D-3D U-Net	

Figure 4.4: Effect of reformulating 2D-3D mapping into 3D-3D.

The optimization of the network d involves finding a function that yields the minimal cost associated with transporting mass from each point in V_{cat} to each point in V . Overall this allows us to align the distributions of features from ground truth CT samples with the produced samples from g_{θ} attained from the concatenated volumes from X-ray views V_{cat} , through regularized dual OT (Fig. 4.2b).

Even though the L2 distance can be used to approximate the optimal transport plan, as it satisfies the triangle inequality, it does not capture the underlying structure of the distributions or how to transport mass from one to another. In contrast, incorporating the Sinkhorn divergence induces a scaling process that ensures, at each step, the resulting transportation plan satisfies the constraints of being a probability distribution, where the sum of each row and column of the cost matrix equals the probability distribution, thus helps avoiding common training instability issues in GANs.

4.4 Experiments

The models were trained on the LIDC chest dataset [12], which consists of only 916 training CT scans, and tested them on both in-distribution and out-of-distribution inputs from six chest and lung datasets. We followed the experimental protocol in X2CT-GAN and CCXRay-net on the LIDC-IDRI dataset for fair comparison. Paired datasets were created by generating digitally reconstructed radiographs from DICOM images using the open source software Plastimatch. The proposed model’s convergence plateaued after 2,000 iterations, taking only an average of 28 hours to reach 5,000 steps (selected weight). The training runs were performed on NVIDIA TITAN RTX with a batch size of 8 using PyTorch.

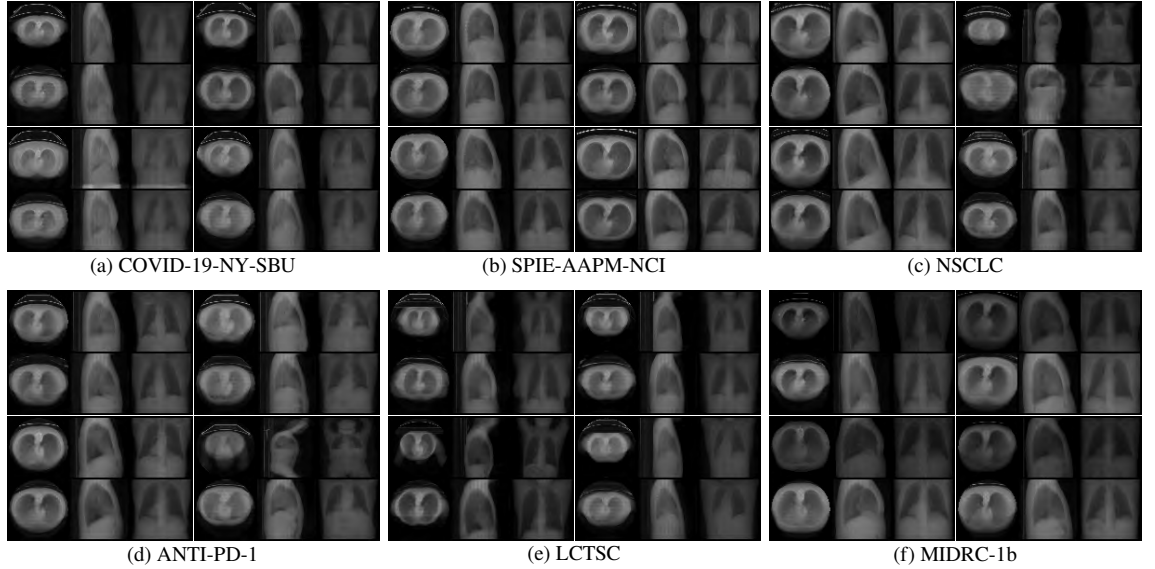


Figure 4.5: CT projections from generated 3D volumes on various out-of-distribution lung datasets. Model weights were selected from iteration 5,000 on the LIDC-IDRI dataset [12]. Ground truth projections are displayed in even rows, while the proposed model’s outputs are shown in odd rows.

Mapping reformulation In Figure 4.4, results from initial experiments are presented comparing an asymmetrical 2D to 3D strategy versus the proposed reformulation as a 3D to 3D mapping. The 2D to 3D approach involves aggregating the individual encoded 2D view features, which serve as input to a 3D decoder. However, this results in highly blurred outputs due to the information content loss from the latent encoding. Despite the simplicity of such an approach in terms

Table 4.1: Primary results of our mapping approach with varying input views. Each model was trained on LIDC dataset [12] and tested for both in- and out-of-distribution inputs from MIDRC-1b dataset [13]. We computed metrics five times with different random seeds and report their average.

In-distribution inputs (LIDC-IDRI dataset [12])				
Input views	↑ SSIM	↑ PSNR	↓ MSE	↓ MAE
$1 \oplus z \sim \mathcal{N}(0, \mathbf{I})$	0.23	20.28	0.040	0.13
1	0.48	23.21	0.021	0.075
2	<u>0.52</u>	24.31	0.018	<u>0.066</u>
4	0.512	<u>24.33</u>	<u>0.016</u>	0.066
8	0.54	24.63	0.015	0.061
Out-of-distribution inputs (MIDRC-1b dataset [13])				
Input views	↑ SSIM	↑ PSNR	↓ MSE	↓ MAE
$1 \oplus z \sim \mathcal{N}(0, \mathbf{I})$	0.13	15.32	0.08	0.23
1	<u>0.40</u>	20.08	0.05	0.15
2	0.40	<u>20.36</u>	<u>0.05</u>	<u>0.14</u>
4	0.37	21.21	0.04	0.13
8	0.20	17.79	0.07	0.20

of feature alignment, important fine-grained details are missing. In contrast, the proposed approach reformulates the problem into a 3D to 3D mapping in a simple architecture, achieving high-fidelity image translations even when using a single input view. Ablations of the proposed ‘repeat and concatenate’ preprocessing pipeline are presented by instead concatenating noise sampled from a normal distribution. When the vector $\mathbf{z}$ is fixed for each patient, the model causes overfitting, while gathering a new $\mathbf{z}$ for each iteration results in mode collapsing, where it produces the same output irrespective of the input.

Ablations and comparisons The proposed approach allows the use of multiple views as input without modifying the model’s architecture. It’s possible to observe that the model benefits from additional views when tested on in-distribution inputs (Fig. 4.3). However, this correlation was found not strictly hold for out-of-distribution inputs. This suggests that when there are disparities in view alignment or style-domain features compared to the training distribution, the model predominantly relies on views that closely resemble the learned patterns. Overall, findings suggest that the combination of two or four views to be empirically effective (Table 4.1).

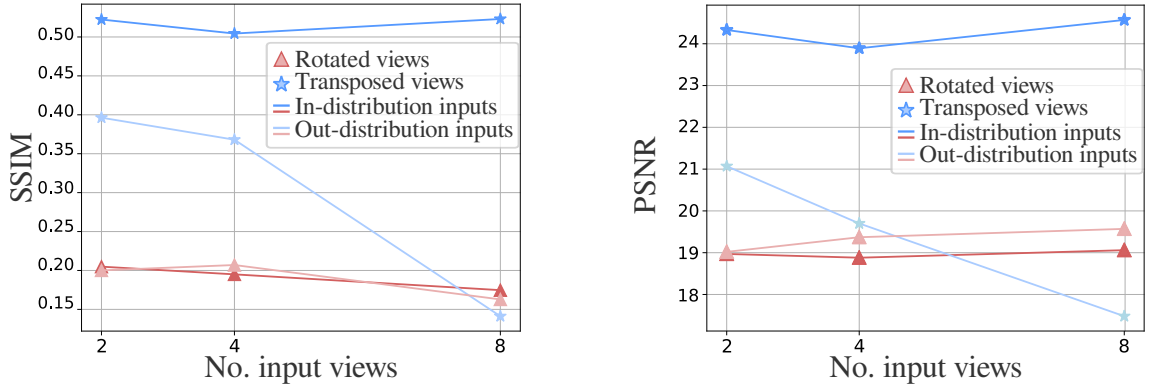


Figure 4.6: Impact comparison between only transposing views with a 90-degree difference (indicated by a star marker) and rotating the views to approximate their ground truth positions in 3D space (triangle marker), using two, four, and eight input views. Smoother colors represent results for out-of-distribution inputs (MIDRC dataset [13]).

In Table 4.2 a comparison of the proposed approach to paired alternative methods is presented in terms of quality of the 3D outputs for both in- and out-of-distribution inputs. Despite variations in datasets originating from different imaging systems, resolutions, and patients’ health conditions, the proposed model demonstrates superior performance across all metrics and maintains consistency across all out-of-distribution datasets. Refer to Figures 4.5 and 4.8 for qualitative results.

Gradients & patient-specific learning In Figure 4.7, we visualize the magnitudes of the gradients of the proposed model with respect to a varying number of input views. It’s possible to observe higher variability across different patients, indicated by the color variations for one, two, four, and eight views independently. On the other hand, if we analyze intra-patient gradients, we notice lower variability, suggesting consistent patterns captured in views coming from the same patient. With the proposed approach, the generative model implicitly learns patient-specific representations, where outputs reflect differences in anatomical structures and/or pathological conditions present in the input data from each patient.

View alignment The effect of aligning the input views during the preprocessing stage is studied prior to the repeat and concatenation operations. Initially, the views are repeated and concatenated without any alignment, which resulted in out-

Table 4.2: Quantitative results on both in-distribution and several out-of-distribution datasets using only two input X-rays. We used our model weights-5k, trained with Swin-U-Net. Other models weights are from 100 epochs ($\sim 90k$ iterations).

(a) In-of-distribution inputs					
Dataset	Method	$\uparrow$ SSIM	$\uparrow$ PSNR	$\downarrow$ MSE	$\downarrow$ MAE
LIDC-IDRI [12]	X2CT-GAN [14]	0.321	19.68	0.045	0.151
	CCX-rayNet [15]	<u>0.386</u>	<u>22.66</u>	<u>0.032</u>	<u>0.108</u>
	Ours	0.527	24.35	0.018	0.066
(b) Out-of-distribution inputs					
Dataset	Method	$\uparrow$ SSIM	$\uparrow$ PSNR	$\downarrow$ MSE	$\downarrow$ MAE
COVID-19-NY-SBU [214]	X2CT-GAN [14]	<u>0.236</u>	16.74	0.089	0.199
	CCX-rayNet [15]	0.205	<u>19.03</u>	<u>0.054</u>	<u>0.144</u>
	Ours	0.400	22.78	0.022	0.077
SPIE-AAPM-NCI [215]	X2CT-GAN [14]	<u>0.174</u>	15.63	0.115	0.245
	CCX-rayNet [15]	0.130	<u>17.87</u>	<u>0.080</u>	<u>0.196</u>
	Ours	0.399	22.04	0.026	0.087
MIDRC-1b [13]	X2CT-GAN [14]	<u>0.220</u>	14.70	0.156	0.360
	CCX-rayNet [15]	0.114	<u>18.06</u>	<u>0.076</u>	<u>0.228</u>
	Ours	0.404	20.36	0.050	0.147
ANTI-PD-1 [216]	X2CT-GAN [14]	<u>0.286</u>	18.04	0.072	<u>0.164</u>
	CCX-rayNet [15]	0.205	<u>18.15</u>	<u>0.067</u>	<u>0.167</u>
	Ours	0.349	20.93	0.040	0.112
LCTSC [217]	X2CT-GAN [14]	<u>0.326</u>	19.20	<u>0.052</u>	<u>0.125</u>
	CCX-rayNet [15]	0.206	<u>19.75</u>	0.062	0.156
	Ours	0.331	20.38	0.040	0.109
NSCLC [218]	X2CT-GAN [14]	<u>0.280</u>	<u>18.13</u>	<u>0.072</u>	<u>0.163</u>
	CCX-rayNet [15]	0.122	16.62	0.093	0.216
	Ours	0.315	19.85	0.047	0.126

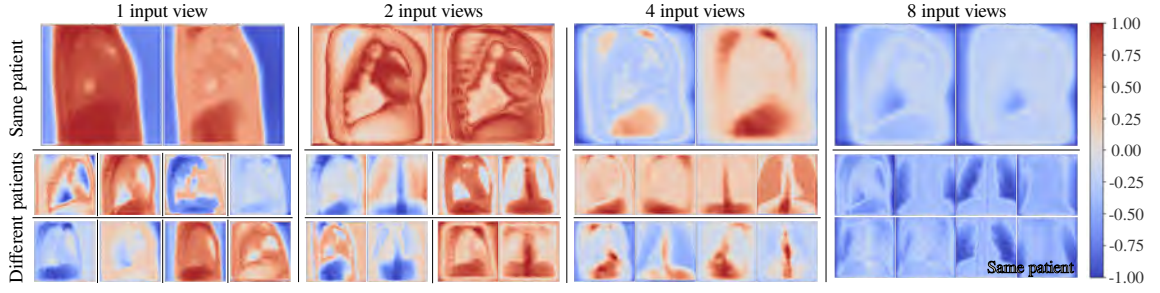


Figure 4.7: Gradient magnitudes with respect to the input views (X-rays) of the proposed image translation model. The *upper row* displays the variability within slices extracted from the 3D input volume of the same patient, while the *bottom row* shows the attribution among slices from different patients.

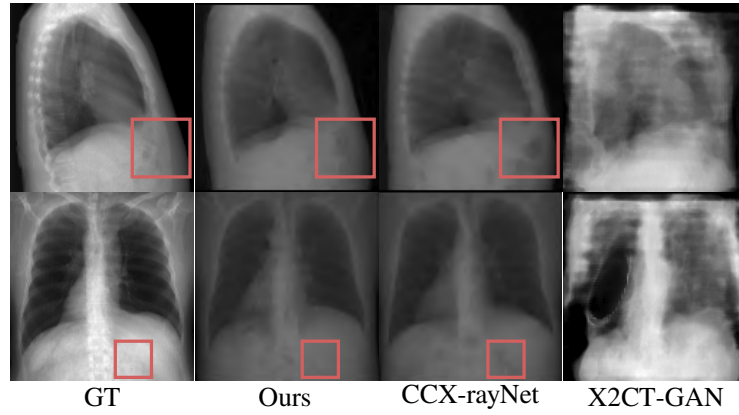


Figure 4.8: Examples of the correlated 3D projections from the proposed approach compared to alternative supervised methods X2CT-GAN [14] and CCX-rayNet [15].

puts containing checkerboard-like artifacts that diminished after an additional 1k training iterations. However, a simple transposition of views differing by 90 degrees along the vertical axis in 3D space alleviated this. Their approximating locations are also investigated through 3D rotation, however this ultimately requires a precise and potentially non-affine transformation which is non-trivial. Overall, findings indicate that the model learns some invariance to this, where a simple transpose consistently yields improved results without apparent artifacts (Figure 4.6). In the future, an iterative 3D alignment approach would be worthwhile investigating a secondary objective, however precise affine alignment might be required for significant improvements.

4.4.1 Implementation Details

The training algorithm is based on the neural optimal transport approach by Korotin *et al.* [219]. In this approach, the mapping network g_φ undergoes $k=10$ iterations while the parameters of the potential network d_ϕ remain frozen. Then, a single training step is performed for d , unfreezing its parameters while freezing those of g_φ . The proposed feature extractor f is parameterized by a 3D convolutional neural network with Kaiming weight initialization [220]. $\lambda = 0.1$ was set to control the degree of regularization of the reconstructed outputs by d . The networks are trained using the AdamW optimizer with parameters $\beta_1 = 0.5$, $\beta_2 = 0.999$, $\epsilon = 0.001$, and a learning rate of 10^{-5} with weight decay. A combination of differential augmentations [68] was applied, including random contrast, rotation, and horizontal flip, to the input 2D X-rays in all experiments. 3D reconstructions are saved in both numpy and .mha formats for visualization using medical image software such as 3D Slicer.

4.5 Limitations and future directions

The approach, while simple and intuitive, has several clear limitations. Firstly, the proposed current architecture attempts the non-linear transformation in a ‘single’ transformative step, resulting in uncertainty and therefore blurriness. It’s possible to expect improved performance through an iterative multi-step transformation, such as a modern probabilistic diffusion generative model. Secondly, although some invariance to input alignment was observed, a further investigation could determine whether better alignment of the concatenated 3D inputs to the target 3D locations might enhance the modeling of high-frequency details in specific 3D slices. For example, optimizing the input affine transformations, particularly their roto-translations, based on a secondary objective at inference, may improve this aspect of the model.

4.6 Conclusion

In conclusion, the work found that simply repeating the 2D inputs into concatenated 3D volumes, then treating the 2D to 3D translation problem as a 3D to 3D

translation task, leads to improved correlation between the synthesized outputs over more sophisticated multi-view latent integration approaches. While it's possible to expect other off-the-shelf 3D to 3D conditional generative modeling approaches to be immediately applicable within the proposed framework, results found particular success by directly applying 3D to 3D neural optimal transport to attain highly correlated 3D synthesized outputs with their corresponding 2D inputs. The overall proposed approach is fast to train, data efficient, stable, and generalizes well to new out-of-distribution views making it applicable in a real-world clinical setting. However it does exhibit blurriness where there is uncertainty in the outputs; in the future it would be worth investigating iterative alignment optimization for the repeated 3D inputs as a secondary objective to the generative modeling, repeated during inference, potentially mitigating this uncertainty.

Unaligned 2D to 3D Translation with Conditional Vector-Quantized Code Diffusion using Transformers

5.1 Introduction

3D imaging is essential in several fields, from clinical applications with Magnetic Resonance Imaging (MRI) and Computed Tomography (CT) modalities [17, 221], to virtual/augmented reality [222, 223], self-driving vehicles [224, 225], and security [226, 227]. However, the diverse range of available imaging devices exhibit differences in cost, quality, and accessibility, which has led to an increased interest in leveraging 2D imaging for 3D acquisition. For instance, in hospitals, CT from biplanar X-rays could minimize patient exposure to a substantial dose of ionizing radiation [20]. Similarly, at airports, volumetric reconstruction from security baggage screening could be more effective at detecting prohibited items [228, 229]. The capability to seamlessly translate between these imaging domains, eliminating their limitations, is a high-impact application; however, this remains a difficult conditional generative modeling problem.

Generating accurate 3D representations from 2D images poses a significant challenge in the field of computer vision. Approaches must account for complex shape

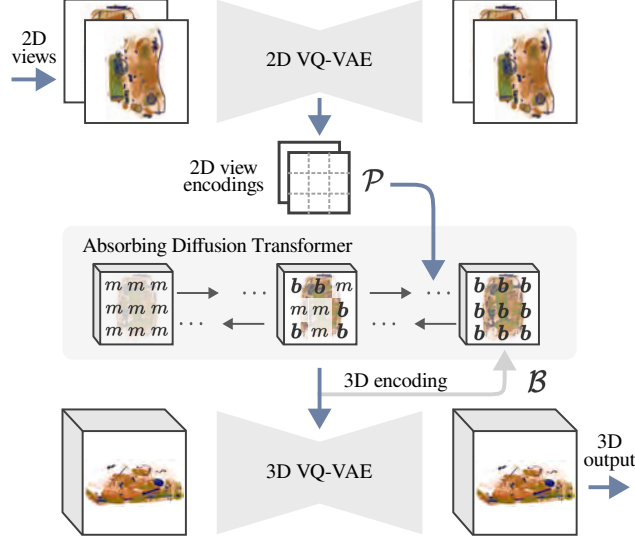


Figure 5.1: The proposed approach independently learns a discrete information-rich codebook for 2D and 3D domains with two VQ-VAEs. New 3D samples are then synthesized conditional on the complete codebook of multiple 2D views with a transformer.

topologies, fine-grained textures, domain differences, and incomplete input information. Current methods aim to solve this under-constrained problem by relying on cycle consistency losses [230] for disentangling 3D shape properties like appearance and viewpoint from a single input image [231], improving fine-grained 3D mesh attributes [232], for 3D-aware face image generation by distilling StyleGAN2 latent space [233] and recovering 3D shapes from videos [234]. Despite these models being unpaired, they require separate convolutional neural network (CNN) designs for predicting each 3D property, and they cannot generate accurate 3D representations without additional 2D supervisory signals.

This chapter presents a unified approach that tackles 3D synthesis by reformulating it as a voxel-to-voxel prediction problem. This was achieved this by conditioning an unconstrained transformer on 2D input views. While some recent methods have used transformers for multi-view 3D object reconstruction, they suffer from computational inefficiency and are sensitive to domain shifts and input view irregularities [235, 236].

This chapter investigates 2D to 3D image translation of complex data that exhibit varying outer and internal topologies with different density properties and domains. To achieve this, a novel two-stage translation approach proposed, which

models the conditional probability of generating a 3D image given 2D input views with a discrete diffusion model. By applying this process in a highly compressed discrete latent space, the proposed approach can extract high level features of complex objects and scale to high-resolution data without requiring paired datasets (Fig. 5.1), which is a desired feature for real-world applications. First, the proposed approach learns information-rich discrete spaces of 2D and 3D distributions independently with two Vector-Quantized Variational Autoencoders (VQ-VAE), removing the need for alignment of both geometries. Second, a diffusion model was used parameterized by an unconstrained transformer that allows bidirectional 2D global context when generating the 3D representation, improving feature learning and speeding up the sampling process.

5.2 Prior Work

Autoregressive Modeling

The following work departs from autoregressive models, as their receptive field is limited to previously generated tokens, resulting in biased representations and restricts then to relatively low dimensional data [237, 238]. Here we emphasize that images do not conform to such sequential manner and proposed resorting to absorbing state diffusion [139] which allows fast and high-resolution image generation.

Vector-Quantized Representations

Following the success of the VQ technique in generative modeling, the proposed method makes use of VQ representations, where a convolutional autoencoder extracts high-level features to an information-rich latent space. VQ image models [239], which compress images to a low dimensional discrete latent space and subsequently model this space with a powerful generative model, have recently been used for a variety of tasks. Chen et al., [147] address the domain gap issue in cross-domain analysis by introducing VQ into the image-to-image translation framework; how-

ever, their approach deals with the two data modalities having the same dimension and relies on spatial correlations.

Generative Adversarial Networks. In our domain’s context, evaluations compare two GAN-based models. The first is X2CT-GAN [17], which utilizes a 3D GAN with skip-connections, along with an additional CycleGAN for capturing style differences. The second model is CCX-rayNet [16], which employs a class-conditioned module to reconstruct a CT from 2D X-ray images. Unlike CNN-based models that require aligned inputs (i.e. consistent resolution, orientation, and transformations) and often rely on skip-connections, the proposed approach allows unaligned inputs via full-coverage attention on an information-rich discrete latent space.

Neural Radiance Fields. The ability of NeRF to generate novel views of complex 3D scenes from a partial set of 2D images [240], inspired MedNeRF [221] for rendering CT projections from a small set or single-view X-rays. While NeRF render 3D-informed images from 2D viewpoints, they differ fundamentally from the proposed method. In particular, they are focused on object and scene representation conditional on coordinate information, rather than novel synthesis and generalizability [241]. In contrast, the proposed approach needs no prior data on camera positions or aligned inputs. It enables direct sampling based on input views while providing control over the generative process.

Hybrid Generative Models

A way to address issues in specific generative models such as long training or poor scaling is by combining two or more approaches [40]. For instance, transformers and their self-attention mechanism use in an encoder-decoder setup [242] to improve both autoregressive models and other generative models due to their stable training and ability to learn long-distance dependencies. This is achieved using the attention scheme,

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right) \mathbf{V}. \quad (5.1)$$

where the values $\mathbf{V}$ represent the encoded inputs, the keys $\mathbf{K}$ are used for indexing and queries $\mathbf{Q}$ to select specific values. Additionally positional information is passed into the function via fixed or trainable positional embeddings [243].

Of particular interest to this work is the two-stage process proposed by Bond-Taylor *et al.* [244], where the discrete latent space is modeled by a diffusion model parameterized by an unconstrained transformer that learns to unmask the data. This process allows faster sampling because multiple tokens can be predicted in parallel. Inspired by this work, the proposed method bridges the domain gap between 2D to 3D translation by combining the infinite receptive fields of attention for representing both data distributions and allowing learning complex topologies with the powerful feature-extraction ability and scalability of VQ-VAE. One of the advantages of the proposed approach is that it does not require spatially aligned 2D and 3D samples, thus avoiding issues with both the geometric and domain misalignment between the two modalities.

5.3 Method

This chapter addresses the task of synthesizing complex 3D data portraying varying topologies and material properties (e.g. CT-like images) given two 2D views taken from different angles (e.g. X-rays). In particular, it aims to avoid dependencies on both the geometric and domain relationships between the 2D and 3D data which are significant sources of real-world data misalignment caused by object/device movement and characteristics. In this setting, learning a deterministic mapping between modalities is impractical due to a large number of possible input/output pairings in real-world data, meaning that outputs would be very blurry and unhelpful [245] to human operators for instance. As such, this work proposes modeling the mapping between 2D and 3D data with a conditional likelihood-based generative model, allowing sampled 3D data to sit at arbitrary positions/rotations relative to the 2D data.

Formally, given 3D data $\mathbf{I} \in \mathbb{R}^{H \times W \times D}$, and multiple corresponding 2D views $\mathbf{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$, $\mathbf{x}_i \in \mathbb{R}^{H \times W}$, the goal is to learn the conditional distribution

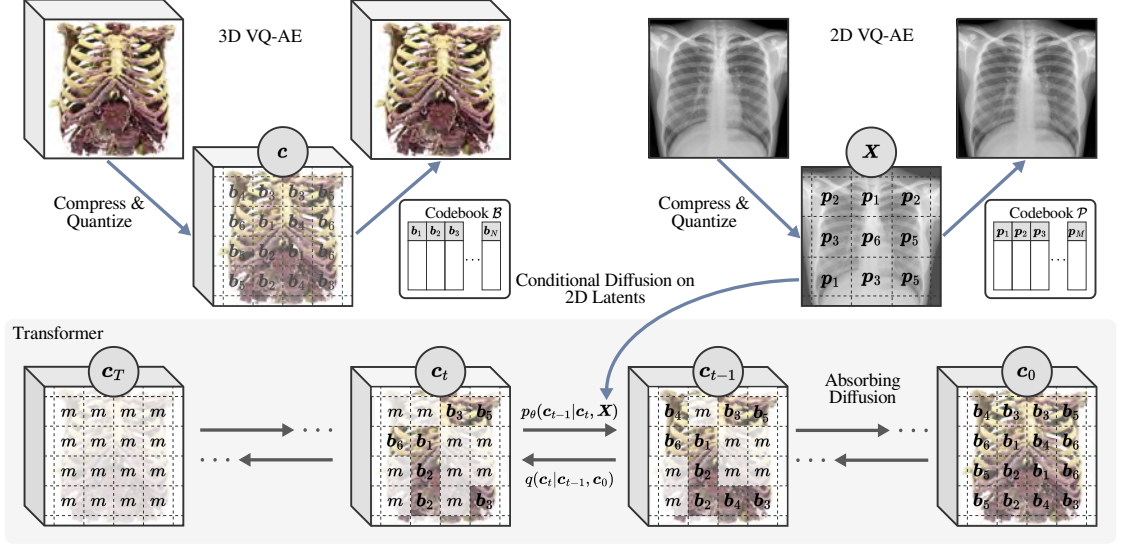


Figure 5.2: Our approach allows 2D to 3D translation from unaligned data by first compressing to an information rich Vector-Quantized discrete space, then modeling the conditional probability with a discrete diffusion model parameterized by an unconstrained transformer.

$p(\mathbf{I}|\mathbf{X})$. Training a transformer model directly on complex high-dimensional pixel data would be impractical, as the computational complexity would increase quadratically from the self-attention mechanism [242, 246]. To overcome this, the task is decomposed into two stages to take advantage of the power and flexibility of transformers. Stage 1 separately learns to compress the 2D and 3D data to small but information-rich discrete latent spaces that accurately represent the data. In Stage 2, a discrete diffusion model, parameterized by an unconstrained transformer that is spatially invariant over the 2D data, models the conditional probability distribution within this learned compressed space. The overall process is illustrated in Figure 5.2, and train it alternatively with a patch-wise discriminator that uses a combination of spatial and style augmentations for both 2D and 3D images. More details in the Supplementary 5.5.1.

5.3.1 Stage 1: Unpaired Compression

Separately for 2D and 3D data, the model learns to compress a single data point $\mathbf{x}$ (each 2D view is also compressed separately) to a relatively small set of integer values $\mathbf{z}$. This is performed using a VQ-VAE [239], which achieves extremely high

compression rates by utilising a codebook for information rich vectors, each of which is able to represent an image patch, while a neural decoder models how these codes interact.

The VQ-VAE approach is of particular interest to our work as it allow a 2-stage scheme where a compact and quantized latent space can be learned through a convolutional autoencoder with a large receptive field. In addition, variational autoencoders (VAE) [47] allow us to provision per-3D-image likelihood estimates, in contrast to other 2D-3D translation models based on GAN, which cannot directly provide a probability interpretation.

In particular, a convolutional encoder downsamples data $\mathbf{x}$ to a smaller spatial resolution, $E(\mathbf{x}) = \{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_L\} \in \mathbb{R}^{L \times D}$. Each continuous encoding $\mathbf{e}_i$ is subsequently quantized by mapping to the closest element in the codebook of vectors $\mathcal{B} \in \mathbb{R}^{K \times D}$, where K is the number of discrete codes in the codebook, and D is the dimension of each code,

$$\mathbf{z}_q = \{\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_L\}, \text{ where } \mathbf{q}_i = \arg \min_{\mathbf{b}_j \in \mathcal{B}} \|\mathbf{e}_i - \mathbf{b}_j\|, \quad (5.2)$$

with the straight-through gradient estimator [247] used to approximate the gradients through this non-differentiable quantization process. Here, the stop-gradient (sg) operation is applied to prevent gradient flow into the quantized representation $\mathbf{z}_q$, and thus treating it as a constant during backpropagation. Then the discrete latents are decompressed with a convolutional decoder $\hat{\mathbf{x}} = G(\mathbf{z}_q)$. The model is trained end-to-end by minimizing the loss,

$$\mathcal{L}_{\text{VQ}} = \mathcal{L}_{\text{rec}}(\mathbf{x}, \hat{\mathbf{x}}) + \|\text{sg}[E(\mathbf{x})] - \mathbf{z}_q\|_2^2 + \beta \|\text{sg}[\mathbf{z}_q] - E(\mathbf{x})\|_2^2. \quad (5.3)$$

which balances reconstruction quality $\mathcal{L}_{\text{rec}}$ against quantization terms that encourage the encoding $E(\mathbf{x})$ to match the closest codes in the codebook.

Our approach eliminates the need for spatially aligned 2D and 3D images, which in practice are difficult to obtain accurately. In addition, the flexibility of our model allows the use of an arbitrary number of 2D input views without the requirement of camera priors or changing the network architecture as this can be effectively

achieved by using smaller or larger latent code sizes. An ablation of these aspects can be found in section 5.5.

5.3.2 Stage 2: Conditional Discrete Diffusion

To translate data from 2D to 3D we model the conditional probability distribution of 3D data given a few 2D views using a discrete diffusion model in the learned vector-quantized space, $p(\mathbf{c}|\mathbf{X})$, where $\mathbf{c}$ represents the VQ codes of the 3D data and $\mathbf{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_n\}$ represents the set of VQ codes of the all the 2D views. Specifically, we use the discrete absorbing diffusion process [139, 244], which is more suitable for this task due to its bidirectional nature, allowing them to outperform autoregressive approaches while also being able to scale to higher dimensional spaces, faster sampling, and being substantially less prone to overfitting [138] which is crucial when training on small datasets (as is generally the case for real-world datasets). More detail on this aspect is presented under section 5.5 (Fig. 5.9).

In this case, the discrete 3D latents are gradually masked out using randomized orders over a number of time steps T such that at time step t , $\mathbf{c}^t$ is defined as the discrete 3D latent $\mathbf{c}$ with each token masked out with probability $\frac{t}{T}$. Formally, this masking procedure is defined by a Markov chain,

$$q(\mathbf{c}^{1:T}|\mathbf{c}^0) = \prod_{t=1}^T q(\mathbf{c}^t|\mathbf{c}^{t-1}). \quad (5.4)$$

The posterior is defined as $q(\mathbf{c}_t|\mathbf{c}_{t-1}) = \text{cat}(\mathbf{c}_t; \mathbf{p} = \mathbf{c}_{t-1}\mathbf{Q}_t)$, where “cat” denotes the concatenation operation. For one-hot $\mathbf{c}_{t-1}$ where $\mathbf{Q}_t = (1 - \beta_t)\mathbf{I} + \beta_t\mathbb{1}e_m^T$ is a matrix denoting the probabilities of moving to each successive state, e_m is a vector with a one on mask states m and zeros elsewhere, and $\beta_t = \frac{1}{T-t+1}$.

The reverse of this diffusion process is another Markov chain that gradually un.masks the latents

$$p_\theta(\mathbf{c}^{0:T}|\mathbf{Z}) = \prod_{t=1}^T p_\theta(\mathbf{c}^{t-1}|\mathbf{c}^t, \mathbf{X}). \quad (5.5)$$

This can be approximated by training an unconstrained transformer to predict the original latents from the masked ones, optimizing the Evidence Lower Bound

(ELBO),

$$\mathbb{E}_{q(\mathbf{c}^0, \mathbf{X})} \sum_{t=1}^T \gamma \mathbb{E}_{q(\mathbf{c}^t | \mathbf{c}^0)} \left[\sum_{[\mathbf{c}^t]_i=m} \log p_{\theta}([\mathbf{c}^0]_i | \mathbf{c}^t, \mathbf{X}) \right], \quad (5.6)$$

where $\gamma = \frac{T-t+1}{T}$ is a reweighting term used to improve convergence [244].

Discrete vs. Continuous Latents Discrete representations are important for the proposed approach because applying self-attention on a compact, information-rich space is more efficient than attending over a larger, continuous space due to limitations in sequence length [246] coupled with less effective integration of information [248]. Moreover, compressed representations have been shown to improve generalization without relying on complex hierarchical architectures [249].

Full-coverage Attention To leverage the compressed latent space from the diffusion model, we employ an unconstrained transformer to parameterize the denoising function. By flattening and concatenating 2D latents, we integrate discrete representations into a 1D manifold, aided by a trainable positional embedding. In contrast to CNN reliance on local alignment and/or deep architectures to increase their limited receptive fields, the proposed method provides global context through information-rich discrete representations, allowing all parts of the 2D inputs to influence voxel predictions [248].

Domain Invariance The proposed approach is robust to differences in the distributions of 2D and 3D data domains as we model distributions over discrete latent representations separately rather than low-level voxels in a joint manner. In contrast, a unified framework such as GAN cannot achieve this unless they explicitly define a domain-invariant joint latent distribution. Domain invariance is of particular interest when dealing with real-world scenarios where data exhibits complex variations, such as different camera perspectives, lighting conditions, or imaging modalities. Examples of this are presented in section 5.5 (Fig. 5.9).

Likelihood Estimation Due to the fact that the VQ-VAE decoder is trained with MAP-inference, within this framework we are able to estimate the conditional

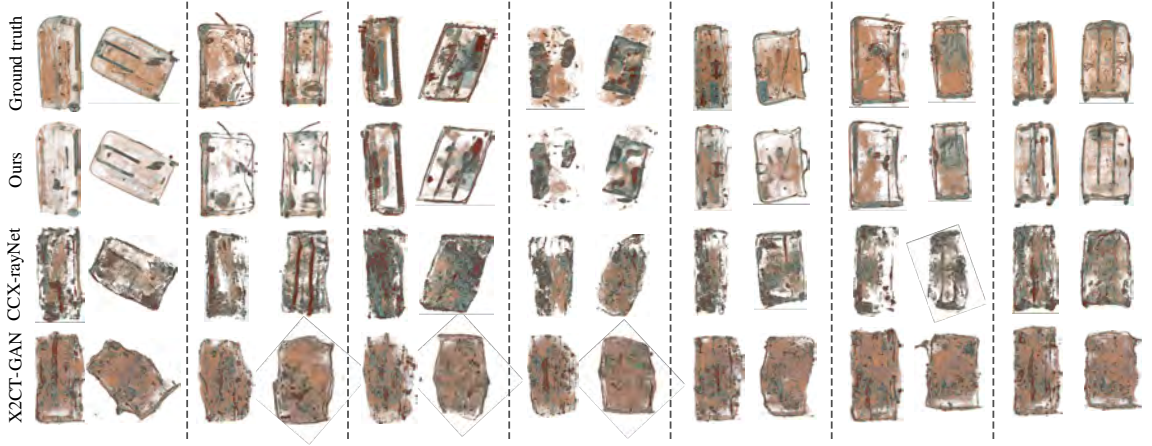


Figure 5.3: Comparison of 3D CT model samples trained on the baggage screening dataset, showing the ground truth, the proposed method, CCX-rayNet [16] and X2CT-GAN [17].

log likelihood $\log p(\mathbf{I}|\mathbf{X}) \approx \log p(\mathbf{I}|\mathbf{c})p(\mathbf{c}|\mathbf{X})$ [239], where $p(\mathbf{I}|\mathbf{c})$ is the 3D VQ-VAE decoder, and $p(\mathbf{c}|\mathbf{X})$ is the conditional discrete diffusion model. Subsequently, the proposed model provisions per-3D-image likelihood estimates, which provide a distance measurement from the true distribution of the data.

5.4 Experiments

This section evaluates the ability of the unconstrained transformer to perform 2D to 3D translation using discrete VQ representations against the SOTA models on this task: X2CT-GAN [17] and CCX-rayNet [16] on two datasets complex volumetric imagery from the publicly available chest CT scans, LIDC-IDRI [12], and security baggage screening dataset that is not publicly available (due to confidential constraints). We followed the experimental protocol in X2CT-GAN and CCXRy-net on the LIDC-IDRI dataset for fair comparison, and for the security baggage dataset 10% of instances were randomly reserved for testing purposes. Additionally, the chapter provides a 2D evaluation compared to the MedNeRF [221] model in terms of MIPs in Sec. 5.4.1. Evaluations are in terms of negative log-likelihood (NLL), Density and Coverage, and distortion metrics, including SSIM, PSNR, MSE and MAE. The best values are in bold, and the second-best values are underlined.

The Density and Coverage metrics are alternative evaluation methods that assess

Baggage Security Screening dataset							
Method	↓ NLL	↑ Density	↑ Coverage	↑ SSIM	↑ PSNR	↓ MSE	↓ MAE
X2CT-GAN	N/A	0.95	0.80	0.655	34.68	0.0014	0.0129
CCX-rayNet	N/A	1.28	0.89	0.886	35.45	0.0012	0.0069
Ours	0.007	2.01	0.97	0.899	39.45	0.0007	0.0049

Table 5.1: **3D fidelity and distortion metrics on the Baggage security dataset.** We compare our method and SOTA models in terms of fidelity and diversity (density and coverage). Additionally we evaluate the quality of generated voxel grids using distortion metrics (SSIM, PSNR, MSE and MAE).

the quality and diversity of generated images by comparing them directly to real images. These metrics are designed to improve upon limitations in FID and KID, particularly by providing a more interpretable breakdown of quality (density) and diversity (coverage):

Density computes the average number of real samples that are within a specified threshold distance (often defined by a k-nearest neighbor approach) of each generated sample in a feature space (often extracted using a pretrained network like Inception):

$$\text{Density} = \frac{1}{kM} \sum_{j=1}^M \sum_{i=1}^N 1_{X_j \in B(Y_i, \text{NND}_k(Y_i))} \quad (5.7)$$

where k is for the k-nearest neighborhoods and $\{B(Y_i, \text{NND}_k(Y_i))\}$ is the binary decision of whether the fake data X_j contained in any neighborhood sphere of real samples.

Coverage is calculated as the proportion of real samples that are within a certain threshold distance of any generated sample in feature space. This metric relies on the same feature embeddings used for density but emphasizes representing the full variety found in the real dataset:

$$\text{Coverage} = \frac{1}{N} \sum_{i=1}^N 1_{\exists j \text{ s.t. } X_j \in B(Y_i, \text{NND}_k(Y_i))}. \quad (5.8)$$

5.4.1 Conditional 3D Modeling on 2D Views

2D to 3D translation is performed by the proposed unconstrained transformer trained on the discrete latents encodings computed by the two VQ-VAE. At inference, the

3D latents are gradually predicted using the conditioning information from two 2D inputs in the form of 16×16 codes, which are concatenated at the start.

Fidelity Metrics. This work also aims to measure the quality as well as the overlap between the manifold of generated samples and the manifold of real data. One of the primary motivations for using likelihood-based generative models is that Negative Log-Likelihood (NLL) allows us to monitor over-fitting effectively. This is a potential issue when training models on real application datasets which are generally much smaller than natural image datasets. In addition, Density-Coverage [250] are also reported, which independently assesses the fidelity and variability of a model. In these metrics, images are first projected into an embedding space, and a scoring function estimates the manifold density in the neighborhood of each embedded data point. Generative models trained with natural images mostly rely on the features from an ImageNet pretrained model for evaluation. Since the target data is distinct from ImageNet samples, to compute these metrics, the features of a randomly initialized convolutional encoder architecture were used, as proposed in the work of [250]. Specifically, embeddings of dimension 2,197 of the real validation data points and samples from the proposed model and competing models were obtained. The embedding size is determined by flattening the feature maps of shape `[batch_size, channels, 13, 13, 13]`, chosen to yield at least 2,000 samples to account for Density’s sensitivity to the number of samples, as opposed to using a `[batch_size, channels, 12, 12, 12]` feature map shape, which further ensures a stable result.

Distortion Metrics. It’s worth noting that commonly used distortion metrics such as Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index (SSIM) correlate poorly with human visual perception, as they ignore global structure and only focus on signal fidelity [251]. A model that simply optimizes pixel-level distortions (e.g. small adjustments in brightness, saturation etc.) can obtain a high score on these metrics. However, values on SSIM, PSNR, Mean Squared Error (MSE) and Mean Absolute Error (MAE) were also reported for the research community as they are widely used for comparison [16, 17].

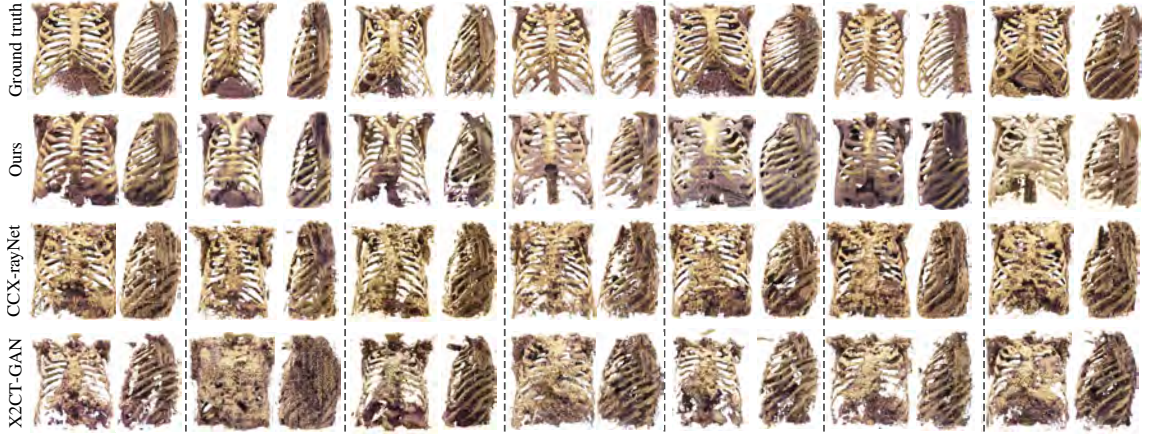


Figure 5.4: Comparison of 3D CT model samples trained on the medical chest (LIDC-IDRI dataset [12]) from different anatomical planes (coronal and sagittal), showing the ground truth, the proposed method, CCX-rayNet [16] and X2CT-GAN [17].

LIDC-IDRI (chest) dataset							
Method	↓ NLL	↑ Density	↑ Coverage	↑ SSIM	↑ PSNR	↓ MSE	↓ MAE
X2CT-GAN	N/A	0.87	0.88	0.321	19.68	<u>0.045</u>	0.151
CCX-rayNet	N/A	<u>1.41</u>	0.98	<u>0.386</u>	<u>22.66</u>	0.032	<u>0.108</u>
Ours	0.10	1.42	<u>0.97</u>	0.436	25.05	0.048	0.013

Table 5.2: **3D fidelity and distortion metrics on LIDC-IDRI (chest) dataset.** We followed the experimental protocol X2CT-GAN and CCX-rayNet with a wider Hounsfield unit range of -1,000 HU to +1,000 HU. We show additional analysis on out-of-distribution inputs in the Appendix A (Fig. 4).

Modeling Objects in Baggage Screening

The proposed approach was evaluated on a set of “*stream of commerce*” CT volumes from an aviation security context. It consists of X-ray images of passenger baggage with instances of prohibited items like knives, guns, and scissors, as well as the presence of cluttered and overlapping stream-of-commerce items. The total dataset consists of 5,964 CT bag volumes. The bag volumes were resampled to $1 \times 1 \times 1 \text{ mm}^3$, and subsequently, resized to a $256 \times 256 \times 256 \text{ mm}^3$ cubic area. This dataset comprises authentic bags and suitcases, depicting both the common and the forbidden items during air travel. A 2D subset of eight views were generated, each taken at 45-degree azimuth intervals per bag. For the model, $|\mathbf{b}_i| = 256$ was set, a codebook of 4,096 and train the transformer to predict sequences of length $16 \times 16 \times 16$.

Results. Table 5.1 reports results for conditional 3D modeling on two 2D input views comparing samples from our model and reconstructions from the competing approaches [17] and [16] for the baggage dataset. This dataset depicts varying topologies as there exists a wide variety of baggage, such as suitcases, and backpacks, in different sizes, styles, and materials. Moreover, the items/objects within passenger bags can be unpredictable and their arrangement and level of compactness could complicate their identification. Despite the challenging complexity of this dataset, results reflect that the proposed approach offers superior performance, with a very large margin in terms of density (+111% improvement over X2CT-GAN [17], and +57% over CCX-rayNet [16]) and coverage (+21% over X2CT-GAN and +9% over CCX-rayNet). In terms of distortion metrics, our samples are of higher quality, with a significant improvement in PSNR, MSE and MAE compared to the competing models [16, 17].

In Fig. 5.3 a set of 3D generated samples from the proposed approach is depicted and 3D reconstructions of the other models for the baggage dataset. It can be observed that the identity of each of the bags and suitcases is accurately modeled by our model, while CCX-rayNet [16] is limited at predicting the shape boundaries, and X2CT-GAN [17] shows a lack of diversity. It is worth noting that airport security officials focus on denser objects as these are more likely to be prohibited compared to less dense items such as clothes. In this context, our model is better able to model latent representations of denser structures, as indicated by darker colors, which aligns with real-world scenarios. In contrast, the other models depict blurring for both dense and soft structures, making it challenging to distinguish them.

Modeling Chest Anatomical Structures

A set of experiments using the publicly available dataset of chest CT scans was conducted, LIDC-IDRI [12]. The corresponding X-ray projections were generated by using digitally reconstructed radiograph technology (DRR). These scans cover a diverse range of ages of both sexes at different contrast types showing both normal and abnormal anatomy. The radiation source and imaging panel are assumed to

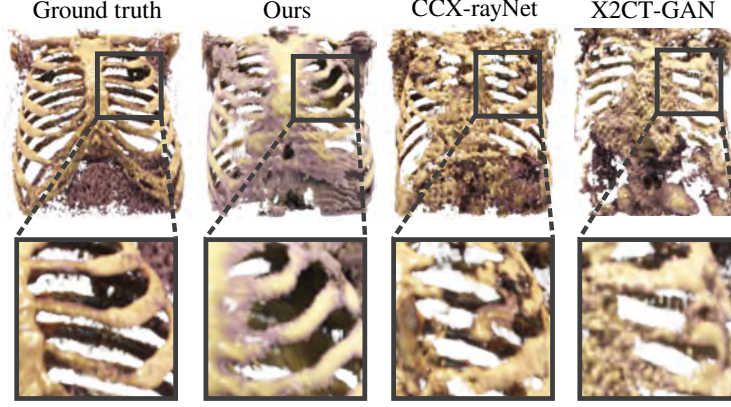


Figure 5.5: Local-level quality comparison of chest CT samples.

rotate in a circular orbit around the isocenter (the Z-axis), generating a couple of DRRs of 128×128 for coronal and axial planes. For this model, $|\mathbf{b}_i| = 256$ was set, a codebook of 1,024 and train our transformer to predict sequences of length $8 \times 8 \times 8$.

Results. As presented in Table 5.2, our method exhibits superior performance across almost all categories for density and coverage and surpasses the other models in SSIM, PSNR and MAE distortion metrics, achieving a 13% improvement in SSIM, 11% in PSNR, and 88% for MAE over the most competitive model, CCX-rayNet [16]. Furthermore, in comparison to X2CT-GAN [17], our model demonstrates an even greater degree of improvement performance.

In Fig. 5.4 a set of 3D generated samples from our approach and 3D reconstructions of the other models for the LIDC (chest) dataset is depicted. Although the sizes of both datasets are relatively small (compared to natural image datasets), our model displays no signs of overfitting and can produce diverse shapes that closely align with the ground truth. In particular, our approach achieves a greater degree of precision in modeling and disentangling soft tissue and bone structures. This is evidenced by the distinctive colors employed in the renderings, with a *purpleish* tone denoting soft tissue and a beige color representing bone. By contrast, the other methods produce blurred images where anatomical structures overlap, and as a result, densities cannot be accurately modeled since they are all merged together. For instance, samples from our approach are less noisy, while the other

models are only able to reasonably predict some parts of the volume and result in spatial blur/uncertainty in others (Fig. 5.5).

Analysis on Maximum Intensity Projections

An additional evaluation in terms of Maximum Intensity Projections (MIP) in 2D was performed, as they provide a representation of the highest intensity values along the spatial dimensions, accentuating low-level structures, which might be difficult to detect on individual slices of the 3D data. MIP are frequently used in real-world settings for clinical diagnosis, planning, object identification, or to simply enhance visibility. For our case, this could provide an additional interpretation of both quantitative and qualitative results.

In addition to X2CT-GAN [17], and CCX-rayNet [16], this section also compares our model with the NeRF-based method MedNeRF [221], which has been designed to handle complex images such as X-rays. For this, MedNeRF [221] was trained on both datasets and test them to render 3D-aware CT projections. For the other models, the MIP of their spatial dimensions was obtained, as well as the calculation of density and coverage, SSIM and PSNR. These results can be found in Table 5.3. Qualitative results of this comparison are presented in section 5.5.

5.4.2 Implementation Details

The 2D and 3D VQ-VAE from the first stage were trained simultaneously on independent NVIDIA A100 cards with a batch size of 32 and 8, respectively. For the VQ-AE, the framework proposed in the work of [128] was used, which substantially improves compression rates by learning a more information-rich codebook, while still permitting likelihood estimates. The 2D model takes approximately half the time of the 3D model to complete 100,000 iterations. Our unconstrained transformer from the second stage takes less than 5 hours to also complete 100,000 iterations; it effectively models 3D tokens very quickly. The weights from 500k iterations were used for the reconstructions (Table 5.4). The transformer can easily fit into memory on a GPU with 12GB of VRAM with a batch size of 20 while the VQ-VAEs can also fit in such a GPU. In practice, this requires the use of small batch sizes making

Table 5.3: **2D (Maximum Intensity Projections) fidelity and distortion metrics on both datasets.** We compare MedNeRF in addition to SOTA models and our approach.

(a) Baggage Security Screening dataset				
Method	↑ Density	↑ Coverage	↑ SSIM	↑ PSNR
X2CT-GAN [17]	0.51	0.68	0.65	34.49
CCX-rayNet [16]	<u>0.96</u>	<u>0.95</u>	<u>0.88</u>	<u>35.49</u>
MedNeRF [221]	0.91	0.46	0.79	25.11
Ours	1.84	0.99	0.91	39.43
(b) LIDC-IDRI (Chest) dataset				
X2CT-GAN [17]	<u>0.96</u>	0.85	0.32	21.38
CCX-rayNet [16]	0.76	<u>0.87</u>	<u>0.40</u>	24.20
MedNeRF [221]	0.90	<u>0.80</u>	0.38	27.02
Ours	1.17	0.91	0.42	<u>25.25</u>

Table 5.4: Training times (each for 100k iterations) and inference times for our approach, all performed on single NVIDIA A100 GPUs. The VQ-VAEs are independent and can be trained simultaneously on different GPUs.

Component	Training time (hs)	Inference time (s)
2D VQ-VAE	~ 23	~ 0.01
3D VQ-VAE	~ 56	~ 0.27
Transformer	~ 4.5	~ 10.9

training in reasonable times less practical.

5.5 Additional Experiments

The following section provides additional experiments including: ablation experiments including latent code sizes, codebook length and number of input views. Next, additional qualitative results showcasing: MedNeRF comparison, using out-of-distribution inputs, training on binary data, data augmentation details, and sampling using the autoregressive method. Finally, subsection 5.5.2 includes code snippets for both the conditional diffusion process and for sampling from the proposed model.

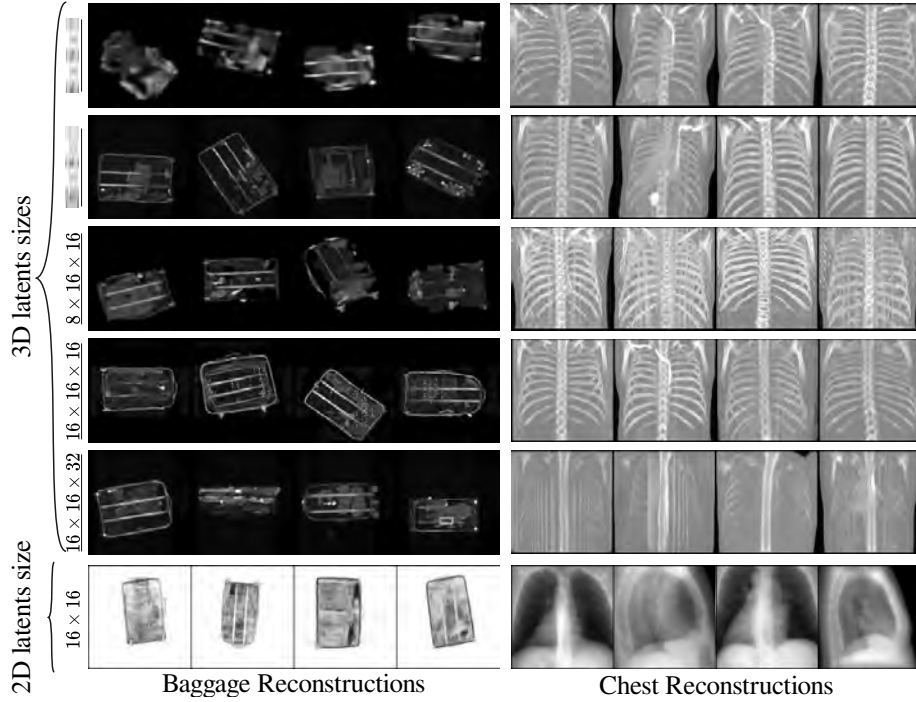


Figure 5.6: **Comparison of reconstruction quality using different latent code sizes on the two datasets.** 3D latents (ablation) correspond to the 3D VQ-VAE and 2D latents (fixed) to the 2D VQ-VAE. The chest dataset shows faithful reconstruction even with a small size, whereas the baggage security screening dataset requires a more complex latent representation. Larger latent code sizes could allow the model to learn more complex structures (baggage case) or mislead to learn unhelpful features (chest case).

2D Comparison Fig. 5.8 contains qualitative results for both datasets of the 2D evaluation that also includes the MedNeRF method, which renders 3D-aware CT projections. This evaluation consists of comparing Maximum Intensity Projections from the spatial dimensions. The proposed method allows faithful samples compared

↓ NLL		
Latent code sizes	Chest	Baggage
$8 \times 8 \times 8$	0.031	0.0087
$16 \times 8 \times 8$	0.032	0.0056
$8 \times 16 \times 16$	0.029	0.0055
$16 \times 16 \times 16$	0.032	0.0044
$16 \times 16 \times 32$	0.067	0.0056

Table 5.5: Quantitative evaluation based on validation Negative Log-Likelihood (NLL) using different latent code sizes for the discrete representations learned by the two VQ-VAE in Stage 1 of our approach.

Codebook, $\text{argmin}(\mathbf{x}, \mathcal{B})$			
Length	Chest (10^{-3})	Length	Baggage (10^{-5})
64	4.01	1024	5.8
128	0.85	2048	4.2
512	0.65	4096	1.7

Table 5.6: Quantitative evaluation based on the codebook loss using different codebook lengths for the discrete representations learned by the two VQ-VAE in Stage 1 of our approach. Our approach allows learning complex data distributions (e.g. baggage security dataset) by increasing the length. For our main experiments we used a length of 512 and 4,096 for the chest and baggage dataset, respectively.

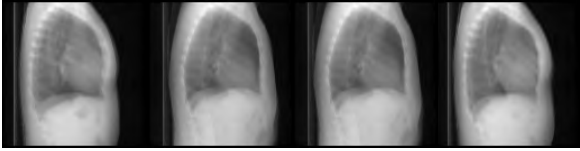
	GT	input views:		
		2	4	8
samples				
↓ re-weighted ELBO		1.163	0.930	0.929

Figure 5.7: **Ablation using different number of input 2D views for conditional 3D modeling on the LIDC-IDRI (chest) dataset.** While increasing the number of inputs views from 2 to 4 brings additional performance, further increments don’t necessarily result in linear returns. For the main experiments only 2 inputs views were used for both datasets.

to competing methods. Specifically, our samples better highlight denser objects which are of particular interest in the detection of prohibited items.

Diffusion comparison The model robustness was tested on out-of-distribution input X-rays and generate samples comparing absorbing diffusion and the autoregressive method using the same transformer architecture. Despite domain differ-

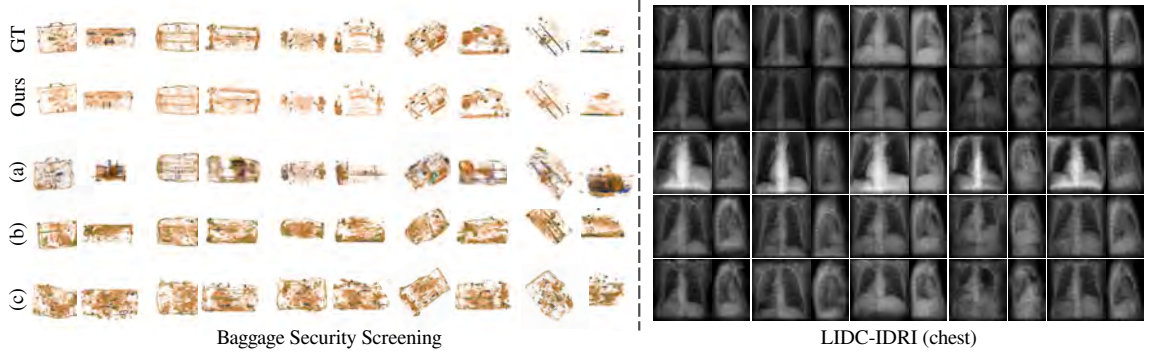


Figure 5.8: Comparison of Biplanar Maximum Intensity Projections (MIP) on the baggage security screening dataset and on LIDC-IDRI (chest) dataset. (a) denotes the MedNeRF model, (b) CCX-rayNet and (c) X2CT-GAN.

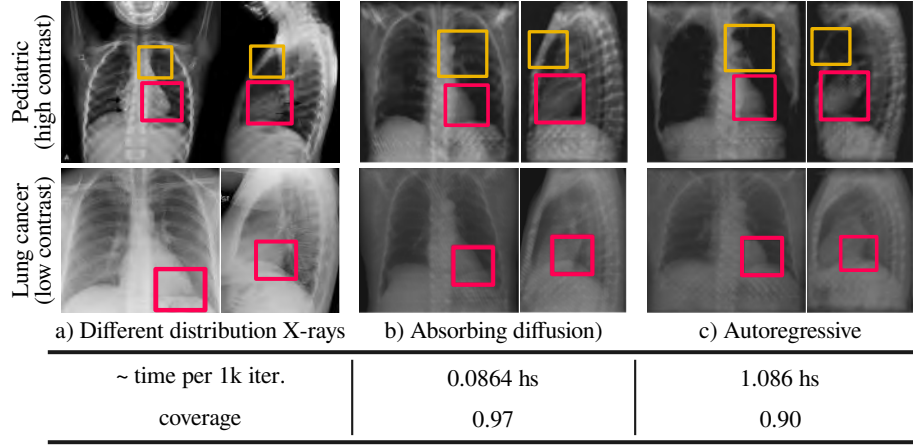


Figure 5.9: **Absorbing diffusion vs autoregressive samples on out-of-distribution input X-rays.** The proposed model was tested on different distributions including pediatric and lung cancer X-rays with different radiographic contrast (high and low contrast).

ences, the proposed model is able to generate accurate samples without requiring any kind of domain supervision, suggesting that our learned discrete representations achieve effective compression to remain invariant to low-level features like contrast while encoding essential structures such as bone and soft tissue (Fig. 5.9).

5.5.1 Controlled Data Augmentations

To avoid overfitting, the stochastic discriminator augmentation framework was extended from StyleGAN2-ADA [252] to handle 3D data. This solution involves augmenting both real and generated data by the VQ-VAE using a set of style and spatial augmentations with a probability $p < 1$. Unlike other data augmentation strategies,



Figure 5.10: **Visualization of additional results on ShapeNet [18].** Training on binary volumetric data in contrast to continuous intensities isn’t trivial as it might lead to instabilities in architectures like simple CNN-based GAN. Voxels with values other than 0 or 1 are flagged as fake by the discriminator, hindering continuous optimization [19]. However, the proposed model can effectively learn this type of data as our approach doesn’t (necessarily) rely on adversarial training.

non-invertible augmentations can be incorporated with an adaptive p -value based on an overfitting heuristic. Results suggest that this prevents the discriminator becoming more confident, and thus both real and fake predictions take more time to diverge. As a result, both VQ-VAE are able to learn richer representations while delaying the drop in its the validation accuracy. Note that our approach doesn’t rely on adversarial training, thus, the incorporation of a discriminator is optional.

5.5.2 Example Code

Python-like code for training a conditional absorbing diffusion process is provided 5.11a, and sampling from our 2D to 3D translation model 5.11b. The use of a Transformer allows the learned distribution to be easily conditioned on arbitrary input shapes, by simply concatenating the conditioning signal with the noisy data. The linear masking schedule allows sampling with smaller numbers of steps, to speed the process up, by simply passing in a smaller value for T .

5.6 Availability & Acknowledgments

The project page of this work along with additional resources can be found at <https://abrilcf.github.io/publications/CodeDiff3D/>. This work was supported by the Mexican Council of Humanities, Sciences and Technologies (CONAHCyT), and by the EPSRC NorthFutures project (ref: EP/X031012/1).

5.7 Conclusion

This chapter proposed a novel 2D to 3D translation approach based on conditional diffusion using transformers. Findings indicate that compressing each domain independently offers several key advantages. The discrete compressed space allows for both fast and high-resolution image synthesis, where the 2D and 3D compression networks can be trained and verified independently without requiring aligned datasets. In particular, full-coverage attention over the complete information-rich 2D codebooks from a few views significantly improves the synthesis of new 3D images, where any part of all 2D inputs can contribute to the voxel predictions.

The proposed approach is surprisingly simple and intuitive in practice as diffusion models are shown to have excellent mode coverage giving diverse samples [138]. In the future, it's possible to consider scaling the proposed approach with larger models trained on more diverse datasets, provide an in-depth study on hallucinated outputs, and see how well it can generalize between very different imaging modalities.

```

def diffusion_training_loss(c_0, Z, T, mask_id):
    c_t, b = c_0.clone(), c_0.size(0)
    # Randomly sample diffusion time steps
    t = torch.randint(1, T+1, (b,))
    # Randomly mask tokens with probability t/T
    mask = torch.rand_like(c_0) < (t / T)
    c_t[mask] = mask_id
    # Calculate  $p(c_0 \mid c_t, Z)$ 
    logits = Transformer(torch.cat(Z, c_t))
    # Calculate reweighted ELBO loss
    loss = cross_entropy(logits, c_0) * (T-t+1)/T
    return loss

```

(a) Python-like code snippet for training a conditional Absorbing Diffusion model.

```

def sample(imgs_2d, T, mask_id, latent_size):
    b = imgs_2d.size(0)
    # Compress 2D images with 2D VQ-Encoder
    Z = vqae_2d.encoder(imgs_2d)
    # Initialise 3D latents with all masks
    c_t = torch.full((b, latent_size), mask_id)
    # Track which latents have been unmasked
    unmasked = torch.zeros_like(c_t, dtype=bool)

    # Loop over sampling steps
    for t in reversed(list(range(1, T+1))):
        # Randomly choose where to unmask
        changes = torch.rand(c_t.shape) < 1/t
        # Don't unmask anywhere already unmasked
        changes = torch.bitwise_xor(changes, \
            torch.bitwise_and(changes, unmasked))
        # Update unmasked
        unmasked = torch.bitwise_or(unmasked, changes)

        # Sample from  $p(c_{t-1} \mid c_t, Z)$ 
        logits = Transformer(torch.cat(Z, c_t))
        dist = Categorical(logits)
        c_0_hat = dist.sample()
        c_t[changes] = c_0_hat[changes]

    # Decompress 3D latents with 3D VQ-Decoder
    imgs_3d = vqae_2d.decoder(c_t)
    return imgs_3d

```

(b) Python-like code snippet for 2D to 3D translation using our approach.

CHAPTER 6

Conclusions

Throughout years of research in deep generative modeling, the problem of image-to-image translation has broadened its scope to specialized applications, including medical imaging. Depending on data availability and the level of precision required for the application, state-of-the-art deep learning models now translate images between different modalities and domains without laborious preprocessing, paired datasets, or geometric alignment. Crucially, the analysis and understanding of the shape and structure of data enable the effective design of hybrid deep learning models. These models can circumvent data constraints and generate accurate images that could be potentially used in clinical settings.

The generation of 3D computed tomography from 2D X-rays remains a challenging problem. However, interdisciplinary collaboration between computer scientists and clinicians has increased in recent years. Additionally, there has been a growing incentive to open source the code for new developments.

As discussed in this thesis, significant progress has been made in 2D to 3D image translation in recent years. However, several questions remain unanswered:

1. How can we strike a balance between generation quality and generalization?
2. Will access to more input data necessarily result in better image-to-image

translation mappings?

3. Are implicit neural representations of data better suited for 3D-related tasks, including 2D to 3D image translation?

I believe these problems are interconnected and open up interesting avenues for future research. The contributions presented in this thesis explore various approaches to address the problem of 2D to 3D image translation for medical imaging, specifically focusing on X-ray to CT generation. However, designing a generative model that can handle data constraints and generalize with minimal hallucination would be truly groundbreaking. The ability to generate accurate medical imaging in a less invasive, cost-effective, while minimizing harm to patients could have significant positive impacts on clinical diagnosis, planning, and treatment.

The next generation of 2D to 3D generative models will probably tackle many data limitations including the need for large datasets. As research discovers how DL models are better able to learn the distribution of data, we expect the community to gradually switch its focus onto compact implicit neural representations instead of relying on the explicit form of data. This has relevant implications for privacy, particularly in medical settings.

6.1 Contributions

To my knowledge, the MedNeRF work presented in Chapter 3 is the first to adapt the NeRF model for medical imaging. This work of research has received significant attention and citations in various published surveys and review papers, including the prestigious International Conference on Computer Vision 2023 (ICCV) [253], the IEEE Access journal [254], among others [255–257]. I believe that the work presented in this thesis gave origin to numerous other applications utilizing NeRF by introducing a protocol to preprocess 3D medical data, such as CT images [200, 258, 259], endoscopic tissue videos [260, 261], ultrasound [262], oral panoramic imaging [263], electron microscopy [264], etc. These applications have seen improvements in terms of representation choices, view encodings [260], and generalization [265].

Since translating images from spaces that differ in dimensionality poses further significant challenges, one can think of the possibility of preprocessing the data from the source domain so it matches the dimensionality of the target domain. In this regard, previous work has resorted to some sort of prior from the target domain or incorporating additional information that constrains the mapping. In Chapter 4 this thesis explored the intriguing research question of whether it’s possible to solve the dimensionality mismatch without accessing additional information. To this end, this chapter proposed a combination of data transformations that provide a solution to this problem while demonstrating notable generalization and accuracy gains. Through extensive ablations and experiments, this chapter showed the applicability of the proposed method in multiple datasets for the task of X-ray to CT generation. In particular, this work opens up future research questions about the way generative models learn and generalize.

Data alignment is an important aspect of I2I as described in Subsection 2.5.5. For same-dimensionality settings, this is a more straightforward concept as it entails that the mapping between objects in images can have a direct correspondence. However, this is not the case for image translation of data with differing dimensionalities, especially because the coordinate systems of both source and target domains will necessarily be different. In Chapter 5, the goal was to circumvent data alignment for 2D to 3D image translation between X-rays and CT scans of two different domains: medical and security. Furthermore, the proposed method not only solves the alignment issue but also provides a generative and unpaired method without the requirement of any kind of priors. In a nutshell, this was achieved by decoupling and compressing both image domains and then modeling the conditional probability of 3D data given a few 2D views using a discrete diffusion model that acts on VQ codes of the data. The proposed method demonstrated SOTA performance for both medical and security datasets compared to competitive methods.

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6.1.1 Code Availability

- Chapter 3 - <https://github.com/abrilcf/mednerf>
- Chapter 4 - https://github.com/abrilcf/3D-3D_repeat-concatenate
- Chapter 5 - <https://github.com/samb-t/x2ct-vqvae>

6.2 Limitations and Future Work

There is a clear limitation when handling 3D data in its explicit form (e.g., volumes, voxel grids). Scaling to higher resolutions or larger batches requires more memory, which can be difficult to obtain. On the other hand, efficiency and fast training have been recent and relevant objectives of research in the field, so we should also aim for faster loading of data. Despite various efforts to implement fast dataloaders for PyTorch, I argue there are alternative data representations that would use both less memory and time.

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